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Prediction of consumers refill frequency of LPG: A study using explainable machine learning

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ABSTRACT

Launched in 2016, the PMUY Programme of the Government of India aimed to provide 8 crore LPG connections to women in rural households over four years. After acquiring a new connection, some households appeared uninterested in ordering subsequent subsidized LPG refills, impacting programme's sustainability, and targeting strategy. We propose a prediction model using "Explainable Machine Learning" to anticipate the beneficiaries' refill frequency with a view to improving LPG-refills and social targeting. In this paper, we suggest an enhanced stacked SVM (ISS) model for classification, which is contrasted with state-of-art ML models: Random Forest (RF), SVM-RBF, Naive Bayes (NB), and Decision Tree (C5.0). Some of the performance matrices that are used to evaluate the models include accuracy, sensitivity, specificity, Cohen's Kappa statistics, Receiver Operating Characteristic curve (ROC), and area under the curve (AUC). The proposed approach, which was validated with 10-fold cross validation, produced the best overall accuracies for data splits of 50–50, 66–34, and 80–20. The "Explainable AI (XAI)" model has also been used to describe how models and features interact, and to discuss the importance of features and their contributions to prediction. The recommended XAI will aid in efficient "beneficiary targeting" and "policy interventions".

1. Introduction

Historically, LPG was primarily utilized for cooking by individuals with medium and high incomes in India until a few decades ago [1]. As time progressed, LPG became the preferred and more cost-effective cooking fuel among the general population. However, those situated just above and below the poverty line were unable to access LPG due to its prohibitive cost, leaving them with no choice but to rely on other solid fuels such as firewood, kerosene, dried cow dung, and coal etc. Unfortunately, using these alternatives to LPG leads to indoor air pollution, poor health, and deforestation [2] etc.

The Indian Government has implemented a range of social inclusion initiatives, including programs such as "Pradhan Mantri Jan

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Dhan Yojana" (PMJDY) which promotes financial inclusion for all. Additionally, the "PAHAL or Direct Benefit Transfer for LPG" (DBTL) initiative was introduced to transfer subsidies directly to the bank accounts of beneficiaries by integrating Aadhar number, Bank Account and Mobile number of individuals [3]. The "GiveItUp" campaign also provided citizens with the option to forgo LPG subsidies, while the "Pradhan Mantri Ujjwala Yojana" (PMUY) aimed to provide clean cooking fuel to poor households, particularly in rural areas based on the Socio Economic and Caste Census 2011. The Indian government worked to establish a technological framework alongside these social inclusion programs to ensure that their goal of providing inclusive social schemes for the poor was realized.

The Pradhan Mantri Ujjwala Yojana (PMUY) was introduced in 2016 by the Indian government as a flagship initiative. Thanks to its efficient implementation, the program successfully enrolled 4 crore households in under two years. By 2020–2021, PMUY had expanded to over 8 crore households and over 9 crore customers by 2023. As of July 2023, LPG usage in India has reached 99.8 %, with a total of 9.59 Crores PMUY connections. Since then, both central and state governments have launched various versions of the PMUY scheme [4]. Its primary objectives are to eliminate indoor air pollution, reduce fuel collection for household cooking, curb deforestation, and promote women's empowerment and social inclusion. To meet the significant increase in domestic LPG consumers from 19.88 Crore in 2016 to 31.5 Crore in 2023, the government has taken massive steps through Oil Marketing Companies (OMC) to improve the supply chain infrastructure for LPG distribution. During this period, new LPG distributors were commissioned, increasing the number from 17916 in 2016–25386 in 2023. The number of bottling plants also increased from 188 in 2016 to 208 LPG in 2023.

The sustainability of the PMUY scheme is a concern for the Oil Marketing Companies and the Indian Government due to the low refill consumption by beneficiaries. Sales data analysis showed that PMUY beneficiaries consumed an average of 2.98 refills annually, while non-PMUY customers consumed 6.73 refills in 2018–19. The audit report by Comptroller and Auditor General of India (CAG) in their report published in 2019 had highlighted that 17.61 % of PMUY beneficiaries never returned for the second LPG refill cylinder, while 33.02 % consumed only 1–3 refills after the initial purchase. This trend has persisted, with a slight increase in per-capita consumption in 2019-20 and 2020–21, followed by a dip in 2021-22 and 2022–23. The prohibitive cost of LPG refills is the primary reason that deters PMUY beneficiaries from lifting refills, which they cannot afford. Additionally, inadequate LPG supply chain in rural areas, non-existent home delivery of LPG refill by distributors in rural areas, single bottled connections, easy access to conventional fuel in rural areas, etc., contribute to poor upliftment of LPG refill by PMUY consumers. The low refill upliftment by PMUY beneficiaries jeopardizes the objective of GOI for launching the PMUY scheme and its associated frameworks. The fall in expected consumption of LPG affects the utilization of the augmented infrastructures and has direct implications for the achievement of the Sustainable Development Goals (SDG) objectives on clean and affordable energy for all mandated by the United Nations in India.

In the past, studies have looked at why PMUY beneficiaries are not buying refills, but these studies were limited and did not give a nationwide view. To understand why PMUY beneficiaries are not buying refills across India, we can use the Theory of Customer Churn in the social sector. We believe this type of study has not been done before for the PMUY beneficiaries. To do this analysis, we need to classify PMUY customers into High-frequency and Low-frequency segments. Once we have this classification, the Indian Government and OMCs can create targeted policies to help encourage more refill uptake by the consumers.

This research paper introduces a model called XISS (Explainable Improved Stacked SVM), which aims to classify customers into High-frequency and Low-frequency LPG segments. The model is designed to provide answers to various research questions, as follows.

RQ1. What are the notable features / factors which contribute to the classification of customers as High-frequency and Low-frequency customers?

RQ2. How can Explainable Artificial Intelligence (XAI) assist in understanding the relative importance of each factor?

RQ3. How good is the proposed XISS machine learning predicting model in classifying customers into High-Frequency and Low-Frequency segments?

2. Literature review

Studies have shown that energy usage has a significant impact on pollution, health, and the environment. In particular, the use of traditional cooking fuel has been studied for its effects on health due to the increase in household air pollution. Various solutions have been suggested based on studies such as the Global Burden of Disease Study, WHO Indoor Air Pollution and Health Database, The Energy Sector Health Impacts Model (E-SHIP), The HAPIN Study (Household Air Pollution Intervention Network), and The Global Alliance for Clean Cookstoves Research and Publications. The impact of conventional fuels, like biomass (wood, crop residues, and dung) and coal, as cooking fuels varies across regions and countries. These fuels are commonly used for cooking in many parts of the world, particularly in low and middle-income countries. Here are some of the key impacts of conventional cooking fuels: 1. Health Impact: The use of biomass and coal for cooking in poorly ventilated indoor spaces can lead to indoor air pollution, which can cause respiratory diseases, especially among women and children who are exposed to smoke on a daily basis. 2. Environmental Impact: Burning biomass and coal for cooking contributes to deforestation because trees are often cut down for fuel. Furthermore, coal combustion releases greenhouse gases and other pollutants that contribute to air pollution and climate change. 3. Time and Labor Intensive: Collecting firewood or biomass for cooking can be time-consuming for women and girls who are often responsible for this task in many cultures. This can limit their educational and economic opportunities.

In the study published in The Lancet journal [5], has highlighted the impact of household air pollution (HAP) in low- and middle-income countries resulting in a rise in respiratory diseases [6]. in the article published in Global Heart has listed the effect of Household Air Pollution (HAP) due to solid fuel use in cardiovascular diseases (CVD) and respiratory illness. Although respiratory illness is the primary fallout of HAP, the study also indicates that CVD also accounts for the substantial number of mortalities due to

HAP across the world. The study reiterates the need for policy formation for the promotion of fuel-efficient low-emission cooking mediums in low-income countries around the world.

Governments worldwide have been promoting the use of LPG as a cleaner cooking fuel through policy interventions, such as India's PMUY, Indonesia's PKH, Ghana's LPG Promotion Policy, Mexico's Programa de Sustitución de Energéticos, Kenya's National LPG Project, Brazil's Bolsa Família, Nepal's Subsidized LPG Program, and Bangladesh's LPG Conversion Program are a few prominent policy interventions.

According to a recent study conducted by Ref. [7] using the Probit Estimation model, implementing clean energy in households yields a range of benefits and long-term effects. These benefits include reducing energy poverty, improving health, increasing household income, and lowering medical expenses. However, the study notes that rural populations are slow to transition to cleaner cooking fuel, particularly the adoption of LPG. In a separate study by Ref. [8], it was found that while LPG is a clean cooking fuel with positive impacts on users, the cost of fuel remains a significant obstacle for rural households in India. A broader policy framework is needed to encourage households to fully embrace clean cooking fuel, and LPG may be an ideal solution as a cleaner cooking fuel source. Additionally [2], highlighted the negative consequences of using traditional cooking fuel on health and the environment in rural areas of India. Other studies by Refs. [9–11] also showed why rural populations continue to use traditional cooking fuel sources, as LPG as a clean cooking fuel source is not affordable for them.

2.1. PMUY – Pradhan Mantri Ujjwala Yojana

The Pradhan Mantri Ujjwala Yojana (PMUY) is a significant initiative of the Government of India that was launched on May 1, 2016, under the leadership of our Prime Minister. The primary aim of the PMUY scheme is to provide access to clean cooking fuel, specifically Liquefied Petroleum Gas (LPG), to women belonging to below-poverty-line (BPL) households across India. The scheme is designed to improve the health and well-being of women and their families by reducing their reliance on traditional and more polluting cooking fuels such as biomass (wood, crop residues), cow dung, and kerosene.

Notable features of the PMUY scheme include the provision of free LPG connections to eligible BPL women, which removes the initial financial barrier to acquiring an LPG connection. This includes a free LPG connection kit that typically consists of a gas cylinder, regulator, pipe, and stove. Beneficiaries of PMUY also receive subsidies on LPG cylinder refills, making them more affordable for continued use. These subsidies are directly provided to beneficiaries' bank accounts through the Direct Benefit Transfer (DBT) system. The scheme aims to expand the reach of LPG distribution networks to rural and remote areas, ensuring that even households in hard-to-reach locations can access clean cooking fuel. PMUY emphasizes the health benefits of using LPG for cooking, as it reduces exposure to indoor air pollution and the associated health risks. The scheme also promotes the use of safe and modern LPG stoves. Furthermore, the scheme is focused on empowering women by making them the primary beneficiaries of the initiative. It recognizes the significant role women play in household cooking and their exposure to health risks associated with traditional fuels. Encouraging the use of LPG, PMUY contributes to reducing deforestation, as it reduces the demand for firewood and other biomass fuels. It also helps mitigate greenhouse gas emissions associated with traditional cooking methods.

The Pradhan Mantri Ujjwala Yojana has been instrumental in promoting clean cooking practices in India, improving the health and quality of life for millions of BPL households, and reducing the environmental impact of traditional cooking fuels. It has been a significant step towards achieving the government's goal of universal access to clean cooking fuel for all thus also achieving the SDG objectives.

A recent study conducted by Ref. [12] explored how Business Model Innovation (BMI) can help bridge the gap in public-sector companies, particularly in the LPG business in India, through supply-side and demand-side strategies and technological innovation via digital initiatives. The research analysed the distinct phases of the LPG business from 1955 to 2013, outlining the transition from LPG subsidy to the dismantling of the subsidy regime, driven by OMC (Oil Marketing Companies) and MOPNG. Another study conducted by Ref. [13] assessed the perception of beneficiaries regarding the benefits of the PMUY schemes, an initiative by the government aimed at improving quality of life, social inclusion, and ecological benefits. The study found that education, LPG refill subsidies, and changes in the use of other fuel sources for cooking were responsible for improving the perception of the scheme.

A recent study examined numerous factors such as social class and gender to suggest that policies promoting greater gender equality and social inclusion could lead to more successful implementation of social benefit programs like PMUY [14]. Another study analysed the use of different fuels across the country, considering sociocultural factors and their effects on health, including the use of LPG. This study also examined the impact of the PMUY scheme and the adoption of LPG as the primary cooking fuel, identifying various methods to improve the sustainable use of LPG in households [15].

A study by Ref. [16] highlighted the benefits of the PMUY scheme, including its affordability and health benefits for rural households. The study also described the implementation procedure of the scheme. Similarly, another study by Ref. [17] examined the impact of cooking fuel choices on the health of urban and rural populations.

A study conducted by Ref. [18] in two states named Raipur in Chhattisgarh and Ranchi in Jharkhand used the TOBIT regression model to analyse survey data and determine the adoption of LPG and the factors that affect it. The study found that the adoption of LPG increased due to PMUY, but continuous usage depended on a range of factors, such as behavioural aspects, education, cost of the cylinder, and supply chain of LPG delivery.

Another study by Ref. [19] used the MILP and a DSS to determine how many distributors are needed to cover the policy objective and increase the adoption and penetration of LPG coverage. The study primarily focused on factors such as the caste, class, and gender of beneficiaries to determine the implementation, effectiveness, and adoption of the PMUY policy, which is based on the Socio-Economic Caste Census SECC (2011) data. The study also found that affordability, availability of LPG, and doorstep delivery by

distributors were crucial in improving LPG adoption through the PMUY scheme.

A study by Ref. [16] considered various important aspects, including Economic, Technological, Institutional, and socio-cultural factors, to outline the adoption of cleaner fuel by rural households across India. The study also highlighted the initiatives by the Govt of India in the adoption of cleaner fuel through LPG and various other schemes for the promotion of solar and biogas, as outlined by Ref. [20].

A study in rural household of Rajasthan post PMUY Scheme implementation [21] has shown several benefits to the beneficiaries such as environment friendly cooking, additional time for the family, entertainment & hobby, relieve from time consuming firewood collection process as fuel etc. These benefits of the PMUY scheme have also made women folks socially and economically empowered. Usage of Internet by the rural population in adoption of cleaner cooking fuel and slow transition to cleaner cooking fuel over a period of time was outlined by Ref. [22].

Table 1
Related research.

Year	Objective	Technique	Findings	Citations
2023	Socio economic survey in a remote village of Gujarat and penetration of PMUY scheme.	Sample Survey and Statistical analysis	Awareness of govt schemes is lacking among the population who are mostly in below poverty line and need financial assistance in elevating their conditions	[34]
2023	Study to identify the constraints in adoption of LPG by consumers and distributors	Exploratory Research	The study recommends policy interventions to irradicate the constraints in sustainable adoption by PMUY beneficiaries.	[35]
2022	Study of PMUY benefits extended to women empowerment through sustainable use of LPG refills	Empirical Study	The study identifies the issue pertaining to low refill uplift and suggests various steps that could be taken.	[36]
2022	Adoption of the PMUY scheme and factors that influence beneficiaries on using LPG as cooking fuel as a primary fuel	Empirical Study	Household income and education were the primary reason/factor the use of LPG as primary cooking fuel by the beneficiaries.	[37]
2021	Study the benefits of PMUY scheme and clean energy transition by the beneficiaries based on the perception of the benefits of the PMUY scheme.	Regression Modelling	The study found that the beneficiaries find no direct corelation of the intended benefits such as standard of living and LPG connection from PMUY scheme.	[13]
2021	To report exact penetration of LPG across the country and the detailed consumption pattern	Statistical analysis	Provided hard facts of LPG usage and its exact number of beneficiaries in each of the states across the country	[38]
2021	Study the benefits of PMUY scheme rolled out from 2016 to 2019 and its effectiveness as per its initial objective of the scheme	Semi-structured interviews and data analysis	The study found out value of time, speed and flexibility, value of saved time as primary benefits of the LPG usage from the PMUY beneficiaries rather than the objective of social inclusion, clean cooking fuel medium etc in the PMUY scheme	[39]
2020	Case study to discuss in detailed about the PMUY scheme and its benefits and its achievement till 2020	Sample Survey and Statistical analysis	Generic details of the PMUY scheme and various schemes that lead up to the launch of PMUY scheme.	[4]
2020	Study intended to find the adoption of LPG as clean cooking fuel by the PMUY beneficiaries	Empirical Study	Found that the LPG consumption by PMUY beneficiaries was lower than the non-PMUY beneficiaries. Subsidy for the PMUY connection was a success but continued adoption of subsequent refill of LPG cylinder was concern for the BPL households	[26]
2020	Analyse the impact of the PMUY scheme on the socio-economic status of women in rural areas	Survey based approach	The study found that there was a huge socio-economic impact of PMUY scheme on the women in rural areas and had achieved a lot in women empowerment.	[21]
2019	To analyse the penetration of the PMUY scheme in Nagpur district of Maharashtra & study issues and challenges in adoption of the scheme while switching to clean cooking fuel, LPG	Stratified random sampling technique	The PMUY scheme had achieved its primary objective of providing LPG connections to the poor household but the problem of sustained use of LPG still remained for the beneficiaries	[25]
2019	Case study to determine the fuel use pattern and trend in India and the socio-cultural, economical, and behavioural factors that influence the shift to clearer fuel	Decision support system for Pradhan Mantri Ujjwala Yojana	Major finding was that 90 % of the rural population use non-clean fuel for cooking and only 2 % population has adopted clean fuel for cooking from 1993–94 to 2009-10	[15]
2018	Analyse integration of LPG as clean cooking fuel in rural household in six states in India	Analysis of survey data	Fuel cost is one of the primary factors in clean Fuel adoption; Dual use of various fuel other than LPG is the generic norm; LPG has greater prospect of adoption in rural India	[40]
2018	To explore how PMUY scheme impacted the social inclusion of poor and marginalized section of Indian Society through cleaner cooking fuel adoption using LPG	Empirical Study	Study found that the accessibility to cleaner fuel and affordability is the primary issue for the rural poor in India	[16]
2018	Quantitatively analyse the implementation of the PMUY Scheme	MILP Technique	Estimated the number of dealers required to sustain the LPG penetration based on the govt projections and the PMUY scheme	[19]

2.2. Long term sustainability of the PMUY scheme

In a study on energy consumed by households [23], discussed that the transition pattern from one energy source to another cleaner and efficient energy source primarily depended on the income growth of the individual households.

A study in two districts of Bihar on the beneficiaries of PMUY scheme [24] with its perceived benefits has brought out the direct connection between the education of the beneficiaries and the adaptation and sustainability of the scheme on the long term. Another study of the rural area of Nagpur district of Maharashtra on the PMUY scheme implementation and its sustainability over the longer period to make the rural household move to cleaner fuel for cooking identified various barriers such as affordability, reliability, accountability, and viability for the long-term sustainability of the scheme and outlines steps that need to be taken [25].

Analysis of the PMUY scheme and its applicability [26] was done based on the empirical data arrived at the conclusion that though the scheme was successful as per the initial intended purpose but was lacking when it came to sustained adoption of the benefits over the longer period of time. Despite the upfront benefits of free connection, the beneficiaries were not able to uplift the LPG refill cylinders subsequently. Subsequent studies [27,28] on the issues of sustained upliftment of LPG refill by the poor households shed light on numerous factors primary being the affordability factor, income criteria, home delivery amongst others.

A study was conducted [29] to determine the adoption of cleaner fuel in comparison to biomass which is the most popular cooking fuels in rural households of Maharashtra. It identified that various socio-economic factors such as affordability, availability of refill, education etc. determine the sustainability of the PMUY scheme in long term. Study in tribal regions of Odisha has shown similar trend on PMUY scheme adoption [30] and highlighted the issues in long-term sustenance of the scheme and usage of LPG as cooking fuel on continued basis.

Concerns have been raised about the long-term sustainability of the PMUY scheme by both the Oil Marketing Companies and the Indian Government, as noted by Refs. [31,32] and audit report by Comptroller and Auditor General of India, [33]. Comptroller and Auditor General (CAG) in their report has advised GOI to encourage the use of LPG among the beneficiaries of the Pradhan Mantri Ujjwala Yojana (PMUY). The CAGrecommended that financial support should be provided to these beneficiaries for the purchase of LPG stoves and cylinders, while also educating them about the benefits of LPG use. This would help ensure the sustainability of LPG usage.

Table 1 summarises the related work for this study.

2.3. Research Gap Analysis

Since its launch, the PMUY scheme has been a popular subject of research. Studies have been conducted to analyse its policy matters, benefits, and objectives. Some studies have focused on specific areas to identify the challenges faced by beneficiaries when using LPG sustainably. All studies have emphasized the sustainability of LPG issues and have identified probable reasons such as the prohibitive cost of LPG refill, inadequate supply chain, non-existent home delivery mechanism, and lack of awareness of benefits. However, there has not been a pan-India study using AI/ML to segment beneficiaries into diverse groups based on the theory of customer churn. This research aims to segment beneficiaries and suggest ways to promote sustainable LPG usage among them wherein OMCs and GOI can device policy objectives for the segments thereby enhance the LPG usage amongst the PMUY beneficiaries. Fig. 1 depicts the research gap identified for this research.

3. Methodology

Fig. 2 shows the research framework of this research work. From the framework, it can be seen that the PMUY program generates sales and customer profile data. A summary of the PMUY beneficiary information is obtained and put through a data-cleaning process to prepare it for AI/ML modelling. Once the data cleaning process is complete, it is passed through various AI/ML models and their

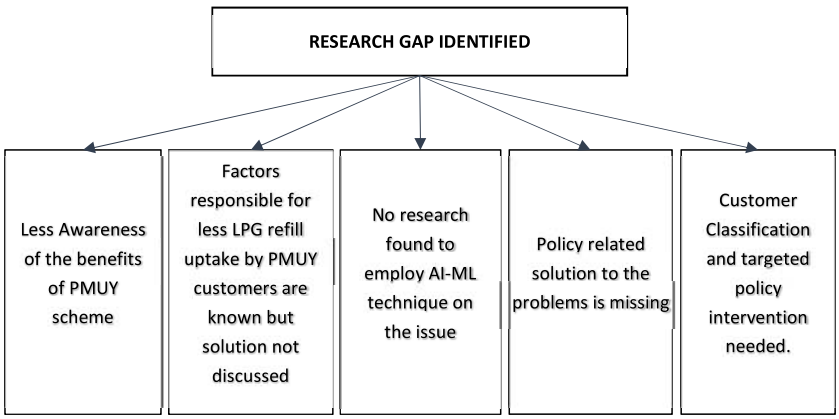


Fig. 1. Gap analysis.

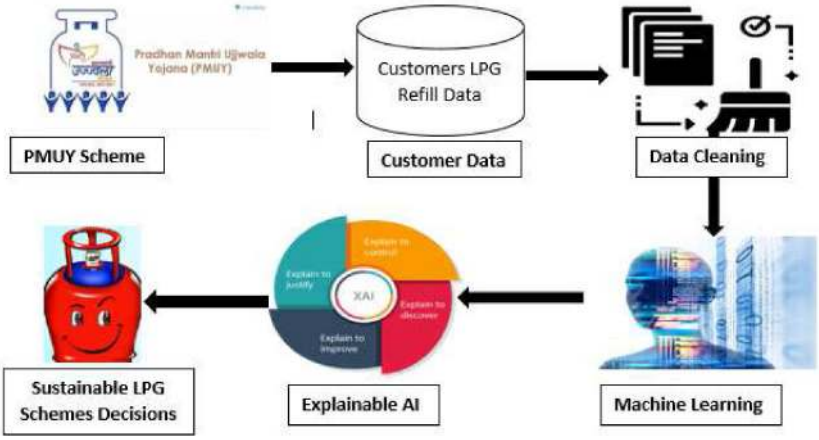


Fig. 2. Framework of this study.

observations recorded. The output from different models is then compared and passed through another phase where XAI (Explainable Artificial Intelligence) analyses the recorded output and provides an interpretation in a human-comprehensible form to aid decision-making.

3.1. Data description

This study has used sample data set having demographic, socio-economic and historical information of refill booking by LPG customers. The primary data for this study has been sourced from researcher who sourced the same from OMCs billing platform which caters to the end-to-end process of Customer Enrolment, Refill Booking, Deliveries of LPG refill cylinder, grant of subsidies etc. and all other allied processes of day-to-day LPG Sales business.

The sample dataset selected for study comprises of randomly chosen 8000 PMUY customers from each of the thirty-five states and Union territories of country. This constituted customers from 329 districts, 3020 different towns and villages spread across 1349 different pin-codes. Representation from all eight eligible sub-types of PMUY categories was observed. The dataset was found to comprise of 16 % of urban populace vs 84 % from rural populace. All the features of the data have been shown in Table 2.

4. Machine learning classifiers

A machine learning classifier refers to an algorithm utilized in supervised learning that classifies data into predetermined categories based on patterns derived from a labelled training dataset. It leverages historical data to predict the class labels of new, unseen data points. The classifier learns the relationships between input features, such as email content, image pixels, or text, and class labels like spam/not spam or sentiment. Once trained, it can generalize this knowledge to make predictions on new data. Machine learning classifiers are indispensable in applications that require data classification, such as image recognition, medical diagnosis, and natural language processing. For this study, we have compared the performance of machine learning models namely Random Forest Classifier, Naïve Bayes, C5.0 Decision Tree Classifier, Support Vector Machine (SVM), and Improved Stacked SVM Classifier.

Machine learning, for a segmentation problem, offers improved accuracy, is flexible and adaptable, and saves time and effort.

Table 2
Data description.

S/N	Feature Name	Description
1	DISTRIBUTORCODE	Code of LPG Distributors
2	CONSUMERNUM	Unique code of a customer
3	STATE	State of Union of India
4	PINCODE	Pincode of a place in India
5	URBANRURAL	Urban and Rural Classification
6	RELATIONSHIPSTARTDATE	PMUY connection issue date
7	LASTREFILLDATE	LPG refill last taken date
8	DELIVERYTYPE	How the LPG refill delivery done
9	SUSPENSIONREASON	Reason for customer's connection suspension
10	SCHEMEONBOARDINGSTATUS	LPG Connection status
11	SUBSIDYSTATUS	DBTL status for Subsidy
12	SCHEMESUBTYPE	Sub-type under which the connection was given
13	CONSUMERTYPE	No of LPG cylinders in the connection
14	FREQUENCY	Target Variable computed by the ISS Model

Businesses can use it to segment customers and markets, while image and text segmentation is useful for medical image analysis, object detection, and information extraction. Machine learning algorithms are superior to classical feature extraction methods in terms of accuracy, flexibility, and robustness. Therefore, they should be used to automatically extract informative and predictive features from data to improve segmentation accuracy.

A brief description of few of the machine learning classifiers is as under.

4.1. Random forest classifier

The random forest classifier is a widely adopted approach for handling high-dimensional and skewed data due to its exceptional performance [41]. The name of this model originates from the fact that the number of decision trees utilized in the model is selected at random. Given a dataset with D data points and m features, each decision tree is trained by randomly selecting rows and features. In cases where the sampled data has fewer rows or columns than D and m, respectively, the decision tree is still trained in the same manner. The random forest model employs multiple decision trees, each with high variance. However, by aggregating all the decision trees, a majority vote is used to convert high variance to low variance. It is noteworthy that random forest remains an accurate model even when there are only a limited number of data points available.

4.2. Naïve Bayes

The Naïve Bayes algorithm is a supervised learning method that employs the Bayes Theorem to address classification problems. As a probabilistic classifier, it predicts outcomes based on an object's likelihood. Naïve Bayes is highly regarded in various domains due to its simplicity. Nevertheless, its fundamental assumption of independence among features is both a strength and a limitation. While this assumption contributes to its robustness, it also results in a loss of accuracy [42] that constrains its applicability in real-world scenarios [43]. Nevertheless, Naïve Bayes remains a desirable choice in situations where large datasets are involved, as it offers faster and less complex performance compared to other machine learning classifiers.

4.3. C5.0 Decision Tree Classifier

C5.0 is a robust machine learning system that employs decision tree algorithms and finds extensive practical applications. In comparison to other decision tree-based classifiers, such as C4.5 and ID3, C5.0 boasts numerous advantages, including faster processing speeds, higher precision, and superior memory utilization [44]. As noted by Ref. [45], C5.0 is an extension of C4.5 and ID3. C5.0 excels in handling data that has noise and missing values, and it is adept at addressing overfitting and pruning errors [44]. Furthermore, C5.0 exhibits no concerns with dimensionality in prediction. The outcomes produced by this classifier are readily comprehensible and visually interpretable, thereby obviating the need for advanced mathematical knowledge.

4.4. Support vector machine (SVM) classifier

Support vector machine, a supervised machine learning algorithm, can be used for both numerical prediction as well as classification and it can accept both linear and nonlinear data as input [46]. This study uses SVM to classify the intention to use social media for education. SVM has several functional areas such as-identifying handwriting, identifying any object, and identifying the speaker etc. [47] used SVM to identify cancer disease in an individual using gene expression data as input. The most significant advantage of SVM is that its training time is extremely short in comparison to other ML algorithms, while producing high accuracy due to its ability to handle nonlinear data by establishing nonlinear decision boundaries. SVM is less prone to overfitting than other methods. SVM uses optimization by looking for the maximum marginal hyperplane determined by support vectors. The hyperplane mathematical equation is $\mathbf{AX^*b = 0}$, where A represents the weight vector, namely A = a1, a2, a3, an, where n represents the number of features, b is a scalar quantity representing a bias in the model, and X represents the feature vector.

4.5. Improved Stacked SVM (ISS) classifier

This study proposes a combined classifier that is built by stacking random forest (RF) and support vector machine (SVM) classifiers. The training subset is used to train random forest classifier, resulting in three distinct trained random forest classifiers. The classification decision of the RF classifiers is used as input for the meta-classifier, which is a support vector machine (SVM) that makes the final classification decision [48,49].

4.6. Explainable Artificial Intelligence (XAI) or explainable machine learning (XML)

Explainable AI (XAI), also known as Interpretable AI or Explainable Machine Learning (XML), is a burgeoning field of research that seeks to increase the comprehensibility of machine learning models for humans. The opacity of many machine learning models makes them difficult to interpret, necessitating the development of Explainable Artificial Intelligence (XAI) or Explainable Machine Learning (Explainable ML). These techniques, methods, and models within the domain of artificial intelligence and machine learning are intended to enhance transparency and provide insights into the decisions and outputs of AI systems, allowing users to make informed judgments about model outputs and fostering trust.

The primary objective of XAI is to generate explanations for why a particular AI model made a specific prediction or decision. This ability, in turn, creates trust and enables users to assess the model's output. XAI's key characteristics include interpretability, transparency, explanations, model types, trust, accountability, and human-centeredness. XAI models are designed to be interpretable, which means that even non-experts can easily understand and make sense of their internal workings. XAI emphasizes transparency in the decision-making process of AI models, making the process understandable and traceable. XAI generates explanations for AI model predictions or decisions in a variety of formats. This feature enhances trust in AI systems by providing insights into the model's reasoning. It also facilitates accountability by allowing users to determine whether the model's decisions align with ethical, legal, and business requirements. XAI prioritizes human-centred design, focusing on providing explanations that are meaningful and relevant to end-users or domain experts.

To explain the results of machine learning models using XAI, the following approach is recommended: select an appropriate XAI technique based on your model and use case, apply the chosen XAI technique to generate explanations for individual predictions or model behaviour, present the explanations in a user-friendly format, and continuously assess the effectiveness of your explanations and gather feedback from users. By integrating XAI techniques into your machine learning workflows, you can improve the transparency, accountability, and user-friendliness of your models. This is particularly important in applications where understanding the reason behind predictions is critical, such as healthcare, finance, and autonomous vehicles.

Fig. 3 shows the working of all the models in the study.

4.7. *Evaluation metrices*

In order to analyse and evaluate the output of AI/ML models on a single dataset, we have to take help of Confusion Matrix, Performance Metrics such as accuracy, precision, recall, F1 score and Statistical Tests such as kappa statistics, AUC of ROC curve, which can quantify and pinpoint the performance of each of the models. A confusion matrix can visually show the performance of each of the model by depicting how many instances are correctly classified and misclassified. Table 3(a) provides the confusion matrix while Table 3(b) explains the performance metrices used in this study.

5. Results and analysis

In order to devise a policy for creating sustainable LPG distribution model, the segmentation of customers is akin to identification of

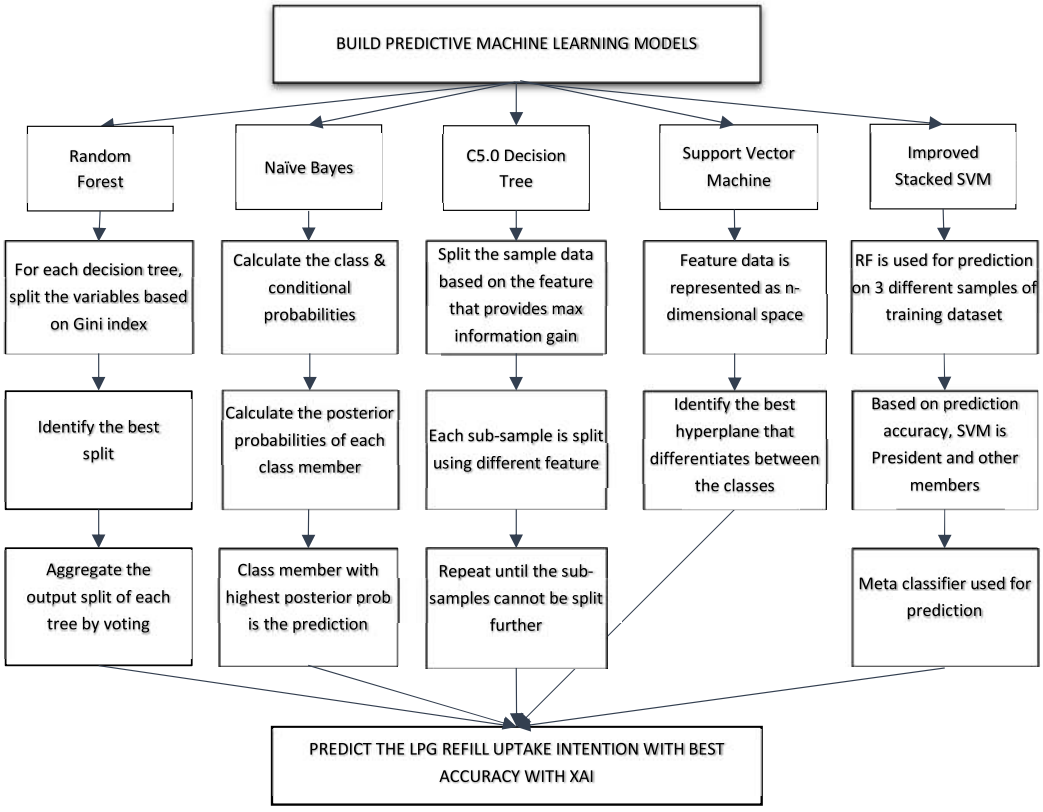


Fig. 3. Working of the ML predictive models.

Table 3(a)
Confusion matrix.

Metrics	Formula/Description			
Confusion Matrix			**Actual**	
			High Freq	**Low Freq**
	Predicted	High Freq	*True Positive (TP)*	*False Positive (FP)*
		Low Freq	*False Negative (FN)*	*True Negative (TN)*

Table 3(b)
Performance metrics.

Metric	Definition	Formulae
Accuracy	Accuracy is the proportion of predictions that are correct. It is calculated by dividing the number of correct predictions by the total number of predictions.	$\frac{TP+TN}{TP+TN+FP+FN}*100$
Precision	Precision measures how accurately the model predicts positive instances. It is calculated by dividing the number of true positives by the total number of predicted positives.	$\frac{TP}{TP+FP}*100$
Sensitivity (Recall)	Sensitivity measures the proportion of positive instances that are correctly identified. It is calculated by dividing the number of true positives by the total number of actual positives.	$\frac{TP}{TP+FN}*100$
Specificity	Specificity measures the proportion of negative instances that are correctly identified. It is calculated by dividing the number of true negatives by the total number of actual negatives.	$\frac{TN}{TN+FP}$
F1-Score	The F1 score is a weighted average of precision and recall. It is calculated by harmonic mean of precision and recall. The F1 score is a good metric to use for classification problems where both precision and recall are important.	$2*\frac{(Precision\ *\ Recall)}{Precision+Recall}$
Kappa Statistics	Kappa statistics is a measure of how well a model performs compared to a random model. It is calculated by comparing the observed agreement between the model's predictions and the actual labels to the expected agreement under a random model.	$\frac{pa\text{-}pac}{(1\text{-}pac)}$ where, pa represents total agreement probability and pac represents probability by chance
ROC Curve	The ROC curve is a graphical representation of the trade-off between sensitivity and specificity. It plots the true positive rate (TPR) against the false positive rate (FPR). The higher the AUC (area under the curve), the better the model's performance.	ROC is plotted between Sensitivity and (1-Specificity). The area under the curve (AUC) measures the degree to which the curve is up in the northwest corner.

customers who are not going to uplift subsequent LPG refills. The prime objective of the presented models in this study is to segment customers (LPG) based on their refill frequency into either a high-frequency (HF) or low-frequency (LF) customer segment.This section presents a comparison of the prediction performance of different machine learning algorithms on different data split and 10-fold cross validation.

5.1. Experimental setup and model comparison

State of the art ML models such as Naïve Bayes, Decision Tree, Random Forest, and SVM have been incorporated for making a prediction model of customer's liquefied petroleum gas (LPG) refill frequency. In addition, this study proposes a model Improved Stacked SVM (ISS) and compares its performance with the other model undertaken in this study. In addition to the development and comparison of these models, the feature selection has been explained using explainable AI method where ISS model has been explained with in-depth features and model interaction. The experimental setup of this research has been made for different training-testing split i.e., 50-50 %; 66-34 %; 80–20; and 10-folds cross validations. After testing, a confusion matrix of each classifier is developed from which accuracy, sensitivity, specificity, ROC, and AUC values are obtained for analysis. In addition, some of the statistical techniques such as Kappa statistics is involved to assess the credibility of the proposed ISS model with other state of art models to predict LPG refill frequency.

The performance accuracy of all the models for different splits is depicted in Table 4, where the proposed ISS model shows the better results with accuracy in all experimental setups i.e., 79.56 % for 50-50 training-testing split, 80.56 % for 66-34 split, 80.95 % for 80-20 split and 77.47 % for 10-fold. Random Forest (RF) classifier turned out to be the second-best classifiers in performance accuracy.

Table 4
Result of performance accuracy of the classification models.

	Random Forest	SVM	Naïve bayes	C5.0	ISS
50%–50 %	77.84	74.47	66.56	76.52	79.56
66%–34 %	80.83	75.92	67.47	78.41	80.56
80%–20 %	79.58	75.92	68.41	79.12	80.95
10-Fold	76.90	73.71	67.10	76.06	77.47

Table 5 demonstrates the Sensitivity of all the models for different split methods. Sensitivity is a performance metric that measures the proportion of actual positive instances that were correctly predicted as positive by the model which is in this research context is High Frequency (HF) of LPG refill.

The proposed ISS model in this case too provides satisfactory results together with Random Forest Classifier. The sensitivity of the proposed model is 61.43 % for 50-50 split, 69.01 % for 66-34 split, 67.40 % for 80-20 split and 66.18 % for 10-fold. For sensitivity Random Forest, Decision Tree and proposed model shows promising results.

The specificity of all the models has been demonstrated in Table 6. Specificity is a performance metric that measures the proportion of actual negative instances that were correctly predicted as negative by the model which is in our case is Low Frequency (LF) refill of LPG and is an utmost concern of this research. The proposed ISS model turned out best among other classifiers for predicting negative instances. The specificity of the proposed ISS model is 89.22 % for 50-50 split, 86.43 % for 66-34 split, 87.76 % for 80-20 split, 83.43 % for 10-fold. Naïve Bayes model here exhibits the best performance here but before it can be adjudged the best model we need to perform test for Kappa statistics test compared to other studied models. Here we also observe that Proposed classifier ISS and Random Forest model exhibit comparable specificity.

The Kappa statistic (also known as Cohen's Kappa) is a measure of inter-rater agreement or inter-annotator agreement for categorical items. It is particularly useful when assessing the agreement between two or more raters or annotators who categorize items into discrete classes. Kappa statistics considers the possibility of agreement occurring by chance and provides a more robust measure of agreement than simple percentage agreement. The proposed ISS classifier exhibits more robustness in terms of Kappa Statistics. The kappa value is considered as a poor agreement if its value is less than 20 %, fair agreement for 20 %–40 %, moderate agreement from 40 % to 60 %, good agreement from 60 % to 80 %, and excellent agreement for 80 %–100 %.

Kappa statistic of machine learning classifiers studied in this research is shown in Table 7. The results show naive Bayes as a poor model with kappa value less than 10 % for all the experimental set ups. Other models show moderate agreement and can be considered further for building prediction models.

Fig. 4 demonstrates ROC curve (Receiver Operating Characteristic Curve) which is the graph showing the training performance of classification models with respect to different classification thresholds and plots a graph between True Positive Rate (TPR) and False Positive Rate (FPR) and AUC (Area under the ROC Curve) provides an aggregate measure of performance across all classification thresholds. The model which has a higher area under curve value performs better than other models under comparison. With respect to AUC, all the models exhibit an AUC score of more than 75 % and hence the training capability of all the models is good in all experimental setups.

5.2. Analysis of explainable ISS model

Going beyond what is apparent from statistical metrics, this study employs Explainable Artificial Intelligence (XAI) to explain the ISS model. All the analysis for explainable ISS has been done for the models trained with 10-fold cross validation. Fig. 5 depicts feature information and their interaction with the different classes (High frequency and Low frequency of LPG refill) of the target variable. It is clear from the figure that both classes are imbalanced (i.e., away from 50 % mark) and is observed that the consumers with high frequency refill are less than consumers with low frequency refill. The same can also be seen in the differences of sensitivity and specificity of the model. It implies that if the classes are balanced, sensitivity of the models may be improved. In addition, the variances of the input features are also explained with respect to both classes of target variable.

Features' importance with respect to ISS model has been shown in Fig. 6. The computed score for each feature of the model simply describes the "importance" of each feature. A high value of feature signifies that the feature has a greater impact on prediction of the instance through the ISS model. It can be inferred from the figure that Last Refill Value plays a significant role in predicting of both classes and is the most significant among the other features. Other features like Urban-Rural, Scheme types etc. are also important.

For assessing the performance of the model and features, two random instances from the test data were taken. The proposed model trained using 10-fold setting was used to predict both test instances (one for High-Frequency and other for Low-Frequency) for in-depth explanation of the ISS model. The combined threshold value is the deciding factor. Based on the combined threshold value, the machine learning algorithm will predict the input data belongs to a particular segment.

Fig. 7(a) is an example of the high-frequency LPG booking class, the threshold value of which is 0.45, which is based on the individual feature value. In figure the red dot indicates each feature's probability value for specific class. The figure depicts the threshold value wherein if threshold greater than 50 % is observed, then there is the high chance that input parameter belongs to the low-frequency segment and if the threshold is less than 50 %, it indicates that the input parameter belongs to the high-frequency LPG segment.

It is observed from Fig. 7(a) that all the feature probability values are equal to or less than 0.5 and the overall combined probability

Table 5
Result of sensitivity of the classification models.

	Random Forest	SVM	Naïve bayes	C5.0	ISS
50%–50 %	67.33	57.43	83.25	63.22	61.43
66%–34 %	67.57	56.23	92.65	75.08	69.01
80%–20 %	69.86	57.26	11.50	70.68	67.40
10-Fold	70.37	53.56	15.70	62.83	66.18

Table 6
Result of specificity of the classification models.

	Random Forest	SVM	Naïve bayes	C5.0	ISS
50%–50 %	83.44	83.55	97.58	83.60	89.22
66%–34 %	87.57	86.03	97.07	80.10	86.43
80%–20 %	84.46	85.28	96.97	83.36	87.76
10-Fold	80.42	83.56	94.75	83.12	83.54

Table 7
Result of kappa statistics of the classification models.

	Random Forest	SVM	Naïve bayes	C5.0	ISS
50%–50 %	50.96	42.13	07.42	47.5	52.92
66%–34 %	56.25	44.02	08.01	53.31	56.04
80%–20 %	54.21	44.01	10.67	53.53	56.31
10-Fold	80.42	83.56	04.70	83.12	83.54

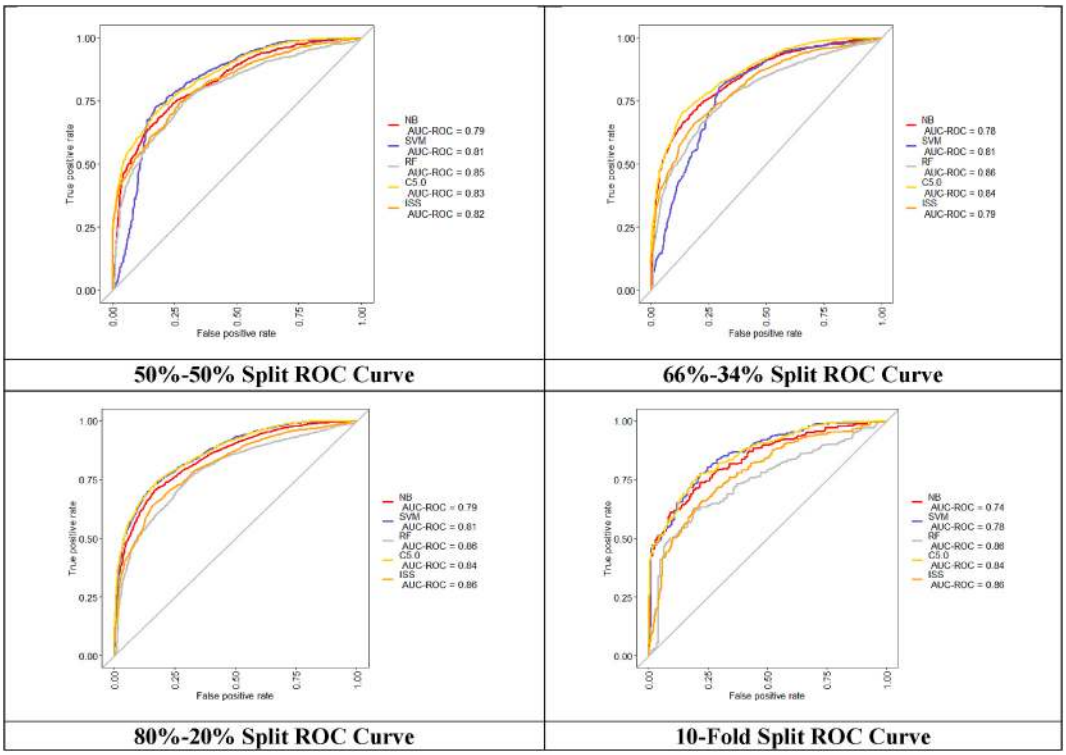


Fig. 4. Roc for ML models for different split and 10-fold.

is 0.45, meaning thereby that this input parameter belongs to the high-frequency LPG booking segment.

Similarly, Fig. 7(b), is an example of a low-frequency booking class. All input features shown in Fig. 7(b) have around 70 % probability for a low-frequency class and the combined probability value is 0.78, which is greater than 0.5. This implies that the input parameter under consideration belongs to the low-frequency LPG booking class. In addition, we can observe that the individual feature interaction and specific threshold value for each feature indicates the strength of the feature for consideration into a particular class.

6. Discussion

CAG in its report had pointed that even though the PMUY scheme has been rolled out to large population of the targeted beneficiaries, the sustainability of the PMUY was still a challenge [50]. Identification and segmentation of customers' LPG refill frequency into high and low segments is a prerequisite for design and development of necessary policy interventions to increase the further adoption and ensuring sustainable offtake of LPG refills by the targeted PMUY customers.

Machine Learning (ML) classification models can be used to detect classification thresholds and automatically classify customers

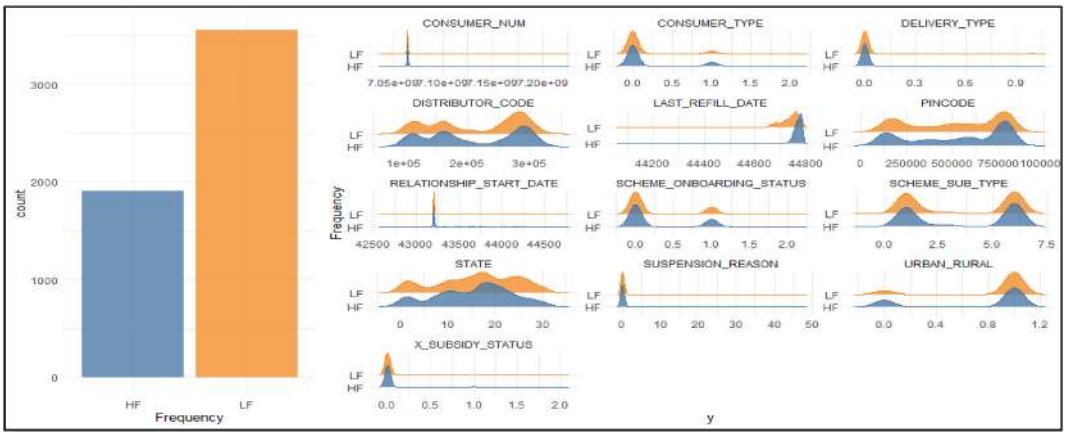


Fig. 5. Features information and interaction with target variables.

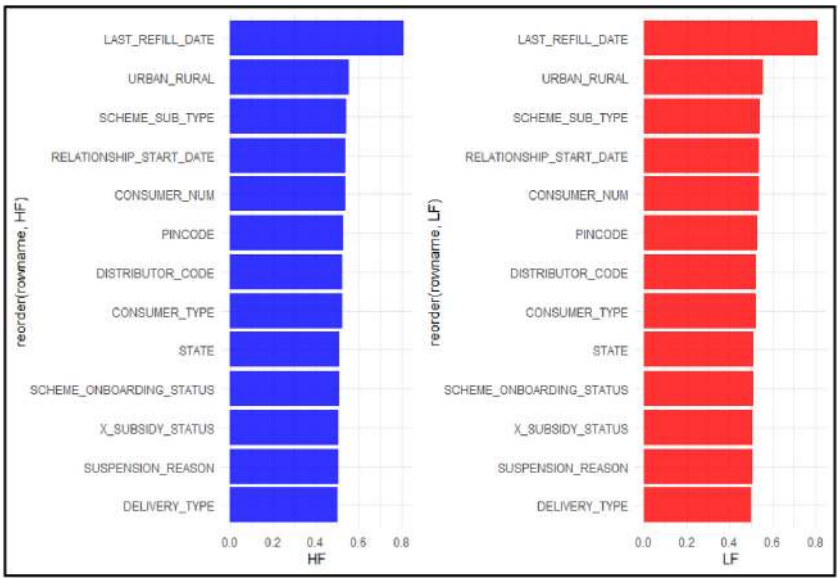
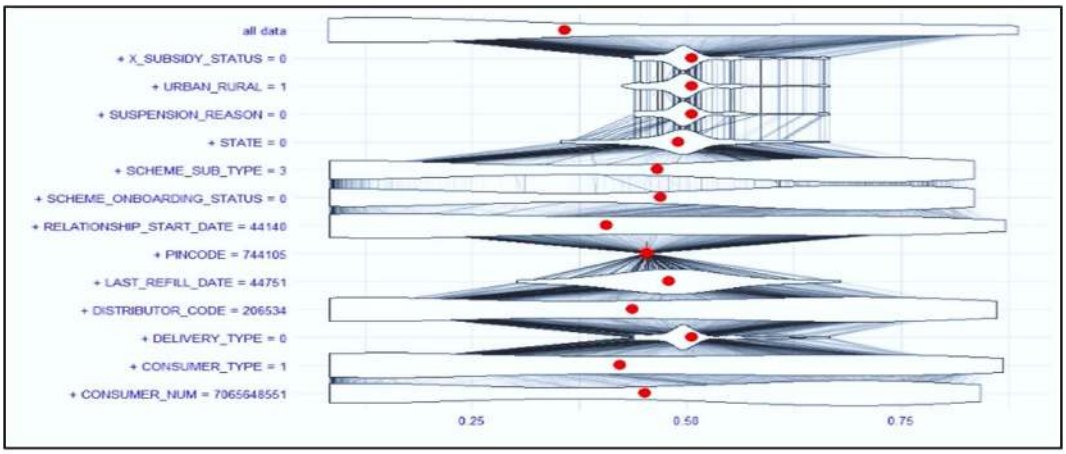


Fig. 6. Feature importance.

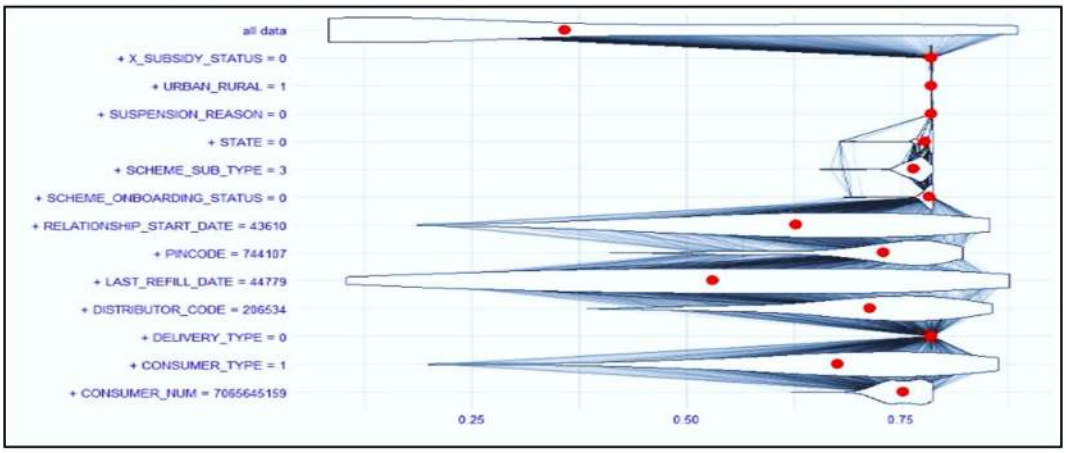
into high frequency and low frequency customers. When framing policies based on data and complex algorithmic procedures to classify customers into desired high and low frequency segments, the ML system may use logic which constantly keeps changing and getting transformed. In the process of encoding into a software, ML systems are expected to comply with specific rules and logics. Reasonability, relevance, and fairness are obscured in the process when multiple data sets are being used. A mechanism is thus required that would de-mystify the process of decision making by ML models to users. Whereas ML models have now become mainstream applications being deployed in all imaginable situations, explainable Machine Learning attempts to take a further leap towards making the ML models more understandable.

This study proposes an explainable combined classifier (namely XISS) by stacking random forest (RF) and support vector machine (SVM) classifiers. The ISS classifier is used to classify PMUY customers into low-frequency and high-frequency customers. Based on accuracy, sensitivity and specificity, the Random Forest and ISS classifiers fare better than the other models. The ROC AUC value of 0.846 for Random Forest and ISS models is also the highest. But in terms of Cohens' kappa value, the ISS model exhibits a higher value when compared to the random forest. On an overall basis, for this dataset ISS algorithm shows slightly higher performance than the random forest.

Also, this research employed Explainable AI to determine and pictorially present the importance of each feature used by the ML model for classifying customers into high and low frequency customers. The probability value for each class for every parameter was calculated and compared with the probability value of each feature to decide to which class it belonged. The overall combined probability for high frequency customer class was 0.45 and for low frequency was 0.78. It is evident from the figures above that for our ISS algorithm, the parameter Last Refill Date plays an especially significant role in decision-making. The other two most significant



(a)



(b)

Fig. 7. A: Features Contribution in Prediction for high-frequency LPG booking class. B: Features Contribution in Prediction for low-frequency LPG booking class.

parameters are Urban Rural customer type and Scheme Sub Type. Thus, the current work employs Explainable AI to get insight into the decision-making process of the ML model.

7. Implications of study

7.1. Practical implications

Unlike the earlier studies by (Kalli et al., 2022 Gill; Wiehl et al., 2021, 2022) and others which have used stratified sampling to arrive at the sample for study and used statistical significance tests to arrive at feature importance and selection, this study employs a novel approach. In the context of PMUY customers with respect to classification of customers into low-frequency and high-frequency customers, this study employs the Improved Stacked Support Vector Model (ISS) Model which is a hybridisation of Random Forest and Support Vector Machine. This study establishes the superiority of ISS model for classification of PMUY customers LPG refill frequency into High and Low, compared to other state-of-art ML models namely the Random Forest, SVM, Naïve Bayes and C5.0. It employs Explainable Machine Learning (XML) techniques to explain the significance of features used for model development based on the features/dimensions of the provided dataset.

This machine learning approach is appropriate for Big-data scenarios where data is generated at scale and repeated human intervention for addressing model development and arresting model drift is next to impossible.

Machine Learning models being probabilistic in nature, there is a fair chance that deserving customers may be classified as otherwise and would be therefore left out in the targeted schemes owing to the misclassification. Policy implementors need to keep this

in mind and build/provision for a separate mechanism/process to bring back the erroneously left out or misclassified customers in this case into the correct segment to ensure the benefits of the schemes flows to the intended beneficiaries.

7.2. Managerial implications

Using the ISS method, desired segmentation of customers can be conducted for detection of PMUY subsidy leakages, identification of areas not adequately covered by LPG Distributors, identification of customers who in-spite of financial assistance are not keen to switch to LPG. The classification technique can be extended to classification of distributors into High selling and Low selling distributors based on the no. of customers being serviced and the nos. of refills being supplied. This prediction can form the basis of demand planning for the LPG supply chain and identification of hot spots with respect to LPG cylinder refill shortages. One of the reasons for the low adoption of PMUY scheme in rural areas has been that the LPG Distributors were far off from the residences of the beneficiaries of PMUY scheme. Mappings of regions/districts in terms of High usage and low usage can be helpful in planning a flexible delivery strategy of LPG cylinders. Instead of the customers coming to the Distributors, it is the Distributors who can plan a scheme when to reach the customers. The gauging of performance of Distributors based on the customer satisfaction can also be conducted using the classification engine.

In addition to the intended purpose of the PMUY scheme, GOI through the ecosystem of PMUY Scheme, Direct Benefit Transfer (DBTL) Platform and the JAN Dhan Yojana has created a roust framework through which it can reach out to a large set of the needy population in times of crisis. During the Covid-19 pandemic, GOI had leveraged this framework to give 3 LPG cylinders free of cost to each PMUY beneficiaries. This implies that the technique of classification of PMUY customers with the help XAI can be further used for fine grained customer segmentation and leveraged for directed service delivery by GOI/other interested parties. Also, classification of customers into high and low frequency will allow for identification of customers with high probability of churn out and accordingly either target them specifically with added incentives or exclude them from subsequent benefits as may be deemed necessary.

8. Conclusion and Future work

This study started with the aim of developing an Explainable Machine Learning (XML) model to predict the customers refill frequency from Ujjwala scheme so that necessary steps for ensuring sustainable offtake of LPG refills can be taken. The study proposed three research questions which successfully solved by developing Explainable Improved Stacked SVM (XISS) model which gave prediction accuracy up to 81 %. To assure the prediction decisions, proposed model and input features interaction was explained using Explainable AI (XAI) methods. The proposed model compared its result with various state-of-art models through various prediction evaluation metrics and found to be best model in current application scenario. Machine Learning models are considered black box models and are opaque to decision makers. Using Explainable Machine Learning (XML) we were able to get better insights into the working of Machine Learning models with respect to assigning feature importance and feature selection. This model also showed the informative features that participated in prediction using XML. Some of most informative features were Last Refill, Urban Rural customer type and Scheme Sub Type etc. The other features were also ranked with respect to their importance. This insight thereby leads to trust in ML models and the impact of bias in models' behaviour can be reduced.

The current research can be further extended by using other Machine learning and Deep Learning models for building prediction model in current application domain. Hybridization of the Deep Learning ANN networks with other ML models can be explored as an extension to this study. A contrarian study on the reasons for high adoption of PMUY using same or extended feature sets can throw interesting insights. Further study with respect to drop out rate after initial connection along with its associated LPG refill uptake pattern and the like can also throw interesting insights into the reasons for low intention of LPG refill uptake. Additional information with respect to respect to geographical proximity of LPG Distributors in the area, the service ratings/feedback, presence of competitors, can also be correlated and studied. The proposed model can also be explored in other application domains.

Data availability statement

Data will be made available on request.

CRediT authorship contribution statement

Shrawan Kumar Trivedi: Project administration, Data curation, Conceptualization. **Abhijit Deb Roy:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis. **Praveen Kumar:** Writing – original draft, Methodology, Investigation, Formal analysis. **Debashish Jena:** Writing – original draft, Software, Resources, Formal analysis. **Avik Sinha:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Long-Term Optimal Delivery Planning for Replacing the Liquefied Petroleum Gas Cylinder

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Abstract In the daily operation of liquefied petroleum gas service, gas providers visit customers and replace cylinders if the gas is about to run out. For a long time, frequent visits to customers were required because they could not determine the amount of remaining gas without a staff visit and observation. To solve this problem, smart meters are started to be employed to acquire gas consumption more frequently without visiting customers. In this study, we construct a system to optimize plans for cylinder replacement, and evaluate it with a large-scale field test. We propose an algorithm to create a replacement plan with three steps: estimating the replacement date, acquiring the customer list for replacement, and determining the delivery route. A more accurate estimation of the replacement date can be acquired with a smart meter, which is used for making a customer list for replacement. The formulation for making a customer list enables the gas provider to replace cylinders some days before the date when the gas would run out. It can suppress the concentration of replacements on certain days. Large-scale verification experiments were performed with more than 1,000 customers in Chiba prefecture in Japan. In the field test, the gas provider incorporated the system into its replacement operations. Moreover, the replacement plans developed by the proposed system were compared with that by the gas provider. Our system reduced the number of gas cylinders with gas shortage, the number of visits without replacement due to plenty of gas remaining, and the working duration per customer, which shows that our system benefits both gas providers and customers.

Keywords: OR practice, transportation, combinatorial optimization, stochastic optimization

1. Introduction

To maintain liquefied gas service, a gas provider visits customers to verify the remaining amount of gas and replace cylinders if gas is about to run out. We denote that the cylinder replacement problem (CRP) is the problem in making a plan to visit customers to replace gas cylinders, including the delivery route. The CRP contains the following subproblems: (1) estimating the replacement date to visit customers, (2) making a customer list for replacement, and (3) obtaining the delivery route with a short working time. Therefore, the CRP is so complex and requires many techniques to solve.

Owing to the difficulty of estimating the replacement date, smart meters have started to be installed for some customers to prevent a gas shortage. They send the gas consumption

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to the gas provider more frequently than conventional meters, such as once a day. For a long time before smart meters emerged, conventional meters were installed in customers' houses. The gas provider is not able to determine the remaining amount of gas without a staff visit and observation. Then, frequent visits by the gas provider are required to prevent a gas shortage, and many nonreplacement visits happen due to the plenty of remaining gas. In this situation, utilizing the smart meter's data instead of conventional meters can suppress the number of nonreplacement visits and gas shortages. However, the gas provider cannot install the smart meter for all customers due to the installment costs. Therefore, we still have to consider the problem of many nonreplacement visits for conventional meters.

In this study, we construct a system to solve the CRP. Our system enables us to make a gas cylinder plan without professional staff (see Figure 1). Moreover, it suppresses the number of gas shortage and nonreplacement visits due to plenty of remaining gas, and decreases the working duration. We can acquire a more accurate estimation of the replacement date with a smart meter, which is input for making a customer list for replacement. Our formulation for making a customer list enables the gas provider to replace cylinders some days before the date about to the gas shortage. Thus, the concentration of replacements on certain days can be avoided.

We briefly explain the proposed algorithm to solve the CRP with three steps, and the overall flow and the comparison with the previous operation are shown in Figure 1. First, we estimate the cylinder's replacement date. (Section 3). The risk of gas shortage per day is calculated based on the current remaining gas and multiple days of forecast gas usage. Beforehand, we extrapolate the conventional meter's gas usage to estimate the current remaining gas because the gas provider does not observe the current remaining gas for the conventional meter. After estimating the replacement date, a customer list for replacement is acquired (Section 4). When we greedy pick up all customers for the predicted replacement date, there can be too many customers to visit or a long working duration. To prevent this situation, the multiple-day customer list is considered simultaneously. Then, we regard the problem as obtaining the smallest rectangle that covers the customers to need replacements, and we formulate the problem and solve it as a mixed-integer optimization problem. Finally, the order of visits to replacement is determined (Section 5). The replacement should be conducted as much as possible to prevent the many customers close to a gas shortage at the same time. We also formulate and solve the problem as mixed-integer optimization problems for obtaining the delivery route with the shortest working duration.

To verify whether the proposed system is useful, two experiments were conducted with more than 1,000 customers, of which approximately one-third did not have smart meters. In a field test, a gas provider followed the output of our system and succeeded in reducing the nonreplacement visits and working duration (Section 6). Furthermore, we performed a verification experiment by comparing the cylinder replacement plan made by the proposed system and that developed by the gas provider (Section 7). The proposed system reduced the number of out-of-gas cylinders, number of visits without replacement due to the plenty of gas remaining, and working duration, which shows that it benefits both gas providers and customers.

2. Related Work

In this study, the cylinder replacement problem, which is a specific case of the gas secondary distribution problem, is addressed. A secondary distribution is the process of transporting refined oil from oil depots to petrol stations or end-users. When we acquire the customer list

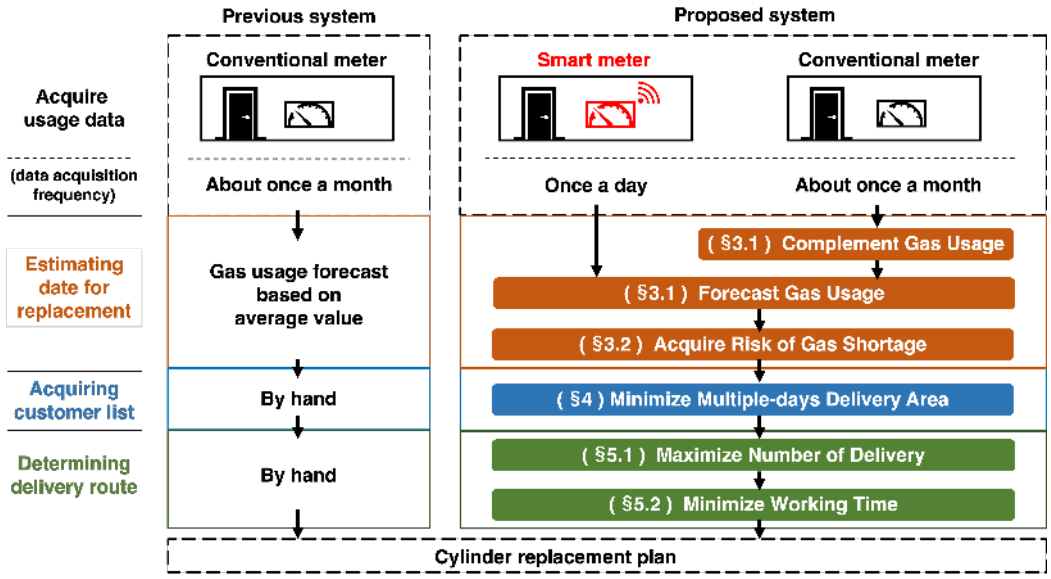


Figure 1: Proposed system for solving the gas cylinder replacement problem

for the replacement and delivery route, the output of the machine learning such as demand forecast is often utilized. In this case, it is necessary to consider the uncertainty of machine learning to prevent an unfavorable situation, such as items out of stock. In this section, we explain the related work about the secondary gas distribution problem and delivery planning with considering demand forecast's uncertainty.

Most studies on secondary gas distribution problems have addressed the gas replenishment problem, whereas only a few have explored the gas cylinder replacement problem. [1] decomposed the final distribution of fuel oil into a weekly replenishment plan for each station and arranged petrol station visiting sequences (vehicle routes) for each day of the week. [2] proposed a multi-depot petrol station replenishment problem with time windows. Multiperiod optimization of the gas is an important property to consider in a sequential decision in [3, 4, 5]. Like the gas replenishment problem, the research for making customer lists and delivery route for other applications has also been widely explored. Note that these problems belong to the inventory routing problem. [6] also studied a tactical delivery plan for large retailers selling and delivering large and heavy items to customers, and [7] modeled a general dynamic multi-period routing problem for postal and courier companies. A similar problem was addressed by [8], who examined a dynamic multi-period vehicle routing problem with stochastic service requests. In this article, the multiperiod plan described in Section 4 is also considered.

Because many delivery plans are made based on the output of the demand forecast, the other important property to consider in optimal delivery planning is the demand forecast uncertainty. [9] used a multiscenario mixed-integer programming model to describe the problem of the transportation process of oil products, considering a stochastic hub disruption and uncertain demands. In addition, [10] presented a two-stage stochastic programming model that determines the replenishment quantity of each petrol station. Moreover, [11] regarded the demands of retailers as discrete random variables to consider the stochastic inventory routing problem with transportation procurement. [12] proposed mixed-integer programming models for medical drug distribution problems under demand uncertainty, such as drug storage. In this study, the demand forecast's uncertainty, which is described

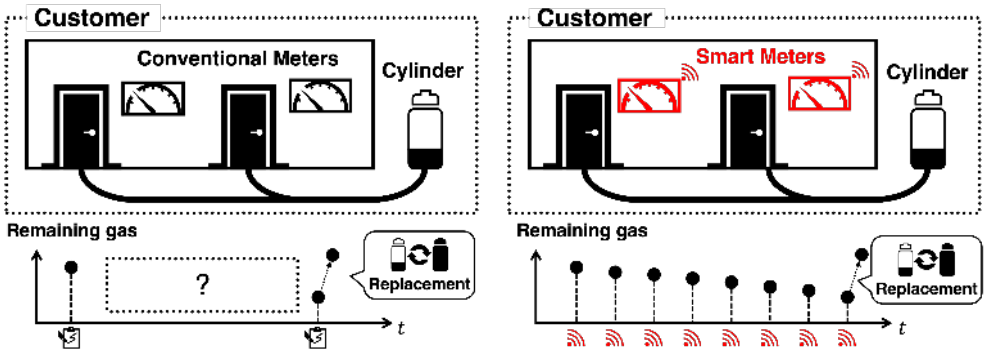


Figure 2: Relationship among meter, cylinder, and customer. Note that the remainder of gas is observed approximately once a month for conventional meters while per day for smart meters.

in Section 3.2, is also considered.

3. Estimating Replacement Date for Customer

In this section, we describe the estimation of the replacement date for customers through three steps. As a result of the estimation, the customers are categorized into three groups per day, high-risk customers, moderate-risk customers, and low-risk customers, based on the emergency of the gas shortage. The categorization is utilized for acquiring the customer list (see Figure 1). First, the gas usage for conventional meters is extrapolated, as described in Section 3.1. In particular, daily gas consumption data from smart meters are utilized to extrapolate gas consumption data for conventional meters. The second step is to forecast gas usage. Well-known machine-learning models are employed for customers with smart meters, such as support vector regression and random forest regression. For conventional meters, daily gas usage is forecasted using a proposed algorithm based on the k-nearest-neighbor algorithm, which is also described in Section 3.1. Finally, we estimate the emergency of replacement per day. The customers are categorized into three groups by considering the forecast gas usage and the demand forecast uncertainty, which is described in Section 3.2.

3.1. Forecasting and Complementing of Gas Usage for Conventional Meters

We propose a complement and forecast method for daily gas usage based on data acquisition from a one-month period. First, the set of customers and meters are defined as follows. Let $\mathcal{M}, \mathcal{M}_s, \mathcal{M}_c$ be a set of meters, smart meters, and conventional meters, respectively. Because every meter is either smart meters or conventional meters, $\mathcal{M} = \mathcal{M}_s \sqcup \mathcal{M}_c$ ($\mathcal{M}_s \cap \mathcal{M}_c = \emptyset$) is satisfied. Note that $M_c \in \mathcal{M}_c$ needs extrapolation to predict the current remaining gas, whereas $M_s \in \mathcal{M}_s$ does not. Some households share the same gas cylinders, and the collection of them is considered a customer. Let $\mathcal{C}$ be the set of customers, and we denote the set of meters $m(C) \subset \mathcal{M}$ corresponds to the customer C. Because each household has a meter to record the gas usage, the total gas usage for the customer is the summation of the gas usage of the meters. The relationship among meters, cylinders, and customers is shown in Figure 2. The gas usage is forecasted per meter $M \in \mathcal{M}$, whereas the replacement is performed per customer $C \in \mathcal{C}$.

The daily gas usage of the meter M for date D is denoted as $\mu_D^{(M)}$. For a smart meter M_s, $\mu_D^{(M_s)}$ can be acquired by the smart meter. However, for a conventional meter M_c, the gas usage can be observed approximately once a month. Therefore, $\mu_D^{(M_c)}$ is acquired through interpolation based on the two successive gas usage observations. Moreover, the

forecast gas usage of the meter M for the date D' is denoted as $\widehat{\mu}_{D'}^{(M)}$.

Daily gas consumption data from smart meters are used to extrapolate and forecast gas consumption data for conventional meters. We propose how to extrapolate and forecast daily usage for a conventional meter whose basic concepts are divided into two steps. First, for every conventional meter, k smart meters are extracted such that the usage tendency is similar to that of the conventional target meter. Second, the extrapolated and forecast value are obtained by the weighted average of their usages.

For a conventional meter M_c, the last observed date is denoted as D_{M_c}. A function $\mathrm{Sim} : \mathcal{M}_c \times \mathcal{M}_s \to \mathbb{R}$ is defined to measure the similarity of gas usage tendency between a conventional meter $M_c \in \mathcal{M}_c$ and a smart meter $M_s \in \mathcal{M}_s$ as follows,

$$\mathrm{Sim}(M_c, M_s) := \frac{1}{\sum_{i=0}^{n_s} \left(\mu_{D_{M_c}-i}^{(M_c)} - \mu_{D_{M_c}-i}^{(M_s)}\right)^2} \tag{3.1}$$

where n_s is the hyperparameter and $D_{M_c} - i$ indicates the date i days before date D_{M_c}. Based on the gas usage similarity, the k-nearest neighbor meters are extracted for every conventional meter M_c and we denote it as $NN(M_c, k)$.

The extrapolated gas usage at D for M_c is obtained by calculating the weighted average of the observed gas usage of $NN(M_c, k)$ as follows:

$$\mu_D^{(M_c)} = \textstyle\sum_{M_s \in NN(M_c,k)} \mathrm{Sim}^*(M_c, M_s)\ \mu_D^{(M_s)} \tag{3.2}$$

$$\text{where } \mathrm{Sim}^*(M_c, M_s) = \frac{\mathrm{Sim}(M_c,M_s)}{\sum_{M_s' \in NN(M_c,k)} \mathrm{Sim}(M_c,M_s')},\ M_s \in NN(M_c, k). \tag{3.3}$$

The gas usage forecast at D' for M_c is also obtained by calculating the weighted average of the forecast gas usage of $NN(M_c, k)$ as follows:

$$\widehat{\mu}_{D'}^{(M_c)} = \sum_{M_s \in NN(M_c,k)} \mathrm{Sim}^*(M_c, M_s)\ \widehat{\mu}_{D'}^{(M_s)} \tag{3.4}$$

where $\widehat{\mu}_{D'}^{(M_s)} (M_s \in \mathcal{M}_s)$ can be obtained by the well-known machine-learning methods, such as support vector regression or random forest regression.

Although the concepts are close to the well-known k-nearest neighbor algorithm for regression, to the best of our knowledge, this application has not been previously explored.

3.2. Estimating Risk of Gas Shortage by Considering Gas Usage Forecast's Uncertainty

Based on the gas usage forecast, we determine whether each customer should be visited or not by obtaining the value of the proposed risk function. The risk function is defined by considering the gas usage forecast and the forecast uncertainty to quantify the emergency of a cylinder replacement. Based on the value of the risk function, high- and moderate-risk customers were extracted as candidates for replacement, which is the input for the problem of making a customer list for replacement.

The definition of the proposed risk function is introduced mathematically. The forecast error of each meter per day is assumed to independently follows a normal distribution. Obviously, the daily gas consumption of one meter does not influence that of the other meters. First, the gas usage forecast and unbiased variances are summed up for some meters belonging one customer per day. In sequence, the cumulative summation is calculated to obtain the cumulative gas usage and unbiased variances for a customer. Based on this

information, the risk function is defined as the probability of falling below a specific gas rate.

Letting D_0 be the current date, we select one customer $C \in \mathcal{C}$. The risk of gas shortage after n_f dates is considered. $\widehat{\mu}_{D_0+i}^{(M)}$ and $\widehat{\sigma_i^2}^{(M)}$ are indicated the forecast daily gas consumption for the i-th day after D_0 and an unbiased variance for the forecast gas consumption for i dates, respectively. Then, $\mathcal{N}\left(\widehat{\mu}_{D_0+i}^{(M)}, \widehat{\sigma_i^2}^{(M)}\right) (M \in m(C))$ is the probability distribution function for forecasting the gas consumption on the i-th day after D_0 for meter M ($i \in \{0, 1, \cdots, n_f - 1\}$), where $\mathcal{N}(\mu, \sigma^2)$ is the normal distribution with the parameters μ and σ indicating the average and the standard deviation, respectively. It is derived from the assumption that the error of the gas usage forecast follows a normal distribution. Based on the assumption that the error in the gas usage forecast of each meter per day independently follows a normal distribution, the reproductive property of the normal distribution can be applied. Therefore, the function representing the amount of gas consumption distribution $p_{D_0,n_f}^{(C)}$ for n_f days from date D_0 can then be written as follows:

$$X_{D_0,n_f}^{(C)} \sim p_{D_0,n_f}^{(C)} = \mathcal{N}\left(\sum_{i=0}^{n_f-1} \sum_{M \in m(C)} \widehat{\mu}_{D_0+i}^{(M)}, \sum_{i=0}^{n_f-1} \sum_{M \in m(C)} \widehat{\sigma_i^2}^{(M)}\right) \tag{3.5}$$

where $X_{D_0,n_f}^{(C)}$ is the random variable indicating the gas usage of customer C from date D_0 to n_f-th day after D_0.

Let $s_{D_0}^{(C)}, \varepsilon_\alpha^{(C)}$ be the remaining gas of customer C at date D_0 and the amount of gas of customer C when the remaining gas rate is equal to α. $s_{D_0}^{(C)}$ can be acquired by utilizing $\mu_{D_0-i}^{(M)}$ $(\forall M \in m(C), \forall i \text{ s.t. } D_0 - i \geq D_M)$ and the latest cylinder replacement date. The function $\hat{r}_{\alpha,D_0}^{(C)} : \mathbb{N} \to [0, 1]$ can be described, which returns the probability that the remaining gas rate of customer C is less than α on given days after date D_0, where P is the probability distribution function.

$$\hat{r}_{\alpha,D_0}^{(C)}(n_f) := P\left(X_{D_0,n_f}^{(C)} \geq s_{D_0}^{(C)} - \varepsilon_\alpha^{(C)}\right) \tag{3.6}$$

Replacement for the customer cannot be conducted every day owing to the staff vacation or the customer's availability. Therefore, it is necessary to consider whether or not to replace the customer's cylinder such that a gas shortage does not occur during the nonavailability dates. Therefore, the risk on n_f-th day after D_0 is assessed based on the remaining gas on the first available date after n_f-th day after D_0. Formally, if the $I_{\text{avail}}^{(C)} : \mathcal{D} \to \{0, 1\}$ is prepared as the indicator function representing the availability of customer C including staff's availability, where $\mathcal{D}$ is the set of dates, the risk function $r_{\alpha,D_0}^{(C)} : \mathbb{N} \to [0, 1]$ representing the emergency of replacement for the customer C on given days after date D_0 is defined as follows:

$$r_{\alpha,D_0}^{(C)}(n_f) := \hat{r}_{\alpha,D_0}^{(C)}(n_f^*) \ \left(n_f^* = \min\left\{i \in \mathbb{Z} \mid i \geq n_f, I_{\text{avail}}^{(C)}(D_0 + i + 1) = 1\right\}\right) \tag{3.7}$$

In this case, two thresholds are prepared for the remaining gas rate: α_{high} and α_{mdr} for categorizing the customers into three groups. When we prepare the thresholds $q_{\text{high}}, q_{\text{mdr}}$, the high- and moderate-risk customers on i days after D_0 can be written as follows:

- High-risk customers $\mathcal{C}_{\text{high},D_0}(i) := \{C \in \mathcal{C} \mid r_{\alpha_{\text{high}},D_0}^{(C)}(i) \geq q_{\text{high}}\}$

- Moderate-risk customers $\mathcal{C}_{\mathrm{mdr},D_0}(i) := \{C \in \mathcal{C} \setminus \mathcal{C}_{\mathrm{high},D_0}(i) \mid r^{(C)}_{\alpha_{\mathrm{mdr}},D_0}(i) \geq q_{\mathrm{mdr}}\}$

Customers except for high-risk customers and moderate-risk customers are categorized as low-risk customers. Its mathematical notation is omitted because they cannot be utilized later in the system. We simply write high- and moderate-risk customers as $\mathcal{C}_{\mathrm{high}}(i)$ and $\mathcal{C}_{\mathrm{mdr}}(i)$, respectively, when it is unnecessary to take care of D_0. When the customer list for replacement is acquired using the above definition, moderate-risk customers are occasionally relatively small in number. To increase the number of moderate-risk customers, we set a parameter D_{lbd}. It ensures that the customer is categorized as a moderate-risk customer D_{lbd} days before becoming a high-risk customer.

4. Acquiring Customer Lists for Multiple Dates for Replacement by Minimizing Delivery Area

In this section, the acquisition of customer lists for replacement is discussed through categorizing customers as high risk, moderate risk, or low risk. We formulate the problem of acquiring the customer list as acquiring the smallest rectangle that covers the customer lists. The formulation enables the gas provider to replace cylinders some days before the date about to the gas shortage, thereby suppressing the concentration of replacements on certain days. Therefore, multiple-day customer lists are obtained simultaneously.

We call a *trip* the process that the truck leaves the delivery center, visit customers to replace the gas cylinders, and returns to the delivery center. When we denote the planning horizon as D_{ph}, we consider the problem for the dates $\{D_0, D_0 + 1, \cdots, D_0 + D_{ph}\}$. There are two types of cylinders, large-sized cylinders and small-sized cylinders, and each customer has one type of cylinder. Note that the formulation can be easily extended to situations where there are more types of cylinders or customers have several types of cylinders. Our formulation is based on the following concepts:

1. Bounding-box based model
   We formulate the problem to acquire the delivery list as obtaining the small rectangle that covers the customers needing replacement cylinders. The rectangle is prepared per trip for each date. In particular, the formulation does not ensure that the rectangle covers the depot, unlike in existing approaches [13]. Figure 3 illustrates the difference in the customer list derived from the formulation regardless of whether the depot is included. When a customer far from the depot must be visited, the rectangle becomes large with the previous approach, and the customer requiring delivery must not be gathered. A numerical experiment about these differences is discussed in Appendix D.
2. Multi-period optimization model
   When the delivery route per day is obtained sequentially, it may harm the future delivery area by resulting in a large delivery area. To prevent such occurrences, we simultaneously acquire a multi-day delivery list.

The formulation is shown in Problem 4.1. We attempt to minimize the maximum value of the summation of the length of edges in a rectangle. Constraints (1)–(4) ensure that the rectangle covers the customers that have been shortlisted for cylinder's replacement. Constraint (5) enables us to deliver consistently to high-risk customers. Without constraint (5), the output rectangle has no incentive to cover moderate-risk customers. Therefore, it is necessary to determine the lower bound of the number of deliveries LB. Constraint (6) ensures at least one visit to a high-risk customer during the target days. Every customer can belong to at most one trip each day, which is ensured by constraint (7). Constraint (8) ensures truck availability in terms of maximum weight loading. Constraints (9) and (10) are prepared to satisfy

the truck availability in terms of space. Let N_{Large} be the maximum number of large-sized cylinders being carried with a single truck, and the constraint (9) describes the maximum number of large-sized cylinders. Let γ_l be the maximum number of small-sized cylinders given the number of large-sized cylinders l ($l = 0, 1, \cdots, N_{\text{Large}}$) being loaded. There is an almost linear relationship between l and γ_l based on the observation. Then, the linear function $al + b$ is prepared with parameters a and b such that the following inequality is satisfied $al + b \leq \gamma_l$ ($l = 0, 1, \cdots, N_{\text{Large}}$). To maintain the search space as much as possible, the gap between $al + b$ and γ_l should be as small as possible among all candidates, approximately a and b. The optimal parameters among them are denoted as a^* and b^*. Finally, a constraint (10) is added for the number of small-sized cylinders with linear inequality. When the problem is infeasible, the situation is derived from constraint (5) because the numbers of high-risk and moderate-risk customers are fewer than the lower bound. Under this situation, we reduce the lower bound by 1, and solve the problem repeatedly until the solution is obtained.

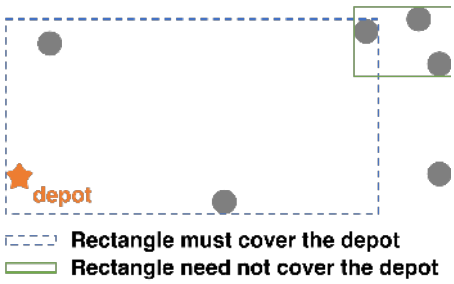


Figure 3: Illustration of the influence whether the depot is included or not in the output

1. Variables
   - $y_{i,j}^{(C)} = \begin{cases} 1 & \text{Customer } C \text{ is selected in the delivery list with } j\text{-th trip on after } i \text{ days} \\ 0 & \text{otherwise} \end{cases}$
   - $v_{i,j}^{lb}, v_{i,j}^{ub}, h_{i,j}^{lb}, h_{i,j}^{ub} \in \mathbb{R}$: Variables representing a rectangle covering the delivery area for j-th trip after i days.
2. Constants
   - $a^*, b^* \in \mathbb{R}$: Hyperparameters representing the available truck capacity
   - $W > 0$: Hyperparameters representing the maximum loading weight of the truck
   - $\mathcal{C}_{\text{Large}} \subset \mathcal{C}$: Set of customers where large-sized cylinders are installed
   - $N_{\text{Large}} \geq 0$: Maximum number of large-sized cylinders being carried with a single truck
   - $\mathcal{C}_{\text{high}}(i)$: High-risk customers after i days
   - $\mathcal{C}_{\text{mdr}}(i)$: Moderate-risk customers after i days
   - Location for customer C: $(\text{lon}^{(C)}, \text{lat}^{(C)}) \in \mathbb{R}^2_{\geq 0}$
   - Number/Weight of cylinders for customer C : $\text{cn}^{(C)}, \text{cw}^{(C)}$
   - Lower bound of number of visiting customer : LB
   - Planning horizon (number of dates) : D_{ph}
   - $\theta_{\text{high}}^{(C)} := \min\{i \mid C \in \mathcal{C}_{\text{high}}(i)\}$
   - $\theta_{\text{mdr}}^{(C)} := \min\{i \mid C \in \mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)\}$

Problem 4.1. Obtaining the smallest rectangle that covers the customers for replacement to acquire multiple-days' customer list

$$
\begin{aligned}
\underset{y,v,w,L}{\text{minimize}} \quad & \max_{i,j} \left\{ \left(v_{i,j}^{ub} - v_{i,i}^{lb} \right) + \left(h_{i,j}^{ub} - h_{i,j}^{lb} \right) \right\} \\
\text{subject to} \quad & \text{lon}^{(C)} y_{i,j}^{(C)} \leq h_{i,j}^{ub} && (\forall i, \forall j, \forall C \in \mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)) \quad (1) \\
& \text{lon}^{(C)} y_{i,j}^{(C)} + \max_{C' \in \mathcal{C}} \text{lon}^{(C')} \left(1 - y_{i,j}^{(C)} \right) \geq h_{i,j}^{lb} \\
& && (\forall i, \forall j, \forall C \in \mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)) \quad (2) \\
& \text{lat}^{(C)} y_{i,j}^{(C)} \leq v_{i,j}^{ub} && (\forall i, \forall j, \forall C \in \mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)) \quad (3) \\
& \text{lat}^{(C)} y_{i,j}^{(C)} + \max_{C' \in \mathcal{C}} \text{lat}^{(C')} \left(1 - y_{i,j}^{(C)} \right) \geq v_{i,j}^{lb} && (\forall i, \forall j, \forall C \in \mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)) \quad (4) \\
& \sum_{C \in \mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)} y_{i,j}^{(C)} \geq LB && (\forall i, \forall j) \quad (5) \\
& \sum_{i=\theta_{\text{mdr}}^{(C)}}^{\theta_{\text{high}}^{(C)}} \sum_{j} y_{i,j}^{(C)} \geq 1 && \left(\forall C \in \mathcal{C} \text{ s.t. } \theta_{\text{high}}^{(C)} \leq D_{ph} \right) \quad (6) \\
& \sum_{j} y_{i,j}^{(C)} \leq 1 && (\forall i, \forall C \in \mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)) \quad (7) \\
& \left(\sum_{\mathcal{C}_{\text{mdr}}(i) \sqcup \mathcal{C}_{\text{high}}(i)} \text{cw}^{(C)} \text{cn}^{(C)} y_{i,j}^{(C)} \right) \leq W && (\forall i, j) \quad (8) \\
& \left(\sum_{C \in \mathcal{C}_{\text{Large}}} \text{cn}^{(C)} y_{i,j}^{(C)} \right) \leq N_{\text{Large}} && (\forall i, j) \quad (9) \\
& \left(\sum_{C \in \mathcal{C} \setminus \mathcal{C}_{\text{Large}}} \text{cn}^{(C)} y_{i,j}^{(C)} \right) \leq a^* \left(\sum_{C \in \mathcal{C}_{\text{Large}}} \text{cn}^{(C)} y_{i,j}^{(C)} \right) + b^* && (\forall i, j) \quad (10)
\end{aligned}
$$

5. Determining Delivery Route

In this section, we determine the delivery route based on the following aspects with the input data: the division of customers, and every customer in the division is classified as a high-risk or moderate-risk customer obtained by the Problem 4.1.

- The customers are to be visited for delivery are determined among moderate-risk customers. We deliver to as many moderate-risk customers as possible with ensuring visit to high-risk customers.
- We determine the visiting order to the customers. Because MapBox[1] is utilized to acquire the optimal route between two customers, the focus is only on the order of the visits. The working duration should be as short as possible.

Let $\bar{y}$ be the optimal solution of Problem 4.1. Because determining the delivery route is considered per day, the suffix i is omitted. The sets of high-risk customers belonging to trip j and that of moderate-risk customers belonging to trip j are denoted as: $\hat{\mathcal{C}}_{\text{high}}(j) := \left\{ C \in \mathcal{C}_{\text{high}}(i) \mid \bar{y}_{i,j}^{(C)} = 1 \right\}$ and $\hat{\mathcal{C}}_{\text{mdr}}(j) := \left\{ C \in \mathcal{C}_{\text{mdr}}(i) \mid \bar{y}_{i,j}^{(C)} = 1 \right\}$, respectively. Note that the indices of j in $\hat{\mathcal{C}}_{\text{high}}(j)$ and $\hat{\mathcal{C}}_{\text{mdr}}(j)$ do not indicate the order of visiting per trip.

[1] https://www.mapbox.com

The delivery route is obtained by solving two optimization problems sequentially. With the first problem, we determine a delivery route that maximizes the number of deliveries to moderate-risk customers, ensuring deliveries to high-risk customers. When the two routes have the same number of deliveries, the routes can differ in terms of the working time. Therefore, a second optimization problem is prepared to minimize the working time, ensuring the replacement to the customers obtained from the first optimization problem. When the solution is not found within the time limit, the heuristic algorithms shown in Section 5.3 are applied to determine the delivery route for applying the proposed system to the real world.

5.1. Maximizing Number of Deliveries

First, we obtain how many customers we can visit for replacement. Note that high-risk customers must be visited for the replacement, and moderate-risk customers need not, but encouraged to be visited to suppress the concentration of the replacements.

5.1.1. Graph Construction for Obtaining Multi-trip Delivery Order

A graph is constructed for formulating a mixed-integer optimization problem. The path on the graph represents the order of visiting. When we denote the number of trips for the date as n, the sets of vertices and edges are prepared as follows:

1. $V := SN \cup \bigcup_{j=1}^{n} \left(\hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j)\right)$: Vertices
   - SN: Set of supernodes representing the delivery center
2. $E := \cup_{j=1}^{n} \left(E_j^{\text{inter}} \cup E_j^{\text{intra}}\right)$: Edges
   - $E_j^{\text{inter}} \subset \left\{(C, C') \in \left(\hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j)\right) \times \left(\hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j)\right) \mid C \neq C'\right\}$. An edge (C, C') represents the movement of the staff from customer C to customer C', both of which were selected during trip j. To reduce the number of variables, we extract the five-nearest-neighbor customers and create edges among the nodes corresponding to every customer.
   - $E_j^{\text{intra}} := \left(\hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j)\right) \times SN \cup SN \times \left(\hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j)\right)$. An edge (C, sn) represents the staff moving from customer C to delivery center sn, and vice versa.

   The trip index j does not indicate the order of visiting, and the order of visiting per trip is determined after solving the optimization Problem 5.1.

5.1.2. Formulation

We show the proposed formulation to maximize the number of deliveries as follows.

1. Variables
   - $z^{(C,C')} = \begin{cases} 1 & \text{Staff visit customer } C' \text{ next to } C \text{ for replacement} \\ 0 & \text{otherwise} \end{cases}$
   - $u^{(C)} \in \mathbb{Z}$: Order in which customer C is visited during a particular day
   - $t^{(C)} \in \mathbb{R}$: Arrival time to customer C
2. Constants
   - $d : V \times V \to \mathbb{R}$: Duration of movement between two points
   - $\left[T_{lb}^{\text{work}}, T_{ub}^{\text{work}}\right] \subset [0, 24\times60]$: Time window representing staff availability (minute)
   - $T_{\max}$: Sufficiently large value (at most 24×60)
   - $\left[T_{lb}^{(C)}, T_{ub}^{(C)}\right] \subset [0, 24 \times 60]$: Time window representing availability of customer C
   - $rep : V \to \mathbb{R}$: Duration for replacing gas cylinder for customer, and duration for break time for supernode

- n: Number of trips
- $\hat{\mathcal{C}}_{\text{high}}(j)$: High-risk customers in trip j selected by Problem 4.1
- $\hat{\mathcal{C}}_{\text{mdr}}(j)$: Moderate-risk customers in trip j selected by Problem 4.1
- $f_j : (\hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j)) \rightarrow 2^{(\hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j))}$: Five-nearest customers from a customer in trip j

The formulation is shown in Problem 5.1. The objective is to maximize the customers to visit for cylinder's replacement. Constraints (11) and (12) represent the flow conservation for customers. Moreover, constraint (12) ensures visit to high-risk customers for replacement. Constraints (13) and (14) ensure that every supernode satisfies the flow conservation and must be visited. Constraint (15) is the subtour elimination constraint, and it restricts the relationship between variables z and u. Constraints (16) and (17) satisfy the time demands of the customers and staff, respectively. Constraint (18) ensures the relationship between z and t. Constraint (19) ensures that the working time is included in the staff availability.

Problem 5.1. Maximizing the visit to the moderate-risk customers for replacement

$$\underset{z,u,t}{\text{maximize}} \quad \sum_{(C,C')\in E} z^{(C,C')}$$

$$\text{subject to} \quad \sum_{C_0\in f_j(C)} z^{(C_0,C)} = \sum_{C_1\in f_j(C)} z^{(C,C_1)} \leq 1 \qquad (\forall j, \forall C \in \mathcal{C}_{\text{mdr}}(j)) \quad (11)$$

$$\sum_{C_0\in f_j(C)} z^{(C_0,C)} = \sum_{C_1\in f_j(C)} z^{(C,C_1)} = 1 \qquad (\forall j, \forall C \in \mathcal{C}_{\text{high}}(j)) \quad (12)$$

$$\sum_{C\in V\setminus SN} z^{(sn_1,C)} = \cdots = \sum_{C\in V\setminus SN} z^{(sn_{n-1},C)} = 1 \qquad (sn_1, \ldots, sn_{n-1} \in SN) \quad (13)$$

$$\sum_{C\in V\setminus SN} z^{(C,sn_2)} = \cdots = \sum_{C\in V\setminus SN} z^{(C,sn_n)} = 1 \qquad (sn_2, \ldots, sn_n \in SN) \quad (14)$$

$$u^{(C)} - u^{(C')} + (|V|-1)z^{(C,C')} \leq |V| - 2 \qquad (\forall\, (C,C') \in E) \quad (15)$$

$$T_{lb}^{(C)} \sum_{C'\in V} z^{(C',C)} \leq t^{(C)} \leq \left(T_{ub}^{(C)} - rep(C)\right) \sum_{C'\in V} z^{(C',C)} \qquad (\forall C \in V \setminus SN) \quad (16)$$

$$T_{lb}^{\text{work}} \leq t^{(C)} \leq T_{ub}^{\text{work}} \qquad (\forall C) \quad (17)$$

$$t^{(C')} - t^{(C)} \geq (rep(C) + d(C,C'))\, z^{(C,C')} - T_{\max}\left(1 - z^{(C,C')}\right) \qquad (\forall\, (C,C') \in E) \quad (18)$$

$$t^{(C)} - t^{(C')} \leq T_{\max} - rep(C') \qquad (\forall\, (C,C') \in E) \quad (19)$$

5.2. Minimizing Working Duration

After solving problem 5.1, the second problem that obtains the delivery route with the shortest working duration is addressed. Letting the optimal solution to Problem 5.1 be $\bar{z}$, the customers to be replaced in trip j are denoted as $\bar{\mathcal{C}}(j) := \{C \in \hat{\mathcal{C}}_{\text{high}}(j) \cup \hat{\mathcal{C}}_{\text{mdr}}(j) | \sum_{C'\in f(C)} \bar{z}^{(C,C')} = 1\}$. We obtain the replacement route under the constraint of visiting all of these customers.

5.2.1. Graph Construction for Obtaining Multitrip Delivery Order

A graph is constructed for formulating a mixed-integer optimization problem. The path on the graph also represents the order of visiting. Because the basic concepts are the same as

in Section 5.1.1, the detailed explanation is omitted in this section. The sets of vertices and edges are prepared as follows:

1. $\bar{V} := SN \cup \bigcup_{j=1}^{n} \bar{\mathcal{C}}(j)$: Vertices
2. $\bar{E} := \cup_{j=1}^{n} \left(\bar{E}_i^{\text{inter}} \cup \bar{E}_j^{\text{intra}}\right)$: Edges
   - $\bar{E}_j^{\text{inter}} \subset \{(C, C') \in \bar{\mathcal{C}}(j) \times \bar{\mathcal{C}}(j) \mid C \neq C'\}$
   - $\bar{E}_j^{\text{intra}} := \bar{\mathcal{C}}(j) \times SN \cup SN \times \bar{\mathcal{C}}(j)$

5.2.2. Formulation

We show the proposed formulation to minimize the working duration as follows.

1. Variables
   - $\xi^{(C,C')} = \begin{cases} 1 & \text{Staff visit customer } C' \text{ next to } C \text{ for replacement} \\ 0 & \text{otherwise} \end{cases}$
   - $\nu^{(C)} \in \mathbb{Z}$: Order in which customer C is visited during a particular day
   - $\tau^{(C)} \in \mathbb{R}$: Arrival time to customer C
2. Constants
   - $\bar{f}_j : \bar{C}(j) \to 2^{\bar{C}(j)}$: Five-nearest customers from a customer in trip j

   Other notations are the same as those in Problem 5.1.

The formulation is shown in Problem 5.2. The objective function represents the total working duration, which should be minimized. Constraint (20) represents the flow conservation for customers. Constraints (21) and (22) ensure that every supernode satisfies the flow conservation and must be visited. Constraint (23) is the subtour elimination constraint, and it restricts the relationship between variables ξ and ν. Constraints (24) and (25) satisfy the time demands of the customers and staff, respectively. Furthermore, constraint (26) ensures the relationship between ξ and τ. Constraint (27) ensures that the working time is included in the staff availability.

Problem 5.2. Minimizing the total working duration

$$\underset{\xi,\nu,\tau}{\text{minimize}} \quad \max_{C}(\tau^{(C)} + rep(C))$$

$$\text{subject to} \quad \sum_{C_0 \in \bar{f}_j(C)} \xi^{(C_0,C)} = \sum_{C_1 \in \bar{f}_j(C)} \xi^{(C,C_1)} = 1 \qquad (\forall j, \forall C \in \bar{C}(j)) \quad (20)$$

$$\sum_{C \in V \setminus SN} \xi^{(sn_1,C)} = \cdots = \sum_{C \in V \setminus SN} \xi^{(sn_{n-1},C)} = 1$$

$$(sn_1, \ldots, sn_{n-1} \in SN) \quad (21)$$

$$\sum_{C \in V \setminus SN} \xi^{(C,sn_2)} = \cdots = \sum_{C \in V \setminus SN} \xi^{(C,sn_n)} = 1 \qquad (sn_2, \ldots, sn_n \in SN) \quad (22)$$

$$\nu^{(C)} - \nu^{(C')} + (|\bar{V}| - 1)\xi^{(C,C')} \leq |\bar{V}| - 2 \qquad \left(\forall\, (C, C') \in \bar{E}\right) \quad (23)$$

$$T_{lb}^{(C)} \sum_{C' \in V} \xi^{(C',C)} \leq t^{(C)} \leq \left(T_{ub}^{(C)} - rep(C)\right) \sum_{C' \in V} \xi^{C',C}$$

$$(\forall C \in \textstyle\bigcup_{j=1}^{n} \bar{C}(j)) \quad (24)$$

$$T_{lb}^{\text{work}} \leq \tau^{(C)} \leq T_{ub}^{\text{work}} \qquad (\forall C) \quad (25)$$

$$\tau^{(C')} - \tau^{(C)} \geq (rep(C) + d(C, C'))\, \xi^{(C,C')} - T_{\max}\left(1 - \xi^{(C,C')}\right)$$

$$\left(\forall\, (C, C') \in \bar{E}\right) \quad (26)$$

$$\tau^{(C)} - \tau^{(C')} \leq T_{\max} - rep(C') \qquad \left(\forall\, (C, C') \in \bar{E}\right) \quad (27)$$

$\alpha_{\text{high}}, q_{\text{high}}$	$\alpha_{\text{mdr}}, q_{\text{mdr}}$	D_{lbd}	D_{ph}	Working time
$0\%, 0.9$	$5\%, 0.1$	3	3	7:00 – 19:00

Table 1: Hyperparameter settings in the field test. Because the experiments described in Appendix C were conducted after the field test, hyperparameter tuning was not performed in the field test. Note that the staff worked only on weekdays.

5.3. Postprocessing

When a feasible solution cannot be obtained within the time limit, we stop solving the optimization problem using the solver. Then, simple heuristic algorithms are used to obtain the replacement plan for any situation, which is essential for applying the proposed model to the real world. The concepts of the heuristic algorithm are as follows.

- The 2-opt algorithm acquires the shortest path for visiting all the given high-risk customers.
- The greedy algorithm extracts as many customers as possible for visiting based on the value of the risk function while satisfying staff availability.

6. Field Test

A field test was conducted in Chiba prefecture in Japan with more than 1,000 customers for approximately two weeks. In the field test, the gas provider operated the replacement based on the output of our system. Evaluation metrics were calculated after the field test. The metrics for the field test were compared with those for the replacement record for the same season in the previous year.

6.1. Experimental Settings

The hyperparameter settings in the field test are summarized in Table 1. Gurobi Optimizer v9.1.0 was utilized to solve the mixed-integer optimization problems, and the time limit for computing was set to 30 min. Because the system was operated in a cloud computing environment in the field test, it was not possible to determine which computational environment was utilized per problem. The candidates are shown in the Appendix A.

The total number of customers was 1,366, and the number of customers with smart meters was 861. The locations of the customers and the delivery center are shown in Figure 4. In the figure, the dots and the star represent the customers and the delivery center, respectively. For the target date, the replacement plan for the next working date was created. The driver works Monday through Friday. The driver loads the cylinders on the truck after the delivery of the previous day, but loads the cylinders to be delivered on Monday after delivery on Friday. Our system was allowed to run for five hours, and the calculation was performed on the business day immediately preceding the delivery date.

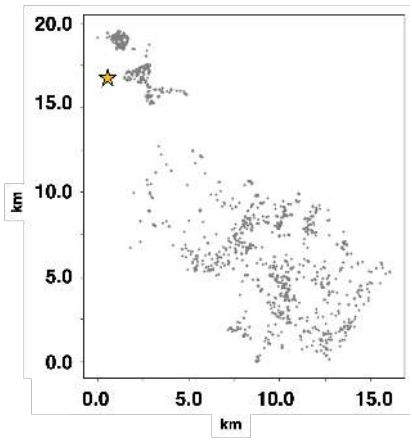


Figure 4: Scatter plot of the customer locations and the delivery center (depot location)

In the gas usage forecast for estimating replacement dates, the support vector regression was used for the smart meter based on the pre-experiment. The proposed method (Section 3.1) was applied for most of conventional meter's gas usage esti-

Name	Gas provider	Proposed system
Period	13/3/2020-27/3/2020	1/3/2021-15/3/2021
Average temperature	9.7 °C	9.8 °C
Travel-time/day(↓)	2 h 52 m 32 s	**2 h 06 m 44 s**
Travel-time/customer(↓)	3 m 46 s	**2 m 34 s**
Nonreplacement(↓)	78	**0**[*1]

Table 2: Experiment setting and evaluation metrics of the field test. Owing to the missing data of the amount of gas remaining, it was not possible to evaluate gas shortage. Note that the gas provider occasionally changed the proposed system outputs, utilizing 83.9% of the output of the algorithm.

mation based on the pre-experiment shown in Appendix B.

Based on the knowledge of the gas provider, the replacement plan operated about one year before the field test was picked up to evaluate the proposed system. Because the gas usage heavily depends on the temperature, the comparison periods were extracted based on the average temperature. The temperature data were obtained from the Japan Meteorological Agency website[2].

6.2. Evaluation Metrics

The quality of the replacement plan was investigated using the following evaluation metrics.

- Average traveling time per day (travel-time/day)
- Average traveling time per customer (travel-time/customer)
- Total number of nonreplacement visits (nonreplacement)

These metrics are different from those described in Section 7. "Travel time per day" and "Travel time per customer" do not include the time required to replace the cylinders at the customers' locations and nonreplacement customers who are visited, that is, the driving time of the truck. The abbreviations in the parentheses indicate the evaluation metrics listed in the Table 2. The average number of customers whose gas has run out cannot be compared because of missing data.

6.3. Results and Discussion

The evaluation metrics are shown in Table 2, and the observations made are as follows.

- The number of nonreplacement visits was determined to be zero. The staff was penalized by the gas provider when the average remaining gas rate at the replacement exceeds 8% per month. During the field test, to determine the number of cylinders that do not require nonreplacement based on the remaining gas rate, the gas provider was asked to replace the gas cylinder even when a large amount of gas remained. After the field test was finished, the average remaining gas rate was below 8%. Therefore, it was determined that there were no nonreplacement visits, although the number of visits to customers in the proposed system was greater than that of the gas provider (see *1 in Table 2).
- The proposed system reduced the duration of travel-time and the number of nonreplacement events. It shows that the proposed system effectively determines the customer list and delivery route.

[2] http://www.data.jma.go.jp/obd/stats/etrn/index.php

7. Verification Experiments

After the field test, further experiments were conducted to investigate the influence of the replacement plan for each part of the proposed method. The entire process of solving the CRP was divided into two parts: 1) estimating the date for replacement and 2) acquiring the customer list and determining the delivery route. Because each of them is performed by the gas provider or the proposed system, four types of experiments were conducted.

7.1. Experimental Settings

The basic hyperparmeter settings were as same as in Section 6.1. These settings are shown in the upper part of Table 3. The target dates were from March 16–24, 2021.

The gas usage forecast model of the gas provider used in Exp A and Exp C had the following features.

- The daily gas usage was assumed to be constant.
- Five different periods were prepared, and the average value was calculated per period. Then, the maximum value among them was obtained.
- The forecast value was calculated by multiplying the above value and the hyperparameter representing the usage tendency of the next period.

Moreover, the workers manually created the replacement plan based on their experience with Exp A and B. Based on the result of pre-experiment shown in Appendix C, the hyperparameters were set, α_{high} as 0%, q_{high} as 0.3, α_{mdr} as 7%, and q_{mdr} as 0.3, respectively.

Gurobi Optimizer v9.1.1 was used to solve the mixed-integer optimization problems, and the time limit was set to 30 min. In the computing environment, the CPU was an Intel(R) Xeon(R) Gold 6240R CPU with a 2.40GHz CPU frequency, and the memory was 768GB.

7.2. Evaluation Metrics

The following evaluation metrics were calculated to investigate the quality of the replacement plan.

- Average remaining gas rate at replacement (rate-average)
- Median of remaining gas rate at replacement (rate-median)
- Total success replacement rate, excluding failures (success)
- Rate of failure in replacement owing to a cylinder being out of gas (fail-out)
- Rate of failure in replacement owing to an overabundance of gas where the threshold was set as 15%. (Fail-over) Note that the fail-over is different from the nonreplacement visit because nonreplacement visit is determined by staff's sense. Therefore, the fail-over metric is calculated for quantitative evaluation.
- Number of failures in replacement owing to the time window of the customer (fail-time)
- Average delivery time per customer (time/customer)
- Average travel distance per customer (distance/customer)
- Average number of customers whose gas has run out (run-out)

The delivery time included the duration of replacing the cylinders in addition to the driving duration. The abbreviations in parentheses are used to summarize the results listed in Table 3.

7.3. Results and Discussion

We summarize the results of the evaluation metrics in Table 3. In summary, when the proposed system was utilized for the entire process of making replacement plans (Exp D), the highest replacement plan quality was achieved. This was because Exp D achieved the best run-out result and second-best fail-over result. Moreover, it succeeded in creating efficient

	Marker	Estimating replacement date	Acquiring customer list	Determining delivery route
Exp A		Gas provider	Gas provider	Gas provider
Exp B		**Proposed system**	Gas provider	Gas provider
Exp C		Gas provider	**Proposed system**	**Proposed system**
Exp D		**Proposed system**	**Proposed system**	**Proposed system**

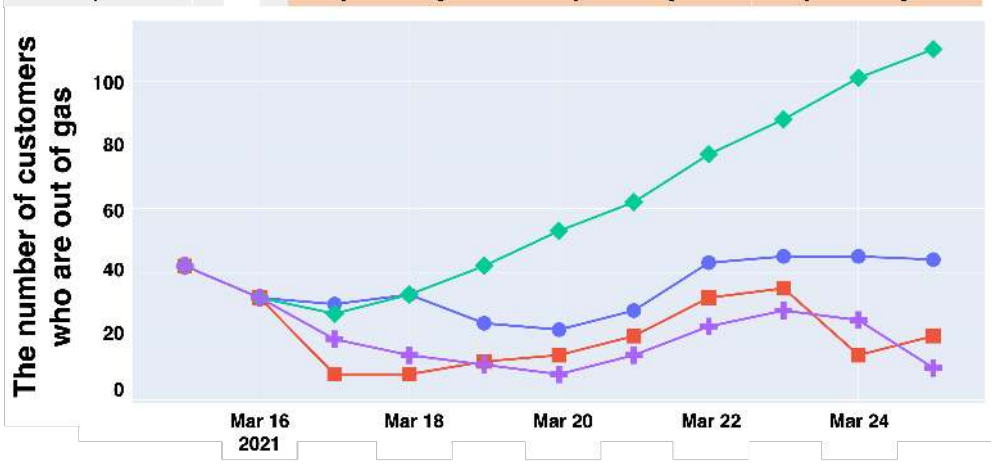


Figure 5: Number of customers whose gas has run out during the target period. Note that the lines overlap if the numbers are the same for the first two consecutive days.

replacement plans that can be expressed through the time/customer or distance/customer. Because approximately one-third of customers did not have a smart meter in the experiment, these results indicate that the proposed system can make the replacement plan more effective even if not all customers have a smart meter. The following aspects are considered based on the results.

1. (**Overall**) In Exp D, the proposed system was completely applied to make replacement plans.
   - Exp D was more successful than Exp A in terms of all evaluation metrics. Exp A recorded the least number of fail-out visits; however, this was mainly because they left out the existence of customers with a gas shortage (see Figure 5). By contrast, the figure shows that in Exp D, more high-risk customers were visited. These tendencies can be observed by comparing the run-out events between Exp A and Exp D.
   - Exp D achieved a smaller number of customers with run-out events and competitive results in fail-over events compared to Exp B. It indicates that the proposed system can realize a better replacement plan than the gas provider. Moreover, in Exp D, considerably better performance for time/customer and distance/trip was achieved than those in Exp B. We prepared for a future increase in high-risk customers by visiting moderate-risk customers, which can be interpreted as the slope occurring near March 22, as shown in Figure 5. The figure shows that Exp D achieved the smallest increase in the number of customers experiencing a gas shortage. In other words, a larger number of failure events indicates that the visit for replacement was assured to the customer after gas ran out. Because the rate-median of Exp D was lower than that of Exp A and Exp C, Exp D tended to achieve visit to customers for replacement who were about to run out of gas.
   - Through the proposed system, more efficient replacement planning was realized

than in Exp C for every evaluation metric. The number of run-out events could be dramatically reduced in Exp D based on a better gas usage forecast.

- In Exp D, more fail-out visits were observed. By contrast, Exp D succeeded in reducing the number of run-out events compared with Exp A. Exp D focused more on delivery to high-risk customers, which caused a decrease in the accumulation of run-out events. The smallest rate-median was reported in Exp D, indicating that the focus was on replacement high-risk customers' cylinders during Exp D.

2. (**Estimating replacement date**) The proposed system was compared with a conventional method used for estimating replacement date. In both Exp A and B, the replacement plan was made by the gas provider. In Exp A, the gas provider executed the gas usage forecast, whereas the proposed system in Exp. B conducted.
    - Rate-median in Exp B was lower than in Exp A. With the proposed system for gas usage forecasting, the gas provider could deliver to customers with a low remaining gas rate. The results show the superiority of the proposed system in terms of estimating the replacement date.
3. (**Acquiring customer list and determining delivery route**) The proposed system and the conventional system for acquiring a customer list and determining a delivery route were compared. Based on the gas usage forecast result of the gas provider, the replacement plan was made by the gas provider and the proposed system during Exp A and Exp C, respectively.
    - The distance/trip in Exp C was longer than that in Exp A. Because the proposed system focuses on visits to high-risk customers with the highest priority, the delivery route occasionally becomes longer than the replacement plan of the gas provider. Exp C recorded the most run-out events owing to the accuracy of the gas usage forecast.
    - In Exp C, as shown in Figure 5, the number of customers with run-out events increased. Many customers were categorized as high-risk customers owing to the imprecise gas usage forecast. This caused many high-risk customers and some moderate-risk customers, who were also considered high-risk customers. When there were too many trips, some trips were extracted to calculate the route for replacement. Based on the results, there were the categorization error that high-risk consumers actually had enough gas and should not be replaced. Thus, the number of moderate-risk customers increased over time. It implies that the quality of the replacement plan can be improved by modifying the selection of many high-risk customers, which is planned as future research.

8. Conclusion

In this paper, we consider the CRP that aims to create a plan to visit customers for replacing gas cylinders, including the delivery route. One of the difficulties for CRP derives from the data acquisition of data on the remaining gas because the gas provider cannot determine the remaining gas without visiting customers and observing the meter. To address this problem, smart meters can be installed for customers, enabling the gas provider to acquire gas usage data daily. In this study, we construct a system to solve CRP. Our system comprises three steps: estimating the date for replacement, acquiring the customer list, and determining the delivery route. Our formulation for making the customer lists enables the gas provider to replace cylinders some days before the date about to be a gas shortage. It can suppress the concentration of replacement on certain days. Numerical experiments were conducted for

Name	Exp A	Exp B	Exp C	Exp D (Exp 3 in Appendix C)
Estimating date for replacement	Gas provider	Proposed	Gas provider	Proposed
Acquiring customer list	Gas provider	Gas provider	Proposed	Proposed
Determining delivery order	Gas provider	Gas provider	Proposed	Proposed
Rate-average	10.3 %	8.83e-05 %	8.73 %	-1.36 %
Rate-median	7.56 %	1.82 %	7.07 %	0.09 %
Success	36.3 %	56.6 %	31.3 %	47.1 %
Fail-out	30.5 %	38.3 %	35.4 %	45.1 %
Fail-over(↓)	33.2 %	**4.7 %**	33.3 %	7.8 %
Fail-time (↓)	3	2	**0**	**0**
Time/customer(↓)	12 m 55 s	13 m 30 s	13 m 19 s	**12 m 05 s**
Distance/customer(↓)	1.7 km	2.7 km	2.7 km	**1.5 km**
Run-out(↓)	34.4	21.7	49.7	**21.6**

Table 3: Settings and evaluation metrics of the experiment. As the criterion for failure, the remaining gas rate at replacement must be greater than 15%. The results of Exp D were the same as those in Exp 3 listed in Table 5 shown in Appendix C.

real gas usage in Chiba prefecture in Japan with more than 1,000 customers. A field test was performed by a gas provider incorporating the output of the proposed system into its operation. Our system succeeded in reducing the gas shortage, nonreplacement visits when an ample supply of gas remained, and working duration. Note that because approximately one-third of customers did not have a smart meter, the results show the potential of utilizing smart meters. As more smart meters are installed to the customer, the gas shortage will decrease because of the acquisition of more-precise replacement date estimation.

Three research directions for the future work are being considered. After the field test, we got feedback from the gas providers that the customer for replacement per day should be more aggregated. Based on the feedback, how to acquire the customer list so that the customers for replacement are more close to each other will be investigated as the first topic. When this issue is addressed, for example, gas providers can replace cylinders for multiple customers without changing parking positions, which reduces the staff workload more effectively. The second direction for future work is how to determine customer priority for installing smart meters. In this study, the cylinder replacement problem after installing the smart meter for some customers was considered. In the future, our system will be utilized to make replacement plans for other areas where no smart meters are installed. In that case, it is necessary to determine the customer's priority for installing smart meters to improve the replacement date's estimation performance. The third direction is utilizing our framework for different industries. Our system creates optimal delivery plans through suppressing the sudden increase in delivery targets, consisting of demand forecasting, acquiring customer list for delivery, and determining the delivery route. The proposed system can be applied to solve related problems of other industries, such as grocery delivery.

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Appendix

A. Computational Environment Used in Field Test

In this section, the computation environment used in field test is described (see Section 6). Different computing environments were prepared for estimating the replacement date and acquiring customer lists, and determining delivery routes. One of the following computational environments with 8-GB memory was utilized for estimating the replacement date.

- Intel(R) Xeon(R) Platinum 8370C with 2.1-GHz CPU frequency
- Intel(R) Xeon(R) Platinum 8272CL with 2.1-GHz CPU frequency
- Intel(R) Xeon(R) Platinum 8171M 2.1-GHz CPU frequency
- Intel(R) Xeon(R) E5-2673 v4 with 2.3-GHz CPU frequency
- Intel(R) Xeon(R) E5-2673 v3 with 2.4-GHz CPU frequency

Moreover, one of the following computational environments with 128-GB memory was utilized for acquiring customer list" and determining delivery route.

- Intel(R) Xeon(R) Platinum 8370C with 2.1-GHz CPU frequency
- Intel(R) Xeon(R) Platinum 8272CL with 2.1-GHz CPU frequency
- Intel(R) Xeon(R) 8171M with 2.1-GHz CPU frequency
- Intel(R) Xeon(R) E5-2673 v4 with 2.3-GHz CPU frequency

B. Numerical Experiment to Estimate the Replacement Date

In this section, we explain the experiment for estimating the replacement date. Two experiments were conducted. The first focused on evaluating the forecast performance for the gas usage of a conventional meter. The second aimed to investigate the influence of hyperparameters for extracting the high-risk customers by considering the gas usage forecast's uncertainty. We denote proposed model shown in Section 3.1 as kNN.

B.1. Evaluation of Gas Usage Forecast for Conventional Meters

The method of the gas usage forecast presented in Section 3.1 was evaluated using a conventional meter.

B.1.1. Evaluation Settings

For the experiment conducted with conventional meters, smart meter data were used to prepare the correct data. Data were generated by intentionally omitting the smart meters' data every 29 days, which is the maximum frequency of the missing data of meters for conventional meters.

- Target dates are 42 days: $\{1/2/2021, \cdots, 14/3/2021\}$
- Target meters : 619 smart meters
- Number of samples k : 10

B.1.2. Evaluation Metrics

The root-mean-square error (RMSE) of each meter was calculated for evaluation. In addition, the *upward rate* was proposed as an evaluation metric. To avoid running out of gas, it

is important that the forecast gas usage be less than the actual gas usage. Therefore, the *upward rate* is defined as the percentage of $prediction \geq observation$, and it was determined that a higher *upward rate* was better.

B.1.3. Baselines

Four baseline models were prepared.

- EachMean: It uses the mean of its own meter data over the past 90 days as the forecast value.
- TQ: It uses the third quartile Q_3 of its own meter data over the last 90 days as the forecast value.
- AllMean: It uses the mean of all smart meter data over the past 90 days as the forecast value.
- MaxModel : It imitates the gas usage forecast model of the gas provider that is described in Section 7.1.

B.1.4. Results and Discussion

The experimental results for conventional meters are presented in Figure 6 and in Table 4.

1. (MaxModel versus kNN/EachMean/TQ/AllMean) The average RMSE values among customers of the proposed models were smaller than those of the baseline Max model.
2. (kNN versus EachMean/TQ/AllMean) The RMSE values of the kNN were smaller than those of the other models. This indicates that similar meters data enables us to obtain the more precise gas usage forecast compared to simple methods.

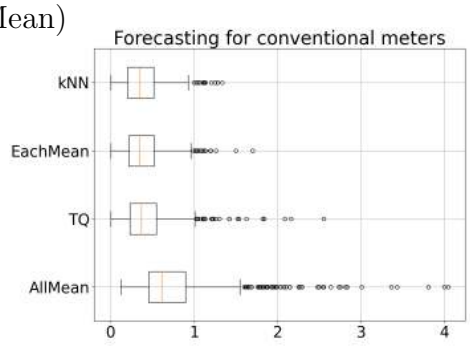


Figure 6: Box plot of RMSE values in the gas usage forecast. MaxModel is eliminated because its performance was much worser than the others.

The kNN uses smart meter data in addition to its own conventional meter value. By contrast, Eachmean and TQ uses only its own conventional meter value, and All Mean uses only the mean of all smart meter data. The numerical results show that the kNN model performs better than the EachMean, TQ, and All Mean models. In other words, utilizing the smart meter's gas usage enhances the performance. However, when the gas usage of the conventional meter is sufficiently larger than that of the other meters, the kNN model does not perform better. This is because the forecast value may be smaller than actual daily gas usage. Under this situation, the TQ model is the best choice because it has the highest upward rate. Therefore, in our experiment, for the gas usage forecast, kNN was applied for most of conventional meters, and TQ was applied for the others with unique gas usage tendency such as a hotel or a restaurant.

B.2. Parameter Sensitivity for Extracting High-Risk Customers

In this section, the effect of the threshold to determine high-risk customers is examined based on the risk functions value shown in Section 3.2.

B.2.1. Evaluation Settings

We extracted meters with no missing values from April 1, 2020 to March 23, 2021. The dates from when the remaining gas rate decreased below the threshold to the remaining gas

Method	Mean	Median	IQR	Maximum	Upward rate
MaxModel	1.3628	0.4227	0.3902	2.165e+02	0.5010
kNN (Proposed)	**0.3795**	0.3478	0.3108	**1.3350**	0.5518
EachMean	0.3895	**0.3440**	**0.2990**	1.7005	0.5916
TQ	0.4243	0.3632	0.3176	2.5515	**0.6708**
AllMean	0.7722	0.6084	0.4431	4.0473	0.4489

Table 4: Evaluation metrics for each forecast model for conventional meters. Mean/Median/IQR/Maximum indicate the mean, median, interquartile range, and maximum of the RMSE values, respectively, in which a lower value is better.

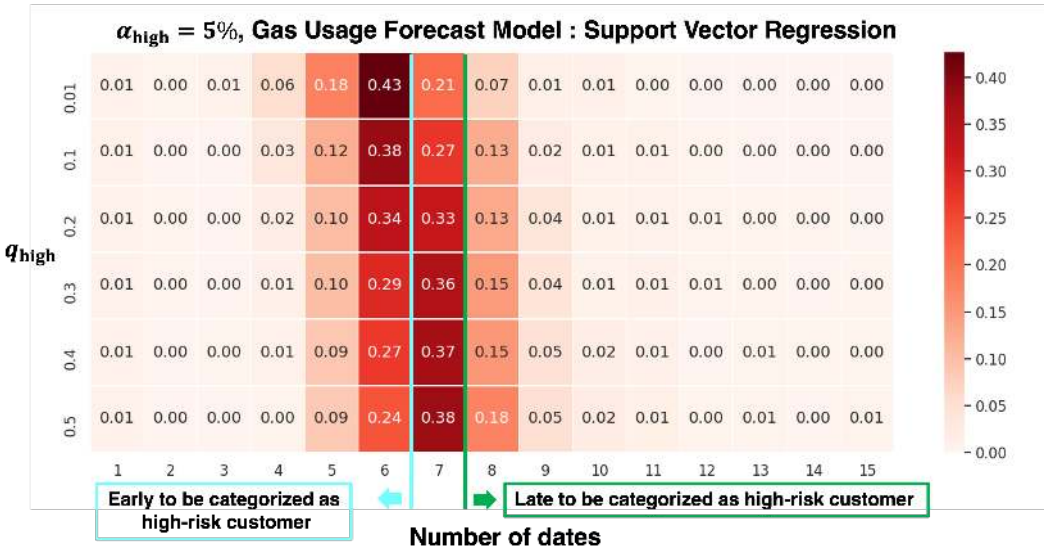


Figure 7: Prediction results for categorizing high-risk customers based on gas usage forecast per day using support vector regression when changing the threshold q_{high}

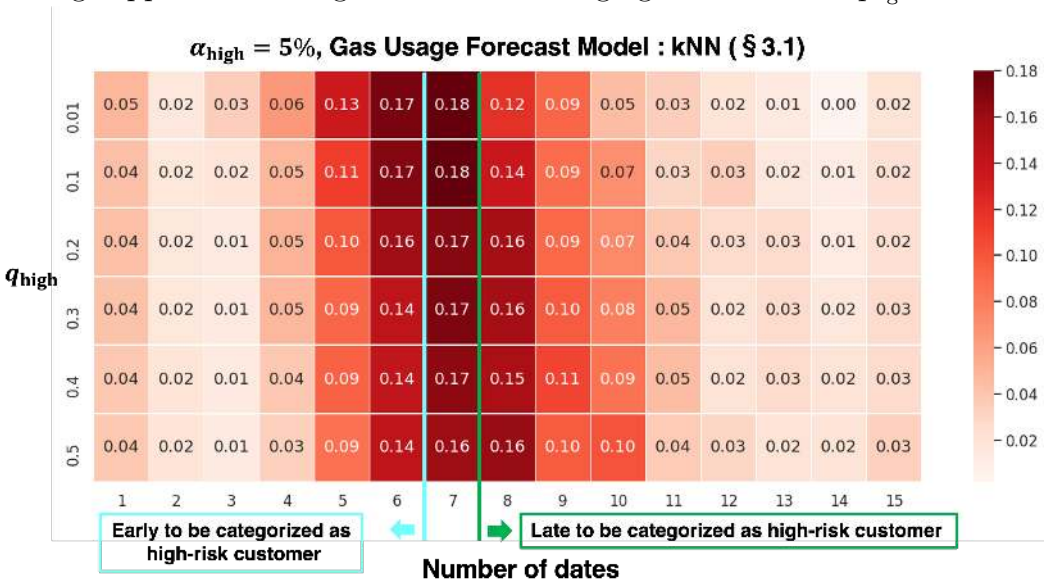


Figure 8: Prediction results for categorizing high-risk customers based on gas usage forecast per day using proposed kNN when changing the threshold q_{high}

rate in the date units were determined. Then, starting from the date seven days preceding the gas usage forecast, the risk function was used to calculate when the customer becomes a high-risk customer. During the experiment, the threshold of the gas rate was set to 5%. Under certain settings, the meter data were intentionally removed by utilizing real missing data. The following two settings were prepared.

- Exp 1: The complete data were utilized without intentionally introducing missing data. Because the setting is assumed for customers with smart meters, we use the support vector regression.
- Exp 2: The data were intentionally removed every 29 days to reproduce the data acquisition of conventional meters. The kNN described in Section 3.1 was used.

B.2.2. Results and Discussion

The results of the experiment on the change in threshold in Exp 1 and Exp 2 are shown in Figures 7 and 8, respectively. Some customers were predicted to be high-risk with an error of one day with respect to the correct days, seven days. To address this situation, the customer can be categorized as a high-risk customer earlier by decreasing the threshold q_{high}. It was experimentally observed that the distribution of the days predicted to be a high-risk was shifted earlier by one day with a threshold q_{high} of 0.01. Under this situation, the rate of customer prediction to be later than the actual date (particularly after seven days) decreases from 28% to 10 % when q_{high} changes from 0.5 to 0.01, respectively. Customers were categorized as high-risk customers earlier as the threshold decreased. Although users can freely choose the threshold value based on these figures, a threshold of 0.3 was chosen for the experiment in Appendix C because it was the lowest value among the values where the most frequent value was seven.

C. Hyperparameter Tuning in Making Customer List

In this section, we investigate the influence of the parameter settings for making the customer list and determining the delivery route.

C.1. Evaluation Settings

The replacement plans are compared when changing the parameters. Only one parameter is changed, whereas the other parameters are fixed. Five hours are allowed to operate our system, and the calculation is executed on the business day immediately preceding the delivery date. The experimental settings are summarized in Table 5. Numerical experiments were conducted under four different settings, as summarized in Table 3. Note that when the hyperparameter q_{high} was set to 0.5, we only focused on whether the average value of the distribution was above the threshold of the gas rate to determine a high-risk customer based on the definition of a risk function.

C.2. Results and Discussion

We use the same evaluation metrics shown in Section 7.2. The results of the evaluation metrics of the experiments are summarized in Table 5, enabling us to consider the following:

Name	Exp 1	Exp 2	Exp 3	Exp 4	Exp 5
$\alpha_{\text{high}}, q_{\text{high}}$	5 %, 0.3	5 %, 0.5	0 %, 0.3	5 %, 0.3	5 %, 0.3
$\alpha_{\text{mdr}}, q_{\text{mdr}}$	7 %, 0.3	7 %, 0.5	7 %, 0.3	7 %, 0.3	7 %, 0.3
D_{lbd}	2	2	2	0	2
D_{ph}	3	3	3	3	0
Rate-average	2.08 %	-0.88 %	-1.36 %	1.16 %	0.49 %
Rate-median	4.10 %	2.57 %	0.09 %	3.87 %	2.42 %
Success	53.3 %	50.1 %	47.1 %	52.5 %	47.7 %
Fail-out	36.6 %	41.1 %	45.1 %	36.9 %	38.9 %
Fail-over(↓)	10.1 %	8.7 %	**7.8 %**	10.6 %	13.4 %
Fail-time(↓)	**0**	1	**0**	1	**0**
Time/customer(↓)	13m06s	13m00s	12m05s	13m38s	11m24s
Distance/customer(↓)	1.8 km	1.8 km	1.5 km	2.0 km	1.2 km
Run-out(↓)	28.1	31.6	**21.6**	26.6	29.6

Table 5: Hyperparameter settings for each experiment setting and evaluation metric of the experiment. The surrounding rectangles highlight the differences in the settings between Exp 1 and the others. The best results are presented in bold.

1. ($\boldsymbol{q}_{\text{high}}, \boldsymbol{q}_{\text{mdr}}$) The difference between Exp 1 and Exp 2 is the threshold for determining high- and moderate-risk customers. The experiments were conducted to observe the effect of considering the risk function defined in Section 3.2.
   - Exp 1 succeeded in increasing the success rate and decreasing the fail-out rate, but it failed to increase the fail-over rate. Because q_{high} and q_{mdr} were smaller in Exp 1, customers were more likely to be included in the customer list for replacement than in Exp 2. A smaller fail-out rate was observed by decreasing the parameters q_{high} and q_{mdr}, which shows the effectiveness of considering the risk function described in Section 3.2.
2. ($\boldsymbol{\alpha}_{\text{high}}$) In Exp 3, the remaining gas rate was set to 0%, whereas it was set to as 5% in Exp 1.
   - Fewer run-out and fail-over events were observed in Exp 3 compared with Exp 1. Because α_{high} was smaller in Exp 3, it was possible to focus more on delivering to a customer who was about to experience a run-out event than with Exp 1. In other words, Exp 1 was more likely to achieve visit to customers whose remaining gas was not below 0%, even if they were high-risk customers. The results indicate that we can focus on delivering to customers with a specific gas rate by changing α_{high}.
3. ($\boldsymbol{D_{lbd}}$) Exp 4 did not force the addition of more moderate-risk customers without considering D_{lbd}, aiming to reveal the effectiveness of such dates.
   - D_{lbd} makes the number of moderate-risk customers increased. Notably, some customers were in the moderate-risk group in Exp 1. However, they were not moderate-risk customers in Exp 4. These customers enabled us to obtain a smaller delivery area than in Exp 4. Such customers even had a low risk of experiencing a gas shortage. Therefore, more run-out events in Exp 1 were reported, with a larger number of visits and shorter distances/trips. However, for few high- and moderate-risk customers, we can obtain more candidates for visit to replace cylinders by setting D_{lbd} larger than zero. This is a promising technique for

realizing workload leveling.

- The two systems were competitive with respect to all evaluation metrics. This indicates that the influence of changing the trip division is slight when compared with the other parameters.

4. ($\boldsymbol{D_{ph}}$) The planning horizon was set to three and zero, in Exp 1 and Exp 5, respectively.
   - In Exp 5, distance/customer was shorter than in Exp 1 because minimizing the delivery area for multiple days was not implemented when formulating the customer list for replacement. As a result, although problems to acquire customer lists in both Exp 1 and Exp 5 provide optimal solutions, the customer list of the first day in Exp 5 is equal to or smaller than that in Exp 1. Therefore, Exp 1 recorded worse results than Exp 5 regarding time/customer and distance/customer.

In conclusion, Exp 3 achieved the best results in fail-over and run-out events and had the best hyperparameter settings between Exp 1 and Exp 5.

D. Ablation Study on the Formulation to Acquire Customer List

The formulations of the proposed methods and when the depot need not be included in the rectangle, such as the formulation in [13], were compared. The hyperparameter settings followed by Exp 3 are described in Appendix C. Figure 9 shows the pairing of the size of the rectangle and the average delivery time, and each dot represents a trip. With both methods, the size of the rectangle represented the cost of geographical delivery costs, and in fact, there was a correlation with respect to the average duration per customer.

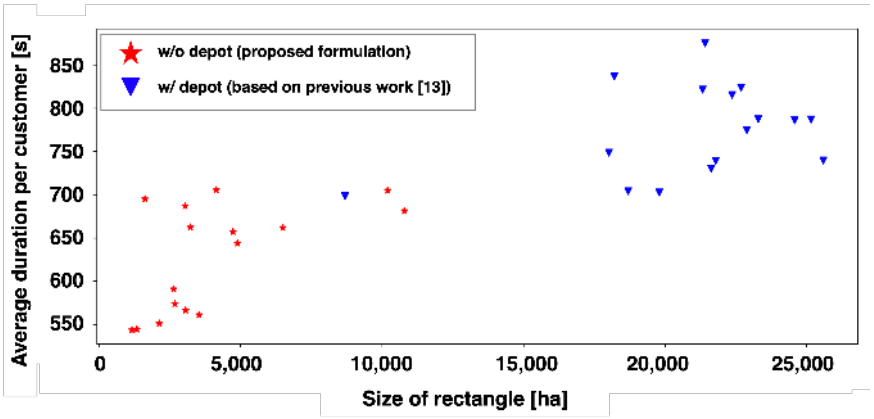


Figure 9: Scatter plot of the size of rectangle obtained by solving Problem 4.1 and average duration of delivery per customer. Blue triangle markers represent the result from an addition constraint in which the depot must be covered by a rectangle.

From Figure 9, we can see that the average duration per customer for the proposed system (without depot) was smaller than that of the method in [13] (with depot). This indicates that the proposed method acquires a more efficient delivery route. This is because the depot is far from the center of gravity of the customers. When a customer far from the depot must be delivered to, the rectangle covering the depot and the customer has already covered most of the other customers.

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Design and Implementation of an Automated Invoice and Billing System

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ABSTRACT

In the modern business environment, efficient and accurate billing plays a vital role in the success of organizations. Billing and invoicing are fundamental processes that directly impact financial management and customer satisfaction. However, many small and medium-scale enterprises still rely on manual or semi-automated billing systems. These traditional methods are time-consuming, prone to human error, difficult to maintain, and lack proper data security. Manual billing often results in incorrect calculations, loss of records, duplication of data, and delayed invoice generation.

This research paper presents the design and implementation of an automated invoice and billing system that addresses the limitations of conventional billing methods. The proposed system automates key billing operations such as customer management, product management, invoice generation, tax calculation, and report generation using a centralized database. The system provides a user-friendly interface that enables users to perform billing operations efficiently while ensuring data accuracy and security.

By reducing manual effort and automating repetitive tasks, the system improves operational efficiency and minimizes errors. The automated invoice and billing system is cost-effective and suitable for small and medium-scale businesses. Experimental evaluation shows improved accuracy, faster processing time, and better organization of financial data compared to traditional billing systems.

***Keywords**: Invoice System, Billing Software, Automation, Web Application, Database Management, GST*

1. INTRODUCTION

Billing and invoicing are essential components of any business organization. Every organization, regardless of its size or domain, requires a reliable system to record transactions, generate invoices, and maintain financial records. Traditionally, billing processes were carried out manually using paper-based methods. In such systems, invoices were written by hand and transaction records were maintained in physical registers. While these methods were suitable for small-scale operations in earlier times, they are no longer efficient in today's fast-paced and competitive business environment.

Manual billing systems suffer from several limitations. Human involvement in calculations increases the chances of errors, such as incorrect totals or tax calculations. Paper-based records are difficult to manage, prone to damage or loss, and require significant physical storage space. Retrieving old records is time-consuming and inefficient. Moreover, manual systems lack proper security mechanisms, making sensitive financial data vulnerable to unauthorized access.

With the advancement of information technology, automated billing systems have emerged as an effective alternative to traditional methods. Automated systems use software applications to perform billing operations such as invoice generation, tax calculation, and report generation. These systems reduce human intervention, improve accuracy, and enhance data security. The proposed automated invoice and billing system aims to modernize billing operations by providing a digital, efficient, and user-friendly solution for small and medium-scale enterprises.

2. PROBLEM STATEMENT

Despite technological progress, many businesses still rely on manual or partially automated billing systems. These systems are inefficient and heavily dependent on human effort, leading to frequent calculation errors and inconsistent data entry. Manual billing makes it difficult to track customer transaction history and generate accurate financial reports. As the volume of transactions increases, managing records manually becomes complex and time-consuming.

Another major issue with traditional billing systems is the lack of data security and backup. Financial data stored in physical form is vulnerable to loss, damage, or unauthorized access. Additionally, manual billing systems do not support real-time data analysis or reporting, which limits effective decision-making.

Therefore, there is a need for a fully automated invoice and billing system that can manage billing operations efficiently, minimize errors, ensure secure data storage, and provide quick access to accurate financial information. The system should be user-friendly, cost-effective, and capable of improving overall business productivity.

3. LITERATURE REVIEW

Several studies have been conducted to improve billing and invoicing systems through automation. Murade et al. proposed a web-based billing system that automates invoice generation and bill submission processes within organizations. Their system reduced manual workload and improved tracking of billing records. However, it lacked scalability and advanced reporting features required by growing businesses.

Pandey et al. focused on electronic billing systems to enhance operational efficiency and sustainability. Their research highlighted the reduction of paperwork and improvement in billing accuracy through automation. While the system improved efficiency, it required enhanced security mechanisms and user adaptability.

Gaikwad et al. developed an online billing and invoice management system for

service-based organizations. The system allowed digital invoice generation and payment tracking, which improved customer satisfaction. However, it did not fully address inventory and product management for retail businesses.

Other research on smart billing systems emphasized GST calculation, inventory tracking, and report generation for small and medium enterprises. Although these systems offered comprehensive features, many involved complex implementation and higher costs. From the literature review, it is evident that existing systems provide partial solutions and often lack simplicity, integration, or cost-effectiveness. This creates a need for a complete, user-friendly, and automated invoice and billing system.

4. PROPOSED SYSTEM

The proposed system is an automated invoice and billing system designed to overcome the drawbacks of traditional manual billing methods. The system provides a digital platform that automates the entire billing process, including customer management, product management, invoice generation, tax calculation, and report generation.

The system allows authorized users to log in securely and perform billing operations through a user-friendly interface. Users can add, update, and manage customer details and product information. During invoice generation, the system automatically calculates the total amount based on selected products, quantities, and applicable taxes such as GST. This eliminates the need for manual calculations and reduces errors.

All billing and transaction data is stored in a centralized database, ensuring data consistency and security. The system also supports report generation, enabling businesses to analyze sales data, customer transactions, and financial performance over a selected period. By automating repetitive billing tasks, the proposed system saves time, reduces workload, and improves overall efficiency.

5. Objectives of the System

The main objectives of the automated invoice and billing system are as follows:

- To automate the billing and invoicing process
- To reduce human errors in calculations and data entry
- To maintain secure and organized records of customers, products, and transactions
- To provide a user-friendly interface for easy operation
- To enable quick generation of invoices and financial reports
- To improve efficiency and productivity of business operations

6. SYSTEM ARCHITECHTURE

The automated invoice and billing system follows client–server architecture. In this architecture, the client side represents the user interface accessed through a web browser. Users interact with the system by entering customer details, selecting products, and generating invoices.

The server side contains the application logic responsible for processing user requests, performing invoice calculations, validating data, and managing authentication. The server communicates with a centralized database to store and retrieve billing-related information.

The database layer stores customer data, product details, invoice records, and transaction history. Centralized storage ensures data integrity, security, and easy access to records. This architecture provides scalability, reliability, and efficient system performance.

7. METHODOLOGY

The development of the automated invoice and billing system follows a structured software development methodology to ensure systematic implementation and reliable performance.

7.1 Requirement Analysis

In this phase, system requirements are identified by analyzing the needs of small and medium-scale businesses. Functional requirements include invoice generation, customer management, tax calculation, and report generation. Non-functional requirements such as security, usability, and performance are also defined.

7.2 System Design

The system design phase involves designing the database schema, system workflow, and user interface. The database is structured to efficiently store billing data, while the user interface is designed to be simple and easy to use.

7.3 Implementation

In the implementation phase, the system is developed using suitable programming languages and tools. The front-end enables user interaction, while the back-end handles business logic and database operations.

7.4 Testing

Testing is performed to ensure correct system functionality. Functional testing verifies invoice generation and calculations, while security testing ensures authorized access. Performance testing checks system response time and reliability.

7.5 Deployment

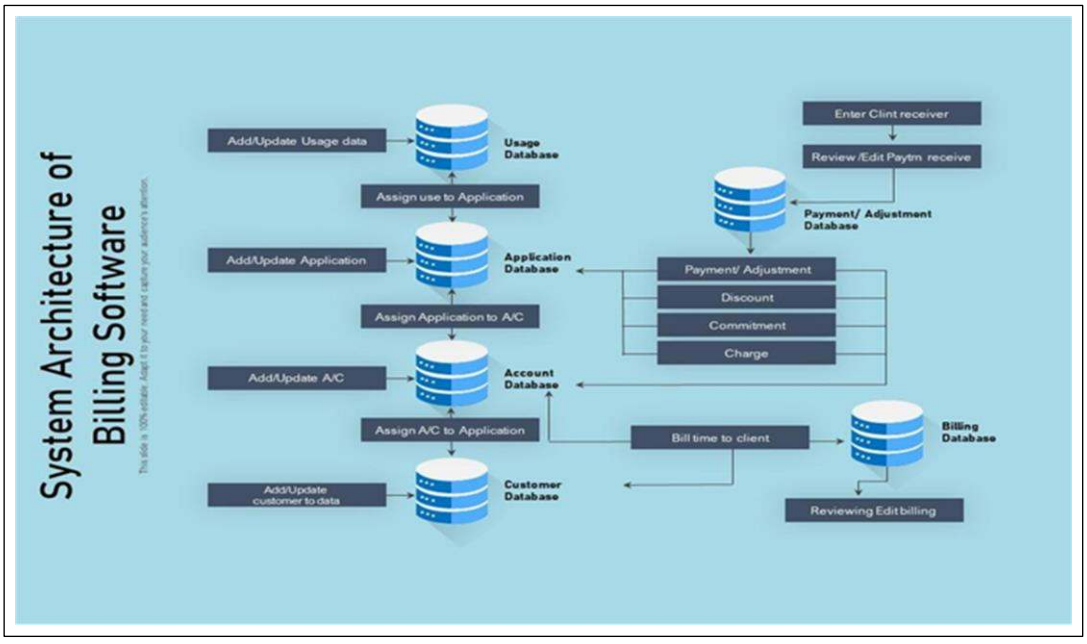


Fig. 1

After successful testing, the system is deployed for real-time use. User documentation is provided to ensure smooth operation.

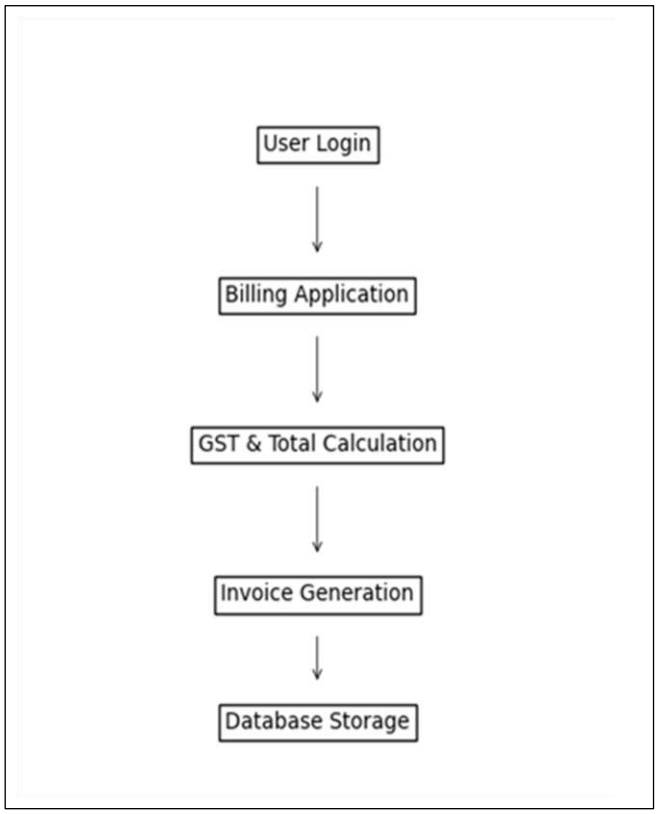


Fig. 2. *Working flow of the system*

8. RESULTS

The automated invoice and billing system was tested using sample business data. The system successfully generated invoices with accurate tax calculations and stored transaction records securely. Compared to manual billing systems, the automated system significantly reduced processing time and eliminated calculation errors.

The results demonstrate improved data organization and faster report generation. Users were able to retrieve billing records easily and analyze financial data efficiently, confirming the effectiveness of the proposed system.

9. Advantages of the System

The automated invoice and billing system offers several advantages:

- Reduces manual effort and processing time
- Minimizes calculation and data entry errors
- Provides secure and centralized data storage
- Enables quick invoice and report generation
- Improves accuracy and consistency of financial records
- Enhances productivity and operational efficiency
- Suitable for small and medium-scale enterprises

10. CONCLUSION

The design and implementation of the automated invoice and billing system successfully address the challenges of traditional billing methods. By automating

billing operations, the system improves accuracy, efficiency, and data security. The centralized database ensures organized record management and easy access to financial information.

The system proves to be a reliable and cost-effective solution for modern business environments, particularly for small and medium-scale enterprises. It simplifies billing operations and supports better financial management.

FUTURE SCOPE

The automated invoice and billing system can be further enhanced by integrating online payment gateways, developing mobile applications, and deploying the system on cloud platforms. Advanced reporting and analytics features can also be added to support better business decision-making.

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Inventory and Bill Management System

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Abstract: *The modern business landscape demands efficient management of inventory and billing processes to ensure smooth operations and enhanced customer satisfaction. In response to this need, this research paper presents a comprehensive case study on the development and implementation of an Inventory and Bill Management System. This system was developed as a desktop-based application to automate various tasks, including invoice generation, inventory management, reporting services, and quotation generation. The case study focuses on the collaboration with Shri Swami Samarth Enterprises, Islampur, who sponsored the project and outlines the methodology used in the development process. Through this case study, valuable insights are provided into the significance and impact of such systems on business operations*

Keywords: Inventory Management, Bill Management, Automation, Software Development, Case Study

I. INTRODUCTION

The ever-evolving landscape of business operations necessitates constant innovation and optimization, particularly in the realms of inventory management and billing systems. In contemporary business practices, the manual handling of inventory tracking, invoicing, and billing processes not only proves arduous but also introduces significant room for error. Moreover, as enterprises strive to scale their operations and cater to a diverse customer base, the limitations of traditional manual methods become increasingly apparent. Recognizing these challenges, there arises a pressing need for the development and implementation of automated solutions that streamline these critical business functions.

The impetus behind the development of an Inventory and Bill Management System stems from a fundamental desire to enhance operational efficiency, accuracy, and scalability. This system represents a paradigm shift from conventional manual approaches, offering a robust framework for automating various tasks associated with inventory management and billing. By leveraging technological advancements and software engineering principles, such a system can revolutionize the way businesses handle their inventory tracking, invoicing, and billing processes.

The significance of this research paper lies in its endeavor to document the development process and outcomes of the Inventory and Bill Management System, presented through a comprehensive case study. Central to this case study is the collaboration with Shri Swami Samarth Enterprises, Islampur, a notable enterprise grappling with the inefficiencies of manual inventory tracking and billing management. Through this collaborative effort, insights into the practical implementation and impact of the system are gleaned, shedding light on its efficacy in real-world business scenarios.

The remainder of this research paper delves into the methodologies employed in the development of the Inventory and Bill Management System, the results and discussions about its implementation, and concludes with reflections on the implications and future directions of such automated systems in the realm of business operations. By elucidating the challenges, strategies, and outcomes associated with this endeavor, this paper aims to contribute valuable insights to the discourse surrounding inventory management and billing systems in contemporary business contexts.

II. OBJECTIVES

The objective of this project is:

- Automate Inventory Management: Develop a system to automate tracking, categorization, and management of inventory.
- Efficient Invoicing: Implement features for accurate and efficient invoice generation, customization, and payment tracking.

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- Streamline Sales Processes: Create modules for smooth handling of sales transactions with registered and unregistered customers.
- Enhance Reporting Capabilities: Develop reporting functionalities to provide insights into sales trends, inventory levels, and financial summaries.
- Improve Customer Management: Establish a database for storing and managing customer information to provide personalized service.
- Quotation Generation: Enable the system to generate quotations for potential customers, facilitating communication and negotiation.
- Track and Manage Imports: Develop a module for importing new stock items into the inventory system, including updating existing records.
- User Authentication and Authorization: Implement secure login mechanisms to control access and ensure data integrity.
- User-Friendly Interface: Design an intuitive interface for easy navigation and efficient task execution.

Scalability and Flexibility: Build the system with scalability and flexibility for future expansion and customization.

III. INDUSTRY SURVEY

Before we settled on our project, we conducted a thorough examination of various industries to pinpoint where our software solutions could have the most impact. What we discovered was quite surprising—many businesses were still reliant on manual processes for tasks like generating invoices, which not only consumed significant time but also introduced the potential for errors.

Among the enterprises we surveyed, Swami Samarth Enterprises, a notable player in the plumbing materials sector, caught our attention. Despite their strong presence in the market, they were grappling with inefficiencies in their billing and invoicing procedures, particularly concerning their regular clientele and retailers. It became evident that they were in dire need of a streamlined solution to alleviate these challenges.

Recognizing the opportunity to address Swami Samarth Enterprises' pain points, we engaged in discussions with them. They were not just interested in our project; they were eager to collaborate and even offered sponsorship to ensure its success. This mutual interest and commitment solidified our decision to partner with them.

In selecting Swami Samarth Enterprises as our project sponsor, we were driven not only by their genuine need for our solution but also by the alignment of their requirements with our capabilities. We were confident that our expertise could be tailored to meet their specific needs effectively. Hence, our collaboration with Swami Samarth Enterprises represented not just a partnership but a promising opportunity to make a tangible difference in their operations.

IV. SCOPE OF THE PROJECT

The scope of the project is as follows:

- User Authentication and Authorization: The system will provide secure login functionality for authorized users, controlling access to different functionalities based on user roles.
- Dashboard and Reporting: Users will have access to a dashboard providing an overview of key metrics such as sales, revenue, and inventory levels. Reporting features will enable users to generate custom reports on various aspects of the business.
- Billing Management: The system will facilitate the creation, customization, and management of invoices for sales transactions. It will also track payments received from customers.
- Inventory Management: The system will track stock levels, product details, and categories. Users will be able to add new products, update stock levels, and categorize products.
- Customer Management: The system will maintain a database of customer information, including contact details, order history, and payment status. Users will be able to manage customer accounts and track interactions.
- Quotation Generation: Users will have the ability to generate quotations for potential customers, including details of products, prices, and terms of sale.

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- Import Module: The system will support the import of new stock items, including functionalities for adding new items, updating inventory records, and managing supplier information.
- Scalability and Customization: The system will be scalable and customizable to accommodate the evolving needs of the business. It will support future enhancements, modifications, and expansions.
- User Training and Support: The project will include provisions for user training and support to ensure effective utilization of the system. Training materials, documentation, and support resources will be provided to assist users.

V. NEED OF THE WORK

- Efficiency Enhancement: The manual handling of inventory tracking, invoicing, and billing processes is time-consuming and prone to errors. Automating these processes through a dedicated system will significantly enhance efficiency by reducing manual effort and minimizing errors.
- Scalability: As businesses grow and expand, the volume of transactions and inventory management tasks also increases. A scalable system is needed to accommodate this growth and ensure that the inventory and billing processes remain efficient and effective.
- Data Accuracy: Manual processes are susceptible to human errors, leading to inaccuracies in inventory records and billing information. A dedicated system will ensure data accuracy by automating data entry and validation processes, reducing the risk of errors.
- Streamlined Operations: The Inventory and Bill Management System will streamline operations by providing a centralized platform for managing inventory, invoicing, and billing processes. This will eliminate the need for multiple disparate systems and manual reconciliations, resulting in smoother operations.
- Improved Decision-Making: Access to real-time data and comprehensive reports will enable businesses to make informed decisions regarding inventory management, sales strategies, and financial planning. The system will provide valuable insights into sales trends, inventory levels, and customer behavior, empowering businesses to optimize their operations.

VI. PROBLEM STATEMENT

The manual handling of inventory tracking, invoicing, and billing processes at Shri Swami Samarth Enterprises, Islampur, is inefficient, error-prone, and time-consuming. The current reliance on paper-based methods and manual data entry leads to challenges in maintaining accurate inventory records, generating timely invoices, and managing billing processes effectively. As a result, the enterprise faces operational inefficiencies, increased risk of errors, and difficulties in scaling its operations to meet growing demands.

Furthermore, the lack of an automated system for inventory and bill management hampers the enterprise's ability to make informed business decisions, track sales trends, and optimize inventory levels. The absence of real-time visibility into inventory status, sales transactions, and financial data also hinders the enterprise's ability to respond quickly to customer needs and market dynamics.

In light of these challenges, there is a pressing need to develop and implement an Inventory and Bill Management System that automates key inventory and billing processes, enhances operational efficiency, and provides valuable insights for informed decision-making. Such a system will streamline inventory tracking, facilitate accurate and timely invoice generation, and improve overall billing management. By addressing these pain points, the Inventory and Bill Management System will empower Shri Swami Samarth Enterprises to optimize its operations, enhance customer satisfaction, and maintain competitiveness in the market.

Therefore, the objective of this project is to design, develop, and implement an automated Inventory and Bill Management System tailored to the specific needs of Shri Swami Samarth Enterprises, enabling them to overcome the challenges associated with manual inventory and billing processes and achieve greater efficiency, accuracy, and scalability in their operations.

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VII. PROPOSED SYSTEM

The implementation of the Inventory and Bill Management System has brought about notable enhancements in operational efficiency and data management for Shri Swami Samarth Enterprises. This section unveils the outcomes of the system implementation, showcasing screenshots and demonstrations of pivotal modules such as login, dashboard, sales, inventory management, reporting, and customer data review. The discussions center around the implications of these results, emphasizing the system's efficacy in automating billing procedures, streamlining inventory management, and furnishing valuable insights for informed decision-making.

Let's delve into the specifics of each module and its impact on the organization's operations:

Login Module:

The login module provides secure authentication for authorized users, ensuring that only designated personnel can access the system.By implementing robust login mechanisms, the organization safeguards sensitive data and maintains confidentiality.

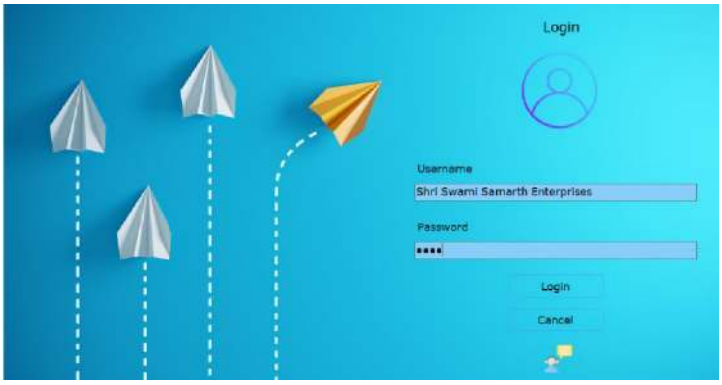


Fig 1. Login Module

Dashboard Module:

The dashboard offers a comprehensive overview of key metrics such as sales performance, revenue trends, inventory levels, and pending invoices.

Through intuitive visualizations and real-time updates, stakeholders gain valuable insights into the organization's operational status at a glance.

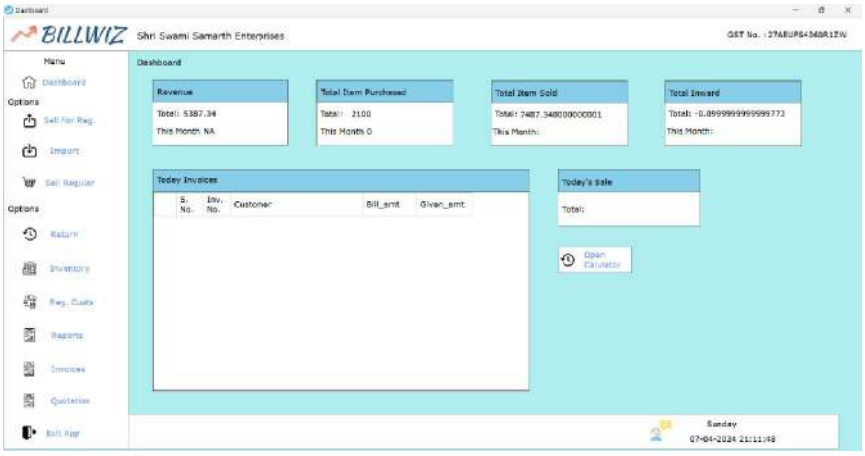


Fig 2. Dashboard Module

Sales Module:

The sales module facilitates the seamless processing of transactions, enabling the generation of invoices for both registered and unregistered customers.

By automating the sales process, the system accelerates revenue generation and reduces manual errors in billing.

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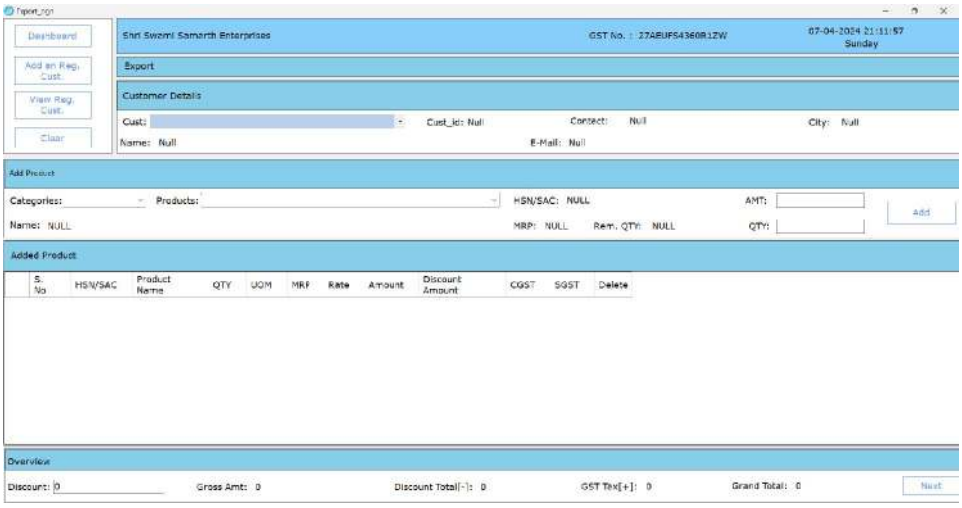


Fig 3. Sale to Registered Customer

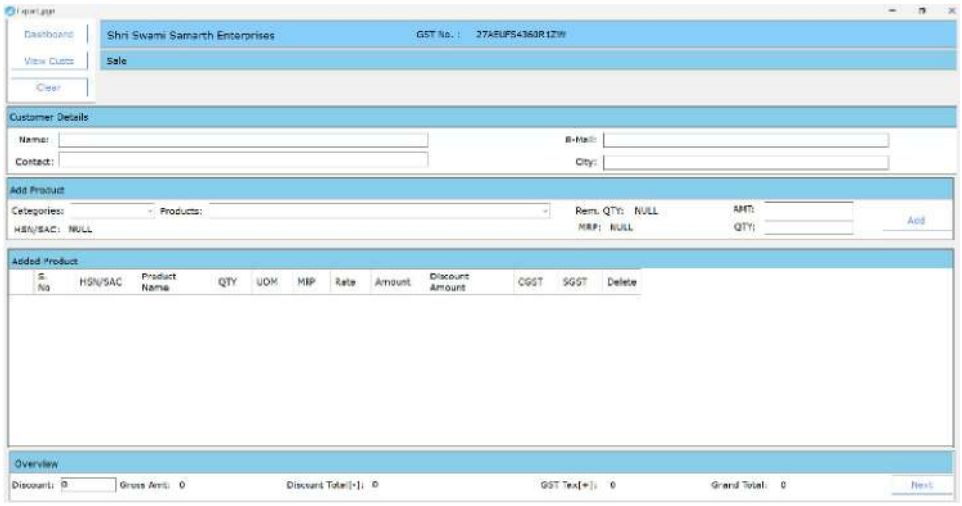


Fig 4. Sale to unregistered Customers.

Inventory Management Module:

The inventory management module empowers users to track stock levels, manage product listings, and monitor inventory movements.

With features for categorization, price manipulation, and stock tracking, the system optimizes inventory control and minimizes stockouts.

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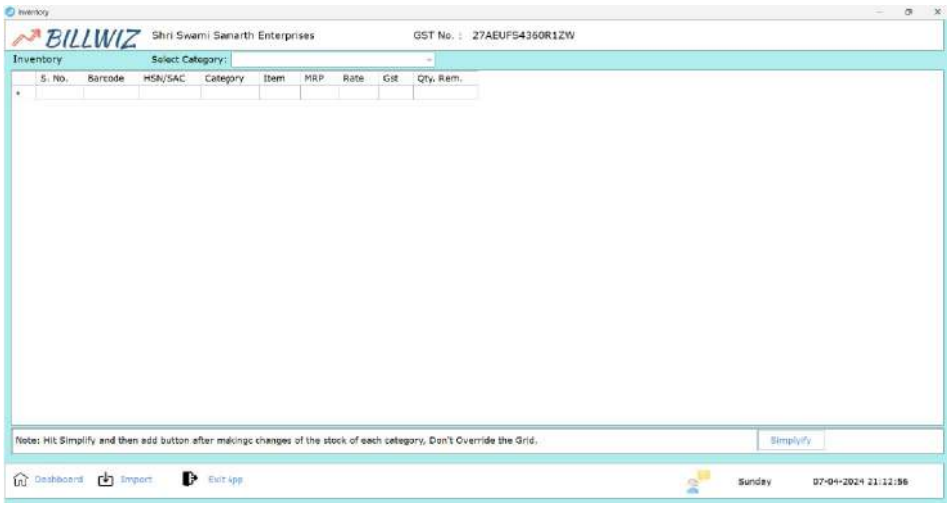


Fig 5. Inventory Module

Reporting Module:

The reporting module offers customizable reporting tools for generating various reports such as sales reports, inventory summaries, and customer analytics.

By providing actionable insights derived from data analysis, the system facilitates data-driven decision-making and strategic planning.

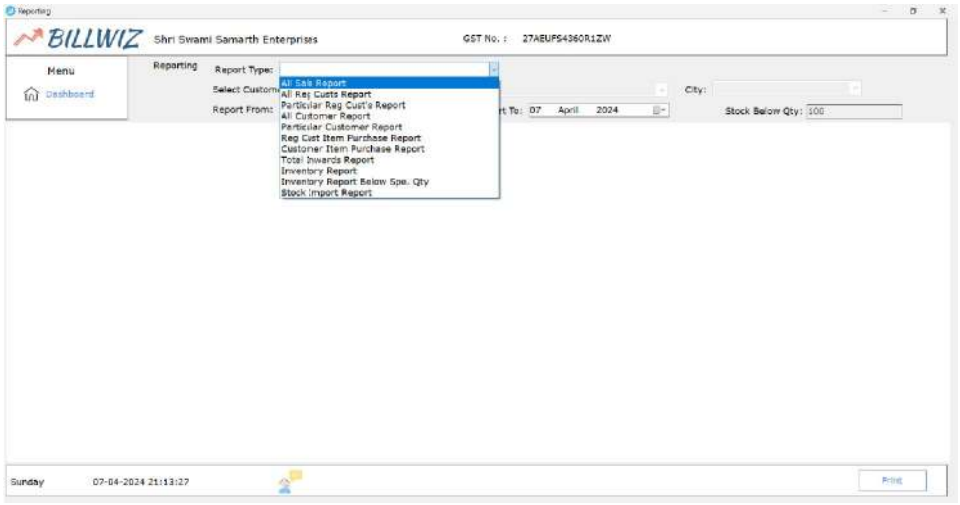


Fig 6. Reporting Module

Customer Data Review Module:

The customer data review module centralizes customer information, including purchase history, preferences, and contact details.

By maintaining comprehensive customer records, the system enables personalized customer service and targeted marketing efforts, fostering customer satisfaction and loyalty.

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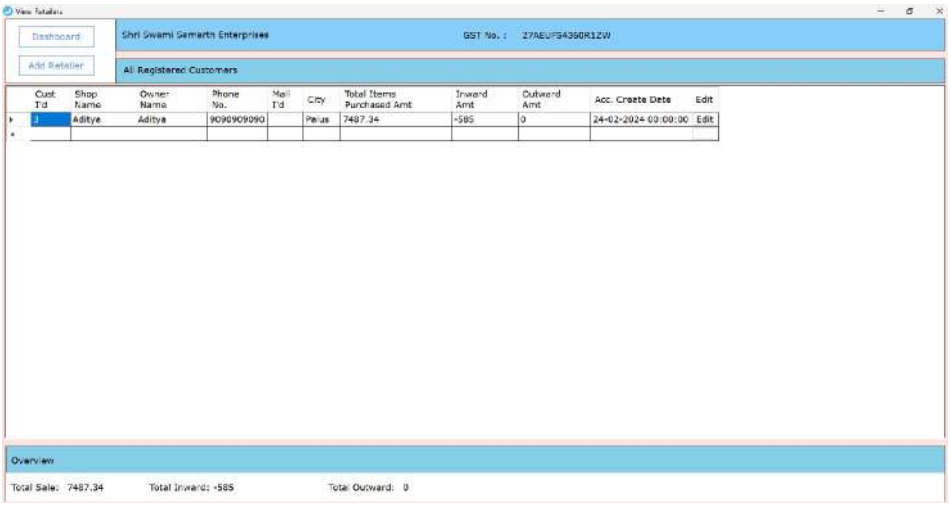


Fig 7. Registered Customer Data

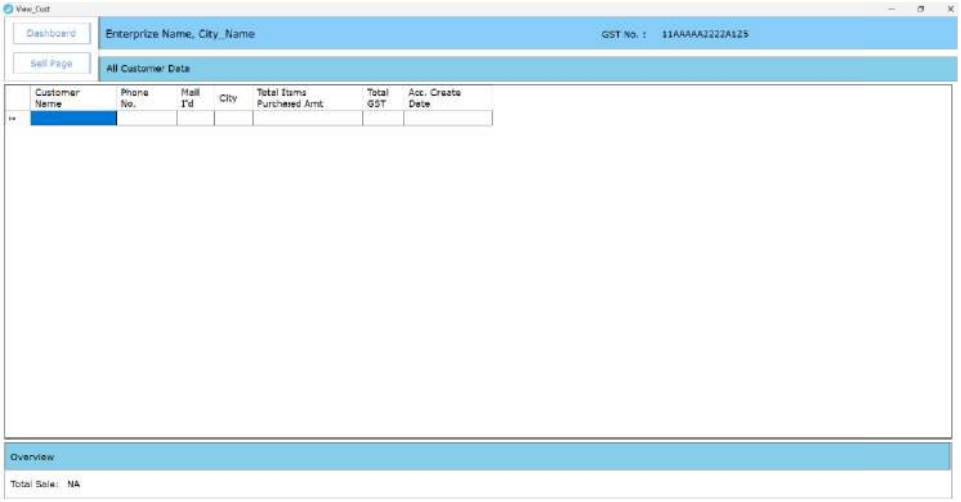


Fig 8. Unregistered Customer Data

Return Module:

The return module facilitates the seamless processing of product returns, allowing customers to initiate return requests and enabling staff to manage returns efficiently.

By automating the return process and updating inventory accordingly, the system ensures accurate stock tracking and inventory reconciliation.

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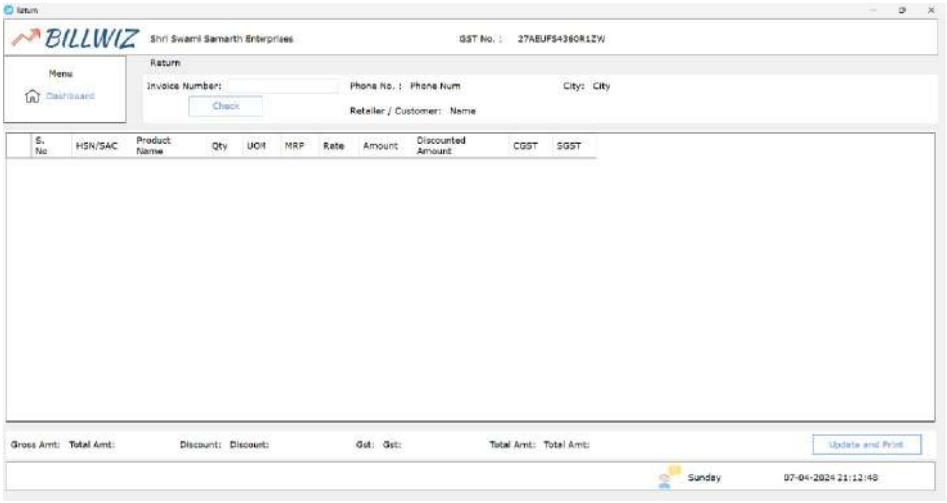


Fig 9. Return Module

Quotation Module:

The quotation module allows users to create and manage quotations for potential customers, providing detailed pricing information and product specifications.

By streamlining the quotation process and enabling quick response to customer inquiries, the system enhances sales efficiency and customer satisfaction.



Fig 10. Quotation Module

VIII. DISCUSSION

The successful implementation of the Inventory and Bill Management System signifies a significant milestone in enhancing operational efficiency and data-driven decision-making at Shri Swami Samarth Enterprises. By automating billing processes, optimizing inventory management, and providing valuable insights through reporting and customer data analysis, the system empowers the organization to streamline its operations and drive business growth. Furthermore, the user-friendly interface and intuitive features of the system contribute to improved productivity and customer satisfaction. Overall, the results demonstrate the transformative impact of the Inventory and Bill Management System on organizational processes and performance.

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VIII. SYSTEM REQUIREMENTS

Software Requirement

- Packages/libraries/frameworks: .NET Core 6.
- Database: MS SQL SERVER.
- Operating system: Windows 11.

Hardware Requirement

- Minimum Disk Space:256 GB or above
- Minimum memory:4GB or above

IX. FUTURE SCOPE

Connecting with other tools: Making software work with other popular business tools like accounting or e-commerce platforms.
Making it work on phones: Creating a version of software that people can use on their phones or tablets.
Using smart technology: Exploring using smart technology like AI to predict trends or automate tasks.
Putting it on the Internet: Moving software to the Internet so people can access it from anywhere and update it easily.
Improving the reports: Keep making the reports better with more useful information and nicer designs.
Keeping it safe: Making sure software is safe from hackers and other security threats.
Following the rules: Making sure software follows all the laws and regulations for businesses.

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INTELLIGENT GAS BOOKING AND LEAKAGE SYSTEM USING WIRELESS SENSOR NETWORKS

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ABSTRACT

Nowadays the technology is rapidly growing, and human life is becoming easier than the past. In our country maximum number of people is using LPG (Liquid Petroleum Gas) in several places like houses, shops and in many industries. To assemble of LPG is too diminutive. But it is very difficult to the illiterate people to understand and many of the people in our society is busy with their works, so that they can't know the status of the gas and if the booking of the LPG gets delay then it will be uncomfortable to the people. In order to overcome this problem, Intelligent Gas booking and Leakage Detection System (IGLS) is proposed. When the LPG gas gets minimum weight then GSM drive the memo to the owner and warns about the LPG gas is getting over, so that the owner tries to book the LPG in advance. Huge numbers of people are facing the gas leakage problem. In order to avoid this drawback, a gas sensor is utilized to sense the gas outflow and switch ON the buzzer and sends a message of outflow of the gas to warn the owner. Proposed system will be benefitted for society.

KEYWORDS

Gas sensor, Gas detector, Wireless Sensor networks, LCD display, Load cell, Buzzer, Arduino board.

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1. INTRODUCTION

LPG is known as Liquid Petroleum Gas. In Arpitha, Kiran, Gupta, and Duraiswamy (2016) consists of saturated propane and saturated butane mixture as well as unsaturated hydrocarbons. In LPG, Ethanethiol is added (Macker, Shukla, Dey, & Agarwal, 2018). So that it becomes odourless gas, which will easily detect leakage of gas. In our country (Varma, Prabhakar, & Jayavel, 2017) there are approximately 40 crore people are using LPG which is nearly 45% of our population. In Shinde, Patil, and Patil (2012), and Mahalingam, Naayagi, and Mastorakis (2012), LPG gas, which is commercially used in domestic area for heating, cooking, shops and industrial purpose, etc., LPG has highly flammable because of the mixture of propane and butane. In Priya, Surekha, Preethi, Devika, and Dhivya (2014) a cylinder is to be found that it is loaded, which will measure the weight of the cylinder. In Ramya and Palaniappan (2012) inserted a minimum weight of the cylinder according to the level of a gas.

2. RELATED WORK

In Shyamaladevi, Rajaramya, Rajasekar, and Ashok (2014) proposed a structure that will make the whole LPG compartment booking system robotized was excluded human intervention. This framework ceaselessly decides the greatness of the chamber drive notice to the insisted LPG agent with the target that they can pass on the LPG chamber in time.

In Cheung, Lin, and Lin (2018) consolidates WSN into an astounding structure which enables the structure site to ostensibly screen the prosperity status by methods for a spatial, concealed interface and empty any perilous gas normally. In Dewi and Somantri (2018) were proposed the inspiration driving this structure is to recognize gas spillage, murder it, and thwart the impact. WSN framework works were reliant on gas sensor MQ-6 and remote module Bluetooth HC-05. Impact neutralizing activity system works reliant on the alarm/ringer, exhaust fan, and customized gas controller.

In Hema, Murugan, and Chitra (2013) the light was unmistakable tangibility and common confinements of different gas identifying developments, experts have been managing different circumstances with overhauled frameworks. In this, they were analyzed the overview of progressed enhancement in leakage of gas recognizing. In Kalaivanan and

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Manoharan (2016) were the essential objectives of the endeavor is to gather a Gas spillage discoverer using LPG gas sensor and furthermore it was interface with IoT using ESP module for security. A solenoid is set such a way, that at whatever point there is a sign, it will close the pipe to stop the movement of gas.

In Padmapriya and Kamini (2013) the structure also is used for other application in the business or plant that depends upon LPG and vaporous oil in their action this endeavor is used to screen relentless load of the LPG chamber. In our country customarily that people don't know decisively the status of the chamber and there is a deferment in lighting up gas office. To keep far from such conditions have completed the endeavor called "Customized Gas Booking Using Embedded System with Safety Guards". In this endeavor used weight cell as a weight sensor. MQ6 as an LPG spillage sensor which will recognize the spillage of LPG and offers security to people. The gas office gets the solicitation of new chamber and owner got the messages regarding the status. In paper (Apeh, Erameh, & Iruansi, 2014) were smoke sensor was used to recognize gas spillage in the home. In case any gas spillage recognized subsequently, it will send SMS to the fire station. GSM is a champion among the most cell frameworks used in India. In our endeavor, we have used weight cell to screen the hold up of the LPG gas routinely. In Bangali and Shaligram (2013) proposes the Internet of Things is considered as the third surge of information development straightforwardly after Internet and flexible correspondence sort out, which is depicted by progressively complete interoperability and learning. By this creative progress, it engages us to screen the activities through our propelled cell phones, to share information.

In Potadar, Salvi, Sathe, and Chavan (2015) proposes the system will recognize the spillage and exhort the client about the gas spillage by SMS Audio yield is moreover made on speaker. As a bit of emergency, the system will immediately temperament executioner the control valve with the help of stepper motor to evade a mishap. The additional good position of the system is that it constantly screens the component of the LPG present in the chamber using load cell and demonstrate the weight always. The programmed gas booking framework will really discard call stopping or dialing such countless numbers while booking the gas chamber.

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In Ravichandran (2017) were LPG will finish without instructing us it can make an inconvenient condition for cooking, etc. Our arrangement relies upon ARM controller, it can pursue LPG void continually in case LPG is close culmination or at void measurement, by then, it can alert us by sending SMS to LPG Agency for mentioning the LPG chamber. In Imade, Rajmanes, Gavali, and Nayakwadi (2018) were analyzed, the fundamental stress of any assignment has not been left impeccable by IoT. In Khan, Prince, Dewangan, and Rathore (2014), Gas Leakages in open or shut districts can show to be dangerous and savage. The customary Gas Leakage Detector Systems have various phenomenal precision; disregard to perceive two or three segments in the field of disturbing the all-inclusive community about the spillage. In Jolhe, Potdukhe, and Gawai (2013) in the unlikely event that LPG is going to complete without teaching us, it can make an irksome condition for cooking, etc. Our structure relies upon the microcontroller, it can pursue LPG void continually in case LPG is especially near completion or at void measurement, by then it can alert us by sending SMS to the owner and it can in like manner move the message to LPG Agency for mentioning the LPG chamber.

3. PROPOSED SYSTEM

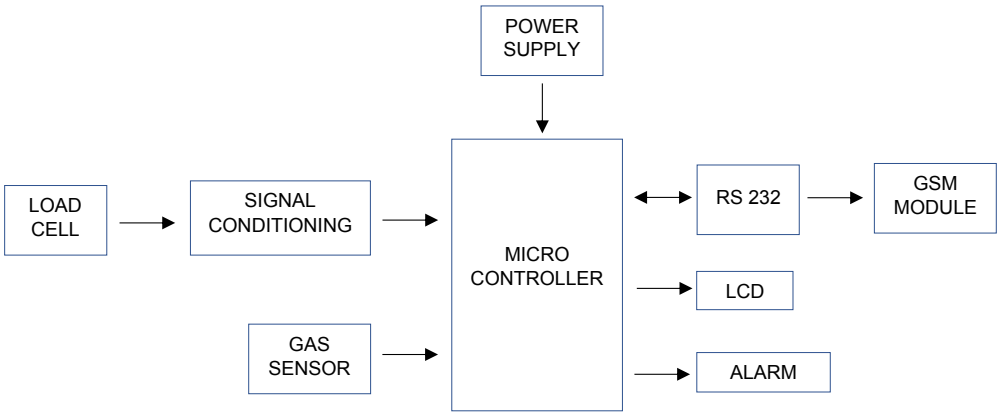
3.1. INTELLIGENT GAS BOOKING AND LEAKAGE DETECTION SYSTEM (IGLS)

Gas leakage means if the gas is transferred from one pipe to another pipe or other container of natural gas or another gas product may not be present. Leaks are very dangerous because a small leak can gradually produce an explosive concentration of gas. This is the main drawback of traditional gas leakage system. In order to overcome the above problem, this work proposes Intelligent Gas booking and Leakage Detection System (IGLS).

When the LPG gets to the base weight, the GSM sends the message to the proprietor by the assistance of Arduino. Arduino is accustomed to sending the message to the client with the assistance of GSM. The LCD shows the heaviness of the chamber consistently. This program is created to send the message through GSM to the given number of the client. The gas sensor MQ6 to detect the gas around the place, after that microcontroller automatically switch ON the signal and sends the message to the proprietor.

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The message sent to the fixed number automatically and access remotely with the assistance of GSM. At the point when the heaviness of the gas chamber gets least; GSM sends the message to the proprietor and cautions the proprietor about the leakage of the gas.



Graphic 1. BLOCK DIAGRAM.

4. METHODOLOGY

The main aim of this system is to estimate the GAS level and leakage of the gas from the room or an industry. The leakage of the gas will be estimated with the help of IGLS system using MQ6 gas sensor. This gas sensor senses the gas and sends the message to the user with the help of GSM module. Liquefied Petroleum Gas (LPG) sensor is simple and easy to detect the gas level. Gas sensor has fast response and high sensitivity module. Liquefied Petroleum Gas (LPG) sensor is simple and easy to detect the gas level. Gas sensor has fast response and high sensitivity.



Graphic 2. PIR Sensor.

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Table 1. Specification of MQ6 sensor.

Model No.			MQ-6
Sensor Type			Semiconductor
Standard Encapsulation			Bakelite (Black Bakelite)
Detection Gas			Isobutane, Butane, LPG
Concentration			300-10000 ppm (Butane, Propane, LPG)
Circuit	Loop Voltage	V_c	≤24V DC
	Heater Voltage	V_H	5.0V±0.2V ACorDC
	Load Resistance	R_L	Adjustable
Character	Heater Resistance	R_H	31Ω±3Ω (Room Tem.)
	Heater consumption	P_H	≤900mW
	Sensing Resistance	R_s	2KΩ-20KΩ (in 2000ppm C3H8)
	Sensitivity	S	Rs (in air) / Rs (1000ppm C4H10) ≥ 5
	Slope	α	≤0.6 (R2000ppm / R1000ppm LPG)
Condition	Tem. Humidity		20ºC±2ºC; 65%±5%RH
	Standard test circuit		V_c:5.0V±0.1V; V_H: 5.0V±0.1V
	Preheat time		Over 48 hours

5. WORKING PRINCIPLE

The working standard of this system has appeared in the square outline in Figure 1. The Arduino board is assuming an indispensable job which is utilized to quantify the parameters of gas and weight. Every one of the parts is associated with the Arduino board. Arduino gets the messages to the purchaser about the gas chamber like gas spillage and gas weight of the chamber. At the point when the gas is spilling from the gas chamber, the gas sensor will detect the gas spillage and after that sends the data to the Arduino then the Arduino will check the code and sends the data to the GSM module and ringer. The GSM module will send the short message too private the client to realize the gas is spilling and the ringer will make the noisy sound that underwear that gas is spilling. This work will screen with the assistance of GSM ceaselessly if the gas weight is low as for the given load in the code. The gas weight is recognized by the heap cell. The heap cell checks the heaviness of the chamber if the heaviness of the chamber is lower than the given code it will send the data to the Arduino board. It will give the direction to the GSM module to send the message to the customer that the gas weight is low and it starts the client to book the gas chamber. By utilizing the LCD show the weight and the gas spillage estimations of the chamber are shown routinely.

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6. RESULT AND DISCUSSION

Intelligent Gas booking and Leakage Detection System is implemented in real time. The gas weight and the gas spillage are estimated by utilizing IGLS, the estimated gas values are showed in the Liquid Crystal Display (LCD). The gas present in the room air before the gas spillage having the level is 0 ppm, which was appeared in the LCD. When the gas was spilled from the gas chamber the gas sensor will detect the gas in the room air. The GSM module will send the message to the individual proprietor of the user where the portable number was in the GSM module and sends the message "GAS WAS LEAKING". On and off messages send to the controller when the gas was not spilling from the gas chamber. Burden cell will detect the heaviness of the gas chamber. On and off message shows that the heaviness of the chamber is not exactly the edge weight, at that point, it will send the message to the user that the "GAS CYLINDER WEIGHT IS LOW BOOK THE GAS CYLINDER" was sent to the versatile number present in the GSM module. This system will be benefit able for society.



Graphic 3. Output for gas booking and leakage system System.

7. CONCLUSION

This project is based on the Intelligent gas booking system using wireless sensor networks and it is working well, and the main parameters of the Intelligent gas booking system is

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based on the safety purpose of the of day today life everyone are using the LPG gas for cooking and other purposes there is no safety in the place. This project will work nicely in any complex stages. During the design as well as the construction on it because to avoid the hiccups at the final stage. The layout was prepared with almost care to avoid wrong things in the circuit is made as simple as possible to understand easily to our knowledge. The components also taken care about these performance and cost effectiveness. It was interesting while preparing the project and some difficult at some stages it will be enthusiastic and easy to do the work on the project. This project is benefitted for society.

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SMART LPG CYLINDER MANAGEMENT SYSTEM USING CLOUD COMPUTING

Project Reference No.: *45S_BE_1235*

College ***:*** *Sri Venkateshwara College Of Engineering, Bengaluru*
Branch ***:*** *Department of Mechatronics Engineering*
Guide(s) ***:*** Mr. Sarvesh Araballi
Student(S) *: Mr.* Ajieeth A
Mr. Bhuvan H S
Mr. Mahesh S
Mr. Vikas B

Keywords:

Web application, Cloud storage, Fleet management, Finance management, Inventory management, Dynamic pricing, Customer authentication, UI/UX, IoT, GPS, NodeMCU, Gas sensor.

Introduction:

LPG gas utilization is very high in India for both commercial and domestic purposes. The global LPG business depended highly on cylinders. LPG companies will often contract out the delivery of these cylinders - through a chain of distributors, dealers, agents and retailers - to reach their customers. Control over the cylinder, after it leaves the filling plant, is a challenge for LPG companies, and this control diminishes as it moves through the distribution chain.

The distribution chain is under threat due to the reasons listed below:

1. In the current system for delivering commercial LPG cylinders, there is no common place where all the data on sales and the customer details are stored and easily accessible.
2. The agent should be trusted to deliver the cylinders to the right customers with requested quantity and promptly hand over the amount to the enterprise.
3. All accounting is recorded on books, which can be easily manipulated by anyone and human errors are possible.
4. Accidentally misplacing the books is also a possibility. This results in significant losses for the company.

Due to the listed activities, it reduces overall ROI of the Gas business and creates a loss. This is high time for a business to review its model and make a substantial change to bounce back to retain its value proposition and off set present losses.

We wanted to develop a system that address the above issues and helps in creating value-added services for distributers/agencies and gain more margin to have sustainable business.

Objectives:

1. To create centralized storage for enterprise operations on cloud.
2. Develop remote inventory management feature.
3. Provide automatic accounting feature for sales and purchases.
4. Create customer authentication model with dynamic pricing.
5. Provide fleet management service for delivery agent tracking using IoT.

Methodology:

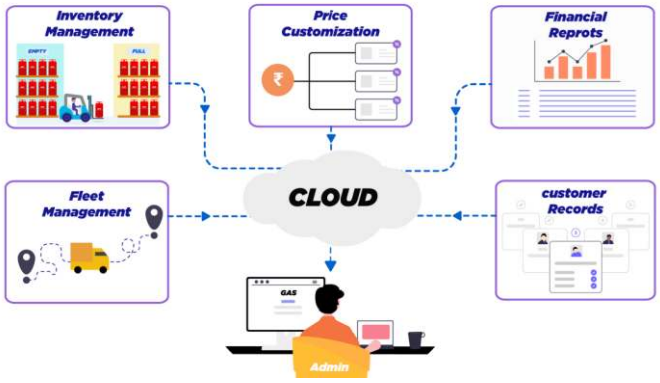


Fig.1: Admin Interface

1. The interface for the system is a web application to access all the features as shown in Fig.1.
2. The application is written in ReactJS.
3. The data is stored in and retrieved from Firestore, a Google cloud service.
4. UI/UX is designed using Figma for a good, user-friendly application.

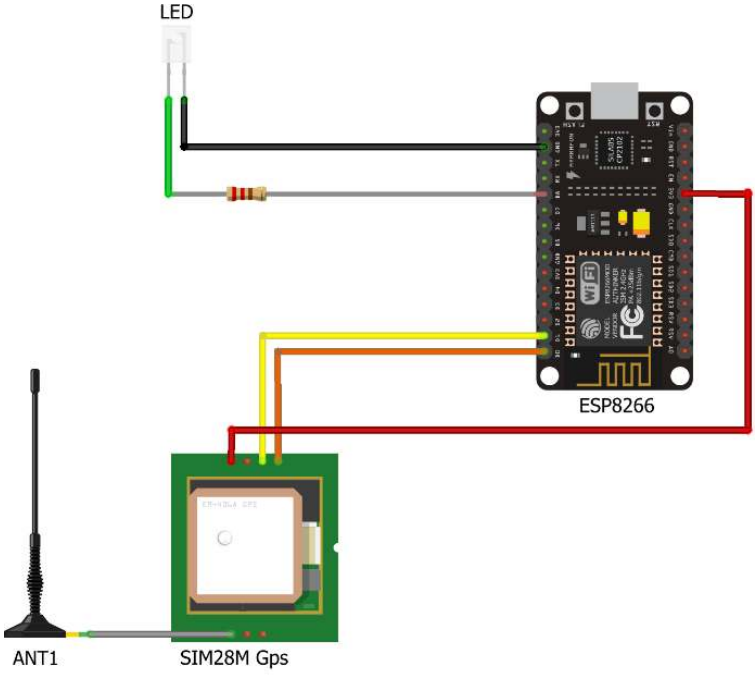


Fig.2: IoT module for geo tracking

5. The fleet management uses IoT module with GPS as shown in Fig.2.
6. The microcontroller used is ESP8266 NodeMCU and programmed using Arduino IDE.
7. The location data is stored in Firebase real-time database, a Google cloud service.
8. The web application is used to retrieve data and displayed using Tomtom maps API.

Conclusion:

Based on the obtained results we can conclude that the proposed solution has several advantages concerning the delivery, data, accounts and finances of the LPG cylinder agency:

1. The system helps to maintain the records of the customer.
2. Agency may verify how many cylinders are in the inventory and manage them accordingly using inventory management service.
3. Accounting is automated on day-to-day basis.
4. Delivery vehicle can be monitored using fleet management services which is integrated in system.
   The delivery chain is more transparent to the enterprise.

Scope for future work:

1. The Ministry of Petroleum and Natural gas to be approached to implement this system.
2. The system to be used for agencies that need delivery management, fleet management, accounting or finance tools.
3. Customer portal to be implemented to order, schedule and track their orders as per their convenience.
4. Better safety protocols can be implemented by using machine learning.
5. Cameras can be implemented to vehicles and warehouses which can be accessed by the admin.

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Survey on Automatic Approach for E-Billing System for Online Application

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Abstract: ***The hassle of queues in shops, human errors in manually generated bills and the wastage of time in the packaging of products after shopping creates inconvenience to shoppers and eventually leads to lower customer satisfaction levels. With the development of digital technology online payments can be made for various purposes. This paper discusses the design of an online shopping system where the generate automatic E-billing system including all taxes. After completion of shopping, customers can exit the shop with their bills deducted automatically from their e-Wallet. A Web Application is hosted on an AWS EC2 instance to allow customers to keep track of their transactions. The system is connected to inbuilt Wi-Fi to communicate with MySQL instance which stores the inventory of the retail store along with the prices of the products. It also stores the current items in the cart and the history of previous transactions of every customer.***
Keywords: Data Security, E-shopping, E-Billing, Maintaince.

I. INTRODUCTION

This paper proposes an E-Billing system on the basis of an E-Invoice Framework. The E-Invoice Framework is an e-invoice solution concentrating on automatically issuing e-invoices, centralized managing e-invoices and providing taxation services. Thus, the efficiency of taxation could be great improved and the labor of taxation could be obviously reduced. Compared with taxpayer-oriented taxation systems, the E- Billing Framework can firmly support data-oriented taxation systems, which are especially tailored for online shopping environment.

A. Problem Statement

The waiting times in billing counters of shops and malls have been increasing, given the growth in human population and personal needs. To build and implement Automated Billing System web application for online shopping.

B. Objectives

1) To provide automatic billing system
2) To implement for online shopping system
3) To provide security using encryption algorithm

II. LITERATURE SURVEY

Electronic Invoicing has been defined as the sending or making available of an invoice and its subsequent processing and storage, wholly by electronic means [4]. In e-commerce, invoice is one of the business documents that are most often exchanged electronically between transaction parties[5]. At present, e-invoice is at the stage of applying and spreading in EU, South America and some nations of North America[6]. For example,

Kaliontzoglou proposed an open invoice system named invoke in 2006, based on XML, XML cryptography and Web Services. The system was established on the basis of trusted third party-provided PKI to seek security and authentication supports[7]. Spanic proposed an invoice system providing multi-services to different users in 2011.

The system can transfer e-invoices among large enterprises through EDI of their ERPs, and provide related web services to SMEs(Small and Medium Enterprises) [8].

Humski proposed an FER e-invoice system in 2012, which owes an extensible and open architecture of e-invoice implementation [9]. The above-mentioned and other e-invoice solutions pay attention to technical details, such as information security, consumer service and system expansibility. This paper, aims at the improvement of taxation systems to focuses more on direct tax-related data (e-invoice data) than taxpayers. In order to achieve this target, the E-Invoice Framework is introduced.

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A. Existing Approach

The users uses MS Excel, and maintains their product list, customer list, and prints the invoice, however it is not possible them to share the data from multiple system in multi user environment, there is lot of duplicate work, and chance of mistake. When the product price are changed they need to update each and every excel file. There is no option to find and print previous saved invoice.

B. Proposed Approach

This project is made for one of the shopping systems, presently they issue their client handwritten invoice and they enter details in manual register. And maintain MS Excel file for product rate. They want computerization of their manual invoice and bill generation process.

There is no security; anybody can access any report and sensitive data, also no reports to find out the sales volume, stock list, and summary report. This E-Billing and invoicing system is used to overcome the entire problem which they are facing currently, and making complete atomization of manual billing including all taxes and invoicing system

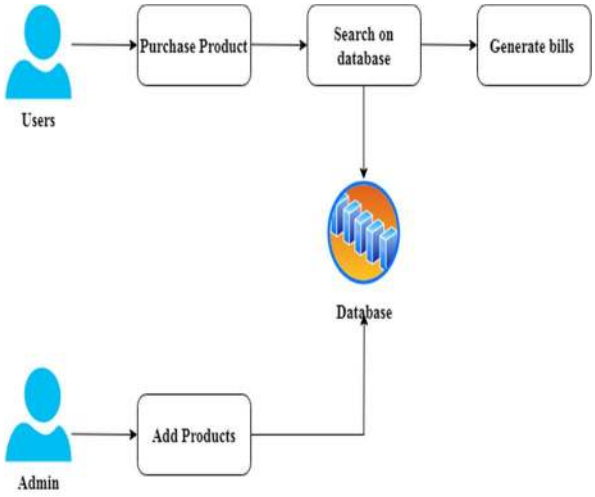


Fig. System Architecture

III. CONCLUSION

This paper proposes an E-billing Framework towards data oriented taxation systems. Compared with current systems, the proposed data-oriented taxation system has the following advantages:

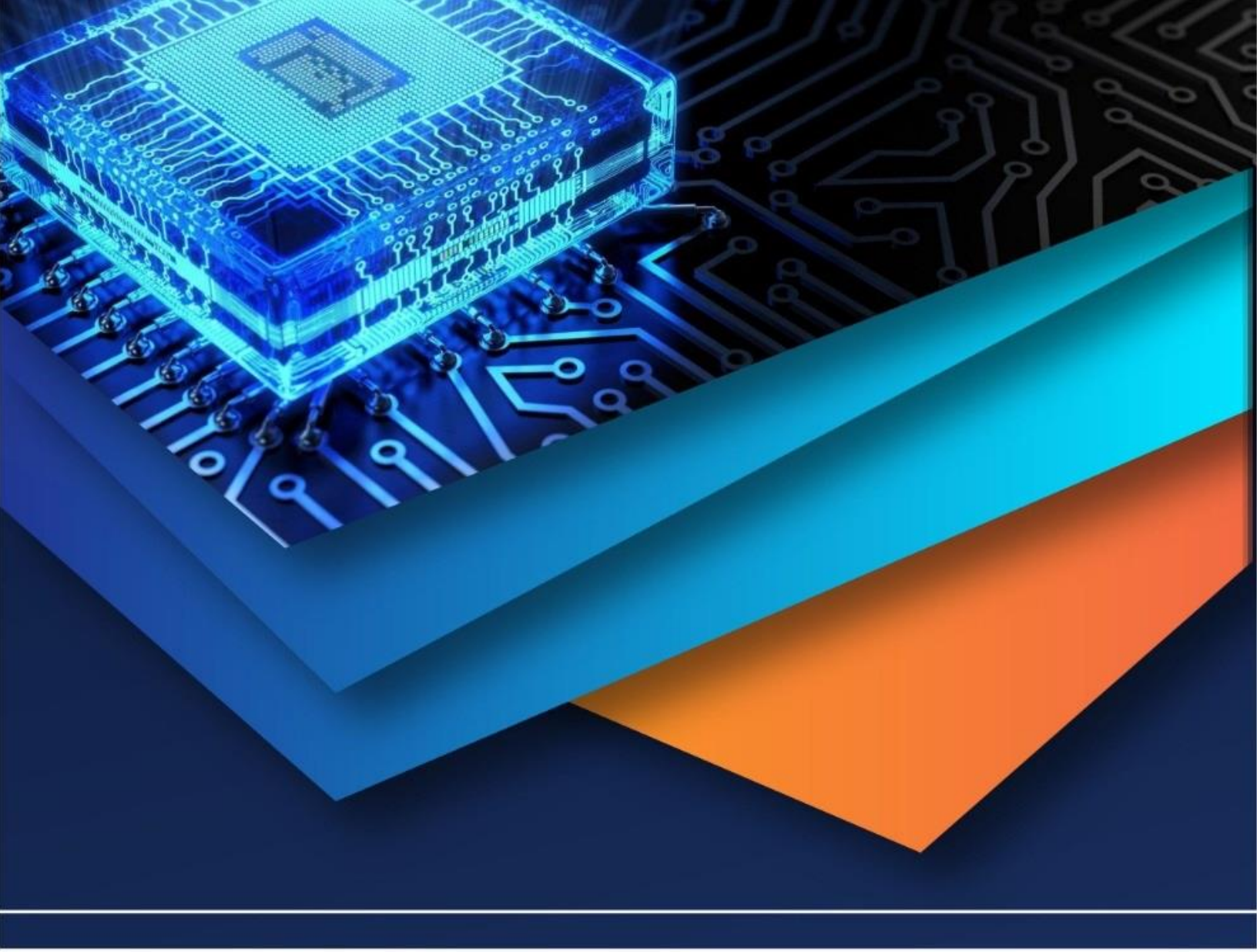
A. Through automated generation and transmission of invoices.

B. Tax authorities could now directly acquire related transaction details from e-invoices instead of communicate with taxpayers. The efficiency of taxation is promoted and the cost is reduced.

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The Impact of *Pradhan Mantri Ujjwala Yojana* on Indian Households

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March 27, 2024

Abstract

This study critically evaluates the impact of the *Pradhan Mantri Ujjwala Yojana* (PMUY) on LPG accessibility among poor households in India. Using Propensity Score Matching and Difference-in-Differences estimators and the National Family Health Survey (NFHS) dataset, the Average Treatment Effect on the intendedly Treated is a modest 2.1 percentage point increase in LPG consumption due to PMUY, with a parallel decrease in firewood consumption. Regional analysis reveals differential impacts, with significant progress in the North, West, and South but less pronounced effects in the East and North East. The study also underscores variance across social groups, with Scheduled Caste households showing the most substantial benefits, while Scheduled Tribes households are hardly affected. Despite the PMUY's initial success in facilitating LPG access, sustaining its usage remains challenging. Policy should emphasise targeted interventions, income support, and address regional and community-specific disparities for the sustained usage of LPG.

Key Words: PMUY, Energy Poverty, Program Evaluation, India, BPL Households

JEL Code: I38, O13, Q48

1 Introduction

Access to clean energy is essential for a high-quality life, yet a significant portion of the global population faces challenges in obtaining clean energy for fundamental needs like lighting, heating, and cooking. While progress has been made in achieving the United Nations Sustainable Development Goal 7 of universal access to clean and affordable energy by 2030, persistent inequalities still exist (UNDP, 2015). Modern cooking fuels, crucial for health and environmental benefits, remain inaccessible for millions, particularly in developing regions. This discrepancy in progress between access to electricity and clean cooking fuels underscores the need for targeted policy interventions to address challenges hindering the adoption of modern cooking fuels. In this paper, we are the first to study one such policy intervention in India, the *Pradhan Mantri Ujjwala Yojana*. We find a causal effect on LPG use that is positive and highly significant but small.

The global pandemic and recent geopolitical events have exacerbated the situation, leading to a surge in energy prices worldwide, mainly for LPG and crude oil (IEA , 2022). The increase in the LPG price has impacted many developing countries severely. The energy price rise is predicted to force up to 100 million people to revert to using traditional fuels for cooking without effective interventions (IEA , 2022). The Government of India, for example, reduced LPG subsidies at the start of the pandemic when global energy prices fell. However, when energy prices surged again, the government could not fully reinstate the subsidies. This added financial pressure on households, as they had to pay more for LPG cylinders (Sharma et al., 2021). Energy poverty has garnered significant attention in development literature over the past two decades (see below). This heightened focus can be attributed to the profound repercussions associated with the absence of accessible, affordable, and modern energy resources for essential needs such as lighting, heating, and cooking.

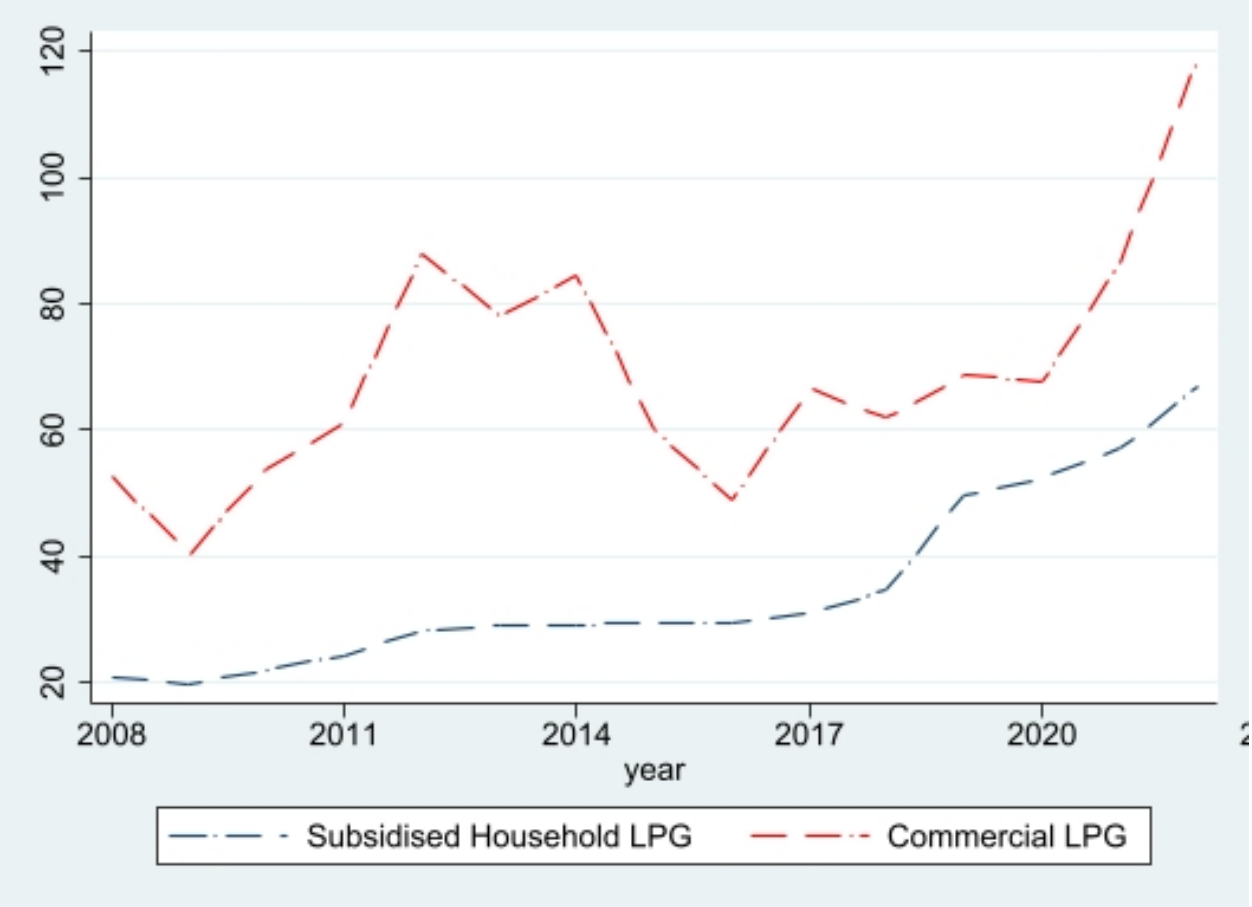
India has nearly completed the universal electrification of households. The electricity coverage in India stands at 97 percent (Agrawal et al., 2020). But around 660 million people lack access to clean cooking fuels (IEA , 2021). This distinction is mainly because access to clean cooking

encompasses more than just technical availability; it also encompasses considerations of dependability, ease of use, safety, and cost-effectiveness (Gould and Urpelainen, 2018). Many policies were introduced to reduce the dependence on traditional cooking stoves in India, but most policies failed in achieving their goal (Khandelwal et al., 2017). Several factors contribute to the limited demand for modern cooking technologies. These include high initial cost and subpar stove quality, inadequately planned and executed policies, a disconnect between stove features and local requirements, and a lack of consideration for environmental and health issues by the general public (Mobarak et al., 2012). This is set against the backdrop of the statistics that approximately 17.8 % of India's total fatalities in 2019 were attributed to indoor air pollution (Pandey et al., 2021). That is, around 1.67 million deaths in India in the year 2019 were due to indoor air pollution. Encouraging households to transition away from traditional cooking stoves has consistently posed a significant policy challenge for successive Indian governments. The cost of LPG has been a contentious issue in political discussions, and the fiscal constraints associated with it have often made it difficult for the government to policy reform measures.

LPG was provided to Indian households at a subsidised price from 1970 onwards. After the year 2000, when the Indian government initiated the deregulation of the LPG industry, there was a notable increase in LPG adoption among households. However, this also led to an upsurge in the sale of subsidised LPG gas cylinders to commercial buyers in the informal market (Agarwal et al., 2021). This period was also accompanied by increased fiscal pressure on the government due to the increased subsidy cost. The significant price disparity between subsidised household LPG and unsubsidised commercial LPG is one of the main reasons for household LPG being sold to commercial users in the informal market (Mittal et al., 2017). This price disparity between the 14.2 Kg subsidised household LPG and the 19 Kg commercial purpose LPG is shown in figure 1.

To address the shortcomings of earlier policies and to increase the usage of LPG, various policies were introduced by the Government of India since 2012. These initiatives include programs such as PAHAL, which involve selling LPG at market prices while directly transferring subsidies to the recipient's bank account (Pena Nelz, 2019). Also, the number of cylinders consumers can

Figure 1: The Evolution Price/ Kg of Commercial and Household LPG

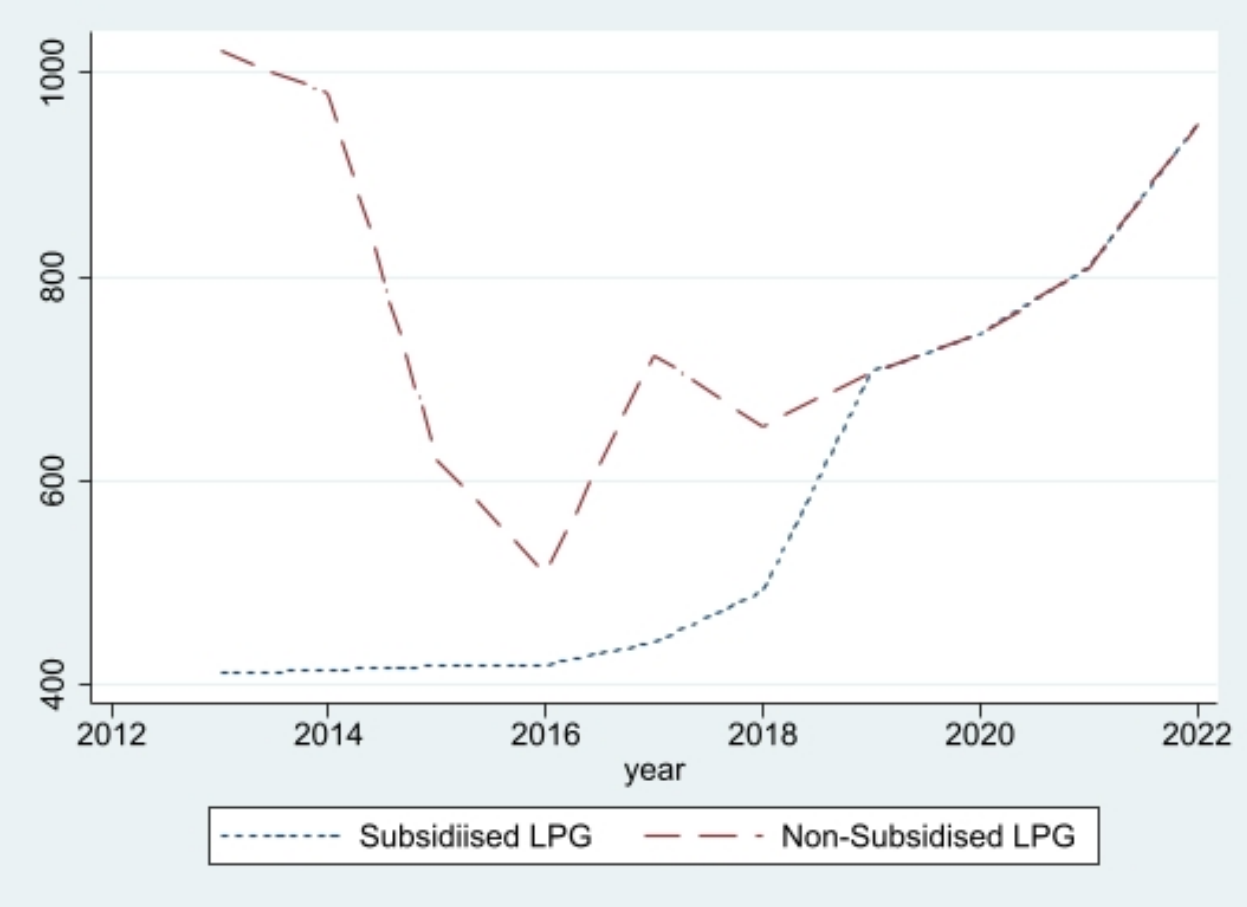


Source: Ministry of Petroleum and Natural Gas (2022), IOCL (2023)
Note: The Rs/Kg of 14.2 subsidised Household LPG and 19 Kg Commercial purpose LPG. The price represents the cost of an LPG cylinder in Delhi at the beginning of April each year

buy a year is capped. The beneficiaries are now determined based on the household's income level and the universal subsidy for LPG consumption is scrapped. But from 2019, the direct bank subsidy transfer to the beneficiaries was completely phased out. This is graphically represented in the figure 2. Despite the drop in crude prices at the onset of the Covid-19 pandemic leading to the discontinuation of the subsidy, it has not been restored even as prices have rebounded. As of August 2023, the price of an LPG cylinder in Delhi is INR 1103, which is roughly equal to USD 13.

Another policy to address the problem of low consumption of LPG among poor households

Figure 2: Price Evolution of Subsidised and Non-Subsidised LPG Cylinders



Source: Ministry of Petroleum and Natural Gas (2022), IOCL (2023)
Note: The cylinder's price is indicated in INR (Indian Rupees). The price represents the cost of an LPG cylinder in Delhi at the beginning of April each year

was the*Pradhan Mantri Ujjwala Yojana* (PMUY henceforth)[1], launched in 2016. According to Government of India (2016), the main goal of PMUY is to provide clean cooking fuel like LPG to rural and underprivileged households that previously relied on traditional cooking fuels such as firewood, coal, and cow-dung cakes. Under PMUY, a deposit-free LPG connection is provided to the adult women of the household. Under this scheme, the government covers the cost of various items, including the security deposit for the cylinder, pressure regulator, safety hose, DGCC booklet, and installation charges, amounting to a maximum of Rs 1600 per connection (Ministry of Petroleum and Natural Gas, 2023a). As of the Ministry's Official Report, there are 95 million

[1]PMUY can be loosely translated as Prime Minister's Cooking Gas Scheme

PMUY beneficiaries, of which 84 million households have refilled at least once in the 2021-22 period (Ministry of Petroleum and Natural Gas, 2023b). The Government has also approved a targeted subsidy of INR 200 to all PMUY beneficiaries, which is credited directly to the beneficiary's bank account starting from March 2023 (CCEA, 2023).In this research study, we aim to evaluate the success of the PMUY in increasing LPG consumption among poor households in India.

The next section provides a brief review of the existing literature on energy poverty and the assessment of programs aimed at mitigating energy poverty. Subsequently, we delve into the discussion of data and the empirical methodology. The findings and their implications are discussed in the conclusion.

2 Literature Review

Energy poverty in India is widely discussed. Balachandra (2011) presents a comprehensive examination of the complexities surrounding energy poverty in rural areas of the country. By adopting a multidimensional approach, this study analyses the temporal, income, and regional aspects of energy access. The study brought attention to an imbalance in addressing energy access issues, where the focus on supply-side factors tends to outweigh the equally crucial consideration of demand-side aspects in academic research and policy formulation. Moreover, their descriptive analysis of NSSO data reveals limited progress in addressing energy poverty, especially among rural households, as negligible improvement can only be witnessed after 23 years of implementing the first policy to address it. According to the author, this is evident from low access levels among rural households and especially in households in the lowest income strata and significant variations in modern energy access levels across states in India. The main reasons for the lack of access among poor rural households are poor policy formulation and implementation, lack of resources and infrastructure and high poverty levels across the country.

Alkon et al. (2016) conduct a detailed analysis using NSSO data from 1987 to 2010, investigating the variation of energy cost burdens among Indian households and the relationship between

household expenditure and access to modern energy sources. The findings shed light on the energy cost burdens faced by different states, revealing that those with modernised power sectors experience lower energy cost burdens. In contrast, rural households encounter increased energy cost burdens despite improved energy access, as their income growth does not parallel urban counterparts. The study identifies access to electricity and LPG as influential factors positively associated with energy expenditure, highlighting those other factors beyond household income, such as government policies and market dynamics, play a role in influencing the adoption of modern fuel sources. Similarly, Chindarkar et al. (2021) delves into the willingness of rural Indian households to pay for LPG, employing a probit model and leveraging data from the ACCESS 2018 dataset. The study assesses the probability of households exclusively using LPG for cooking for a specific bid amount for an LPG cylinder. The results indicate that as the randomly assigned bid for the LPG cylinder increases, the probability of households opting for exclusive LPG use decreases for both LPG and non-LPG households. Moreover, the study also showcased that salaried employees are more likely to accept bid amounts for exclusive LPG use than other households, citing the influence of regular income in the energy transition.

Various studies have analysed the impact of the PMUY on LPG consumption in India such as Dabadge (2018); Swain and Mishra (2020); Gill-Wiehl et al. (2020); Ranjan and Singh (2020). Most of these studies are descriptive in nature or regional analysis. Ranjan and Singh (2020) analysed the impact of PMUY using data from the National Sample Survey Office (NSSO) expenditure data and the Petroleum Planning and Analysis Cell (PPAC) report. The analysis reveals a notable decrease in firewood consumption in rural areas since 2014. Furthermore, the study's findings suggest that, based on the PPAC report, there has been a 15 % increase in the consumption of LPG. However, they also found that the number of inactive LPG connections has risen. According to the study, this phenomenon can be attributed to PMUY's success in expanding access to LPG but its relative failure to encourage consistent and sustained usage among its beneficiaries. Dabadge (2018), in their analysis, arrives at a similar conclusion, drawing upon information from the Petroleum Planning and Analysis Cell (PPAC) reports. They attribute the limited uptake in LPG

refilling, even among those who gained access to LPG cylinders through PMUY, to the increased prices of LPG, including subsidised LPG cylinders.

In contrast, Gill-Wiehl et al. (2020), utilising the ACCESS data from the Council on Energy, Environment, and Water (CEEW), examined the effects of the PMUY on LPG connectivity in six states. They employed an intent to treat Difference-in-Differences (DiD) methodology and arrived at the conclusion that PMUY had led to an increase in LPG connectivity, with gains ranging from 3.3 to 3.6 percentage points. Their control group consisted of households that slightly exceeded the income limit for the Below Poverty Line (BPL) classification. They also found that PMUY beneficiaries are far behind in terms of LPG consumption compared to the control group. Swain and Mishra (2020) conducted an analysis in the Indian state of Odisha to examine the factors influencing the adoption of LPG. They utilised both probit and tobit models for their analysis. Their key finding indicates that the reduction in firewood usage is more pronounced among households with higher levels of education but less significant among households benefiting from the PMUY. This outcome implies that PMUY may not have been entirely successful in expediting the energy transition in Odisha. It's worth noting that their primary survey data, collected from the Puri district in Odisha, comprises a relatively small sample of just 106 observations.

In conclusion, while the prevailing literature effectively underscores the importance of policies like the Pradhan Mantri Ujjwala Yojana (PMUY) within the Indian context, it is essential to acknowledge the predominant descriptive nature of most evaluations, often reliant on reports from the Petroleum Planning and Analysis Cell (PPAC). This reliance on descriptive assessments may hinder the establishment of a robust causal link to PMUY's impact. Additionally, assessments utilising the ACCESS dataset by CEEW are geographically limited to six Indian states with below-average per capita GDP figures, potentially limiting the external validity of the findings to other regions within the country. To address this gap in the literature, we focus on analysing the causal impact of the PMUY on LPG consumption in India.

3 Data and Descriptive Statistics

The research relies on data from two waves of the National Family Household Survey (NFHS) dataset: NFHS-4 (2014-15) and NFHS-5 (2019-20). NFHS is a comprehensive survey conducted nationwide in India to gather crucial information on population, health, and nutrition. The main goal of NFHS surveys, as stated in their official document, is to provide essential data on health and family welfare, including crucial indicators like fertility levels, infant and child mortality, maternal and child health, and other emerging health and family welfare issues at both the national and state levels (IIPS and ICF, 2021). These surveys cover around 707 districts in all 29 states and 7 union territories of India and survey around 600,000 households in each wave. We here exclude the union territories.

3.1 Key variables

The variables used for predicting treatment and matching households into treatment and control groups are given in table 1. These include standard household socio-economic characteristics that are almost unique to Indian households, such as caste religion, along with other characteristics.

Table 1: Key Variables

Variables	Description
hhid	Household Identification Number
State	State where the household resides
Age	Age of the household head
Religion	Religion of the household head
Caste	Caste of the household head
BPL card	A dummy variable indicating whether the household holds a BPL card (1) or not (0)
Wealth Index	Wealth index relative to the state
Educ	Highest educational attainment of the household head
urban	A categorical variable indicating the household is in urban (1) or rural area (2)
Gender	Gender of the household head. (1 for male and 2 for female)
hhsize	Total number of people living in the households

The treatment group is all households with a Below Poverty Line (BPL) card because PMUY

Table 2: Proportion of Household having Access to LPG

Category	2014-15	2019-20
BPL Household	22.07	45.13
Non-BPL Household	44.32	61.88

Source: Based on author's calculation using NFHS-4 and NFHS-5

is essentially targeted at BPL households. The BPL card is provided to every household who lives under INR 1000 in cities or INR 816 in rural areas in India (Gaur and Rao, 2020). From table 2, It is clear that the percentage of BPL households with LPG access has increased by more than two-fold. It was 22 percent in 2014-15 and increased to 45.13 percent in 2019-20. It is a compounded annual growth rate of 15.38 percent. At the same time, the LPG access of non-BPL households increased by only 6.9 percent CAGR annually. So, the main objective of this research is to analyse whether this increase in the growth rate is due to the PMUY.

To simplify the analysis, India's 28 states are grouped into 6 distinct zones, each with its own unique characteristics. These zones are as follows:

1. **North Zone:** Haryana, Himachal Pradesh, Jammu and Kashmir, Punjab, and Rajasthan.
2. **North East Zone:** Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Tripura, and Sikkim.
3. **Central Zone:** Chattisgarh, Madhya Pradesh, Uttarakhand, and Uttar Pradesh.
4. **East Zone:** Bihar, Jharkhand, Orissa, and West Bengal.
5. **West Zone:** Goa, Gujarat, and Maharashtra.
6. **South Zone:** Andhra Pradesh, Karnataka, Kerala, Tamil Nadu, and Telangana.

These zones are characterised by their geographical, cultural, and socio-economic differences. In the regressions, we use *state* fixed effects. The *zones* are used for sample splits. Table 3 indicates the proportion of households having access to LPG in each zone. While all zones have an

incremental increase in LPG access, the West Zone shows a decrease in the proportion of households having access to LPG for cooking. The Central Zone has the highest number of households with LPG access, followed by the South Zone.

Table 3: Proportion of Households having access to LPG by Zone

Zone	2014-15	2019-20
North Zone	40.88	61.51
North East Zone	32.18	44.75
Central Zone	30.43	69.91
East Zone	17.77	58.97
West Zone	46.74	33.32
South Zone	56.23	63.04

Source: Based on author's calculation using NFHS-4 and NFHS-5

Wealth Index is a variable encompassing the household's assets and their composition. The index is determined by assigning scores to households based on the quantity and types of consumer products they possess. These products can include items such as televisions and automobiles, as well as household amenities like access to clean drinking water and sanitation facilities, as well as the materials used for flooring (IIPS and ICF, 2021). The index scores are then estimated using principal component analysis by the DHS. It has five categories: poorest, poorer, middle, richer and richest. More than 50 percent of the rural population in India falls into the lowest two categories: the poorest and the poorer. For the analysis, the wealth index within the state is used as it is more meaningful to compare the poor households within the state rather than comparing with the poor in other states.

Table 4 shows the summary statistics of the model. The treatment group is the households below the poverty line, and the control group is the households above. The total number of observations for the treatment group is around 450,000 and around 550,000 for the control group, which is the non-BPL category. The mean difference between the treatment and control group is not much different for most of the variables except for the cooking fuel.

Table 4: Summary Statistics

Variables	Summary Statistics					
	Treatment group			Control Group		
	Obs	Mean	S.D	Obs	Mean	S.D
Age	459,940	49.370	13.770	564,532	48.576	14.485
HHsize	459,940	4.593	2.208	564,532	4.524	2.223
Urban	459,940	1.817	0.387	564,532	1.644	0.479
Gender	459,940	1.177	0.382	564,532	1.150	0.357
Cooking fuel	459,940	6.247	4.989	564,532	5.314	6.040
Wealth Index	459,940	2.446	1.279	564,532	3.158	1.434
Religion	459,940	3.141	12.816	564,532	2.648	10.450
Caste	444,953	2.452	1.052	542,232	2.782	1.123
LPG Access	459,940	0.337	0.473	564,532	0.506	0.500

Note: Household belongs to the BPL category is the treatment group and not belong to the BPL category is the control group. Summary statistics is estimated before the matching exercise.

4 Empirical Strategy

Due to a lack of information on PMUY beneficiaries in the NFHS dataset, an Intention to Treat (ITT) analysis is utilised, following the approach outlined by Duflo et al. (2007). The ITT approach considers all eligible individuals as part of the treatment group, irrespective of whether they received the treatment. The households with a Below Poverty Line (BPL) card are designated as the treatment group, while those without are the control group. Propensity scores, derived from observable characteristics, group households into treatment and control groups. As the NFHS dataset is no panel, the analysis adopts a repeated cross-sectional approach. A propensity Score Matching Method is employed to estimate the causal impact of PMUY.

4.1 Propensity Score Matching

The Propensity Score Matching (PSM) method is the most commonly used methodology to replicate the randomisation process in policy evaluation exercises where randomised data is not available. This approach pairs individuals from the control and treatment groups with similar propensity scores, effectively matching them based on these characteristics (Athey and Imbens,

2017). The resulting treatment and control groups, formed through propensity score matching, enable a direct and balanced comparison ((Rosenbaum and Rubin, 1983)). The PSM method addresses potential bias and facilitates the creation of more balanced comparison groups. This single score reflects the likelihood of receiving treatment, allowing to mitigate bias by matching treated and untreated subjects more effectively (Rubin, 1997).

The model relies on two essential assumptions: Conditional Mean Assumption (CIA) and overlap. CIA assumes that observed outcome differences between treatment and control groups arise solely from the treatment, effectively controlling for other influences. Overlap, on the other hand, ensures that individuals in both groups have a non-zero probability of receiving either treatment or control, enabling meaningful comparisons. These assumptions are essential in assuming that the treatment satisfies (some form) of exogeneity. The observations are matched into treatment and control groups based on the predicted propensity score. The nearest neighbour matching algorithm pairs units in the treatment and control groups. Further, the quality of treatment exercise (balancing property) is also examined. To examine the balancing property of the matching exercise, Rubin (2001) suggested three measures: analysing the standardised bias and examining Rubin's B and Rubin's R. The results are provided in table 5.

The matching is carried out in both pre and post-treatment periods. A Difference in Differences (DiD) estimator similar to Heckman et al. (1997, 1998) is employed to estimate the treatment effect on the treated. So, the ATT is estimated for the pre-treatment period using PSM. Then, the ATT is estimated for the post-treatment period. Then, the DiD is employed to calculate the difference between the ATTs. In this way, the bias arising out of time-invariant unobservables can be isolated (Smith and Todd, 2005).

The treatment assignment equation employed in the model is

$$prob[Treatment = 1] = \frac{exp(z_i)}{1 + exp(z_i)} \tag{1}$$

where:

$$z_i = \gamma_0 + \gamma_1 state_i + \gamma_2 hhsize_i + \gamma_3 urban_rural_i + \gamma_4 age_i + \gamma_5 religion_i + \gamma_6 caste_i + \gamma_7 bplcard_i + \gamma_8 wi_state_i + \gamma_9 hheduc_i + \gamma_{10} gender_i \quad (2)$$

where subscript *i* represents the household. The treatment is 1 if the household belongs to the BPL category and 0 if it does not belong to the BPL category. All other explanatory variables are explained earlier in the data section.

There are two outcome variables in this model.

1. LPG Access: 1 if the household has LPG, 0 otherwise

2. Firewood Access: 1 if the household uses firewood for cooking, 0 otherwise

Despite the advantage of propensity score matching for emulating randomisation in observational studies, it is essential to recognise that PSM has its own limitations. A significant concern is due to the reduction in dimensionality. Combining various covariates into a single propensity score, although helpful for predicting treatment assignment, can potentially reduce the overall effectiveness of the analysis. Another critical consideration revolves around PSM's exclusive reliance on observable confounding covariates. It is vital to acknowledge that PSM does not accommodate unobservable confounding covariates, a potential source of bias that could impact the accuracy and validity of the estimated outcomes (Rubin, 1997).

4.2 Robustness Analysis

In order to check the robustness of the results computed using the PSM methodology, Inverse Probability Weight and Augmented Inverse Probability Weight (AIPW) Estimators are used to compute the causal impact of PMUY using the same dataset. Even though PSM reduces bias by adjusting the propensity score for treatment and control groups, it decreases the dimensions of the estimate because reducing covariates into a single propensity score is less efficient (Hirano et al.,

2003). To overcome this flaw, IPW estimators give each observation a weight, which is the inverse of the probability of treatment. The idea is that the observation within the treatment group having the highest weight is more similar to the control group because they have less propensity for treatment and therefore more inverse probability weight (Huntington-Klein, 2021). That is, an observation which gets treatment will have a weight of $1/p$, whereas an observation in the control group will have a weight of $1/(1-p)$. The higher weight will be given to the treatment group observation, which has a low propensity to get treated (high inverse probability) but gets treated anyways and to control group observations, which have a high propensity to be in the treatment group but did not get the treatment (thus high inverse probability). This weighting exercise creates a pseudo-population that, once re-weighted, will demonstrate a comparable distribution of observable traits.

The augmented Inverse Probability Weighting (AIPW) estimator is a doubly robust estimator, which is a consistent estimator to estimate ATE when the treatment model is correctly specified or the outcome model is correctly specified (Glynn and Quinn, 2010). It is called a doubly robust estimator because it estimates a consistent ATE if any of the two regression models, one with matched covariates and treatment as the dependent variable or the other regression model with the outcome as the dependent variable, are correctly specified (Huntington-Klein, 2021). Robins et al. (1994), first proposed this estimator based on inverse probability weights, which are asymptomatically normal and unbiased to estimate the unknown parameter in the regression equation when there is a problem of missing data in the model. The ATE of AIPW is expressed in the equation 3.

$$ATE_{aipw} = \frac{1}{n}\sum_{i=1}^{n} E(Y_i|D=1) - E(Y_i|D=0) + \frac{(D_i)(Y_i - E(Y_i|D=1))}{e_i(x)} - \frac{(1-D_i)(Y_i - E(Y_i|D=0)}{1-e_i(x)} \quad (3)$$

The AIPW estimator has many statistical properties similar to other estimators. As mentioned earlier, it is consistent if either the outcome model or the model to predict the propensity score is correctly specified. Because the bias in the first part of equation 3 will be estimated and eliminated

by the second part of the equation 3 or vice versa. The other properties of AIPW include attaining variance bounds if both models are correctly specified, efficient if the outcome model is correctly specified and joint population boundness (Tan, 2010). The treatment assignment equation is the same as equation-2. The same DiD estimator is employed to calculate the difference between the pre and post-treatment ATT to filter out the potential unobservable bias in the treatment effects for both IPW and AIPW estimators.

Further, the sensitivity of treatment effect due to unobserved selection bias is also tested in this analysis using the Rosenbaum (2002) bound test. This test does not test for the magnitude of bias but instead tests how sensitive the treatment effect is to unobservable factors affecting the treatment assignment. Suppose the probability of treatment assignment is given by

$$e(x) = prob(D_1 = 1|x, u_i) = F(\beta x_i + \gamma u_i) \tag{4}$$

where x_i is the observable characteristics, u_i is the unobservable characteristics and γ is the effect of the unobservables on the probability of the treatment. Suppose two units having the same probability of treatment, i.e.; $e(x_i) = e(x_j)$, the odds that two individuals getting treatment is given by $\frac{p_i}{1-p_i}$ and $\frac{p_j}{1-p_j}$. The odds ratio is then given by

$$\frac{\frac{p_i}{1-p_i}}{\frac{p_j}{1-p_j}} = \frac{exp(\beta x_i)exp(\gamma u_i)}{exp(\beta x_j)exp(\gamma u_j)} \tag{5}$$

Since the probability of getting treatment for the two units remains the same, the identical and observable x vector gets cancelled out and the remaining part is $exp[(\gamma)(u_i - u_j)]$. If there is no unobservable selection bias or the unobservable has no impact on treatment assignment, the probability will be the same and the odds ratio will be 1. However, if the unobservable has an impact on the treatment probability or the unobservable affects the units differently, the odds ratio will not be equal to 1. Rosenbaum (2005) suggest that in this case, the odds ratio will be bounded

between

$$\frac{1}{\Gamma} \leq \frac{p_i(1-p_j)}{p_j(1-p_i)} \leq \Gamma \tag{6}$$

where $\Gamma = e^{\gamma}$ and assuming the unobservable factors affecting are binary in nature.

The testing of sensitivity towards the unobservable factors is assigning different values for Γ to examine for what values of Γ the p-value of the odds of participation changes (probability ratio). If $\Gamma = 1$, then there are no unobservable factors affecting the odds of treatment ratio. And if $\Gamma = 2$, it means that the probability of treatment assignment is changed by a factor of two. That is, out of the two units, one unit's probability of getting treatment is doubled due to the unobservable confounders. In effect, the Rosenbaum Bound test manipulates the value of Γ to gauge the extent to which hidden variables affect the odds of treatment assignment and subsequently, the study's conclusions. So if for a small value of Γ (say 1.5) the p-value changes, then the model is less robust (Imbens and Wooldridge, 2009). For binary outcome, the MH-stat similar to (Aakvik, 2001) is used.

5 Results and Discussions

5.1 Balancing Property Diagnostics

Before discussing the treatment effects, it is crucial to analyse the balancing property of the matching assignment. For the matching at the baseline, i.e., for 2014-15, all the variables used for the treatment equation are highly significant except for the gender variable. That may be because only a small proportion of households out of the total households are headed by women. So, the gender variable might not be a good predictor of treatment for the model [2]. Out of 570214 observations for the baseline, only 8 observations were off-support; the rest of the observations were on-support. Since the dataset is large and has 8 variables, kernel-based matching is beyond our computational power. So, the matching was based on the nearest neighbour algorithm without

[2]The results of probit regression for the treatment assignment are provided in Appendix A

any caliper and replacement.

To assess the balancing property of the matching, the standardised mean bias of each variable after matching is analysed. The mean bias indicates how well the distributions of covariates are balanced among the treatment and control groups. After matching, the mean of covariates should be similar for both the treatment and the control group for a fair comparison. Ideally, the standardised bias after matching should lie between -5 and 5. It is evident from table 5 that all variables used to predict the treatment lie under the required bound. Another measure to check for good balancing is Rubin's R, which is the ratio of the variance of the propensity score for the treatment and control group. The recommended value of Rubin's R should lie between 0.5 and 2. And for all the variables in the treatment model, it lies within the bounds.

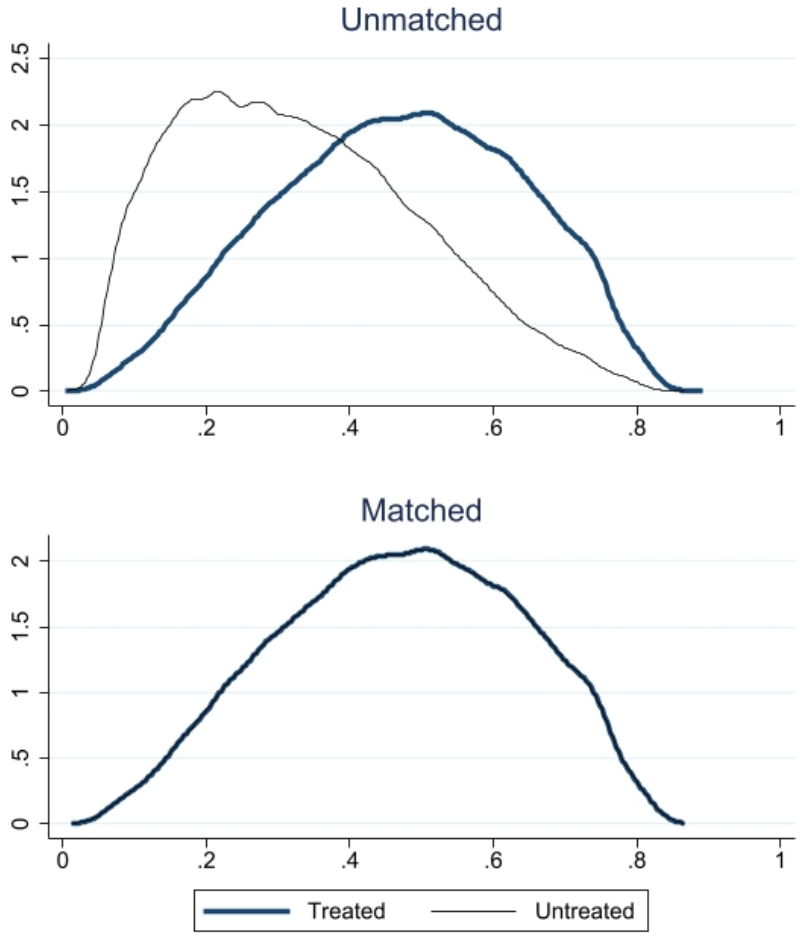
Table 5: Balancing Property Diagnostics

Variable	**Treated**	**Control**	**%bias**	**Rubin's R**
State	17.855	18.021	-2.1	1.06
Age	48.906	48.614	2.3	0.94
Religion	2.870	2.873	-0.3	1.05
Caste	2.475	2.482	-0.6	1.03
Educ	1.021	1.023	-0.1	1.12
Wealth Index	2.417	2.393	2.2	0.96
urban_rural	1.8134	1.825	-2.6	1.04
Gender	1.162	1.154	2.8	1.04
hhsize	4.859	4.873	-1.2	0.95

Note: %bias represents the percentage difference in means between treated and control groups. The ratio Rubin's R represents the variance ratio of the treated group to the control group.

Figure 3 shows the density of propensity score before and after matching for treated and untreated groups. It is evident from the graph before matching that the density of propensity to get treated is different for both groups. However, once the observations are matched, the density is almost the same for both groups.

Figure 3: Probability Density of Treatment and Control Group



5.2 ATT Estimates of PMUY

Table 6 reports the ATT from the PSM matching exercise. The results suggest a causal impact of a 2.1 percentage point increase in the uptake of LPG among BPL households relative to if there was no treatment in the first effect. The effect of PMUY in increasing LPG access to BPL households was not that large. The result is significant at the 1 percent significance level. The standard error is so small because the sample size is very large. Even for the poorest sample, the treatment effect is small: PMUY increases the LPG intake among the poorest households only by 3.8 percentage points. Again, this estimate is significant at a 1 percent significance level. There are only a few observations out of the common support, that is, less than 100. So, the loss of data

Table 6: Impact of PMUY on LPG Access

	Full Sample		Poorest Sample	
	Pre Treatment	Post Treatment	Pre Treatment	Post Treatment
ATT (Treatment Group)	0.220	0.451	0.026	0.194
ATT (Control group)	0.230	0.439	0.035	0.165
Difference in ATT	-0.010	0.012	-0.009	0.029
Standard Error	0.002	0.003	0.002	0.004
t-stat	-4.69	4.5	-4.91	6.82
Treatment Effect	0.021 ***		0.038***	
Standard Error	0.003		0.005	
On support Untreated	349318	194726	62463	36898
On support treated	218753	226180	67979	68824
Mean Bias	1.6	2.1	1.2	1.2
Rubin's B	5.4	6.6	4.5	5.7
Rubin's R	1.01	0.9	1.08	1.07

Note: Treatment effect is the difference in the ATT from the post and pre-treatment period. T-stat for the treatment effect is calculated using computing sample variance and standard error. The significance level is determined based on the t-stat. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

due to the exclusion of off-support observation is not an issue in this model.

The success of the matching exercise lies with the standardised mean bias between treatment and control covariates after matching should be the minimum; less than 10 percent is the widely accepted mean bias limit. As reported in Table 6, the mean bias after matching is only less than 2 percent, suggesting that the matching is free from any major flaw. Other metrics to examine the quality of the matching exercise are Rubin's B and Rubin's R, which are mentioned earlier are also within the recommended bounds.

The results from table 7 suggest a small but significant impact of PMUY on firewood reduction among BPL households in India. PMUY reduced firewood consumption among BPL households by 2.2 percentage points relative to what would have been for BPL households in the absence of any policy intervention and is significant at a 1 percent significance level. The increase in LPG access due to PMUY is also 2.2 percentage points. Since the observation on support is large, the standard error is minute, as the estimate's precision increases as the sample size increases. For

Table 7: Impact of PMUY on Firewood Usage

	Full Sample		Poorest Sample	
	Pre Treatment	Post Treatment	Pre Treatment	Post Treatment
ATT (Treatment Group)	0.618	0.459	0.823	0.703
ATT (Control Group)	0.589	0.452	0.791	0.720
Difference in ATT	0.029	0.007	0.032	-0.016
S.E	0.002	0.003	0.004	0.005
t-stat	12.27	2.56	8.3	-3.26
Treatment Effect	-0.022 ***		-0.048***	
Standard Error	0.003		0.006	
On support Untreated	349318	194726	62463	36909
On support Treated	218753	226180	67984	68828
Mean Bias	1.6	2.1	1.2	1.2
Rubin's B	5.4	6.6	4.5	5.7
Rubin's R	1.01	0.9	1.08	1.07

Note: The treatment Effect is the difference-in-difference of the ATT from pre and post-treatment period. Sampling variance and standard errors are calculated separately.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

the poorest sample, the effect is a reduction of firewood consumption by 2.5 percentage points. The mean bias between treated and controlled covariates lies well below the limit, suggesting the quality of the matching exercise. Both Rubin's B and R fall within acceptable limits, indicating that the matching process successfully balanced observable covariates.

Table 8: ATT of PMUY on LPG Access by Zone

Zone	ATT	SE	$\beta - \beta^*$
North	0.056***	0.010	0.035***
North Eastern	-0.026***	0.007	-0.047***
Central	-0.051***	0.009	-0.072***
East	-0.017	0.012	-0.038***
West	0.039***	0.010	0.018*
South	0.046***	0.014	0.025*

Note: ATT is the difference-in-difference of ATTs from pre and post treatment period. t-stat is computed separately using sampling variance. * represents significance level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8 reports the ATT estimates of PMUY on the LPG access in different zones in India.

The treatment effects are significant in all regions except for the Eastern zone, which includes the states of Bihar, Jharkhand, Odisha, and West Bengal. These Eastern zone states are predominantly economically disadvantaged with low per capita income. In the North Zone, comprising various states, PMUY has had the most significant impact, increasing LPG uptake by 5.6 percentage points. The South Zone, encompassing states like Andhra Pradesh, Karnataka, Kerala, Tamil Nadu, and Telangana, follows closely with a substantial impact of 4.6 percentage points. These results are highly statistically significant at the 1 percent level. However in the Northeastern and Central Zones, PMUY has led to a reduction in LPG uptake. In the Northeastern zone, there is a decrease of 2.6 percentage points, while the Central zone shows a more significant reduction of 5.1 percentage points. This implies that, on average, PMUY has decreased the likelihood of BPL households gaining access to LPG by 5.1 percentage points in the Central zone and 2.6 percentage points in the North East zone. The Central Zone includes states like Chattisgarh, Madhya Pradesh, Uttarakhand, and Uttar Pradesh. Conversely, in the West Zone, there has been a notable increase of 3.9 percentage points in LPG consumption among BPL households compared to if there had been no treatment. This result is also statistically significant at the 1 percent level.

The reason why the PMUY has a negative causal impact in the North East and East zone is mainly because of the high proportion of ST households. From the estimates reported in table 11, it is evident that the ST households are at a disadvantage in access to LPG. To understand the reason why these regions show a negative effect, a separate sub-sample analysis for the North East, East and Central zones has been carried out. The result suggests that for the Eastern zone, PMUY has a small but significant impact of 0.6 percentage points on ST households but not a significant impact for the other category households. The minor effect observed in ST households and the lack of any impact on households in other categories collectively render the overall statistically insignificant impact for the entire region. For the Central zone, the impact is negative for all caste categories but significant only for the non-ST households. A further deep analysis into the reasons why every category of households is at a disadvantage in accessing LPG in the Central region. For the Nort-East sample again, the impact is statistically significant only for SC households. For all

other categories, the impact is statistically insignificant. The result indicates that the SC household has a negative causal impact of 4.2 percentage points in accessing LPG due to PMUY [3].

A t-test is conducted to determine whether the ATT estimated for different zones is statistically different from the ATT estimated for the whole country. This is reported in the fourth column of the table 9. The null hypothesis is $\beta = \beta^*$. And the alternative hypothesis is it is not. The β is the estimated ATT of the different zones and β^* is the ATT for the whole sample which is reported in the table 6. However, based on the t-test, the null hypothesis can only be rejected for the North, North Eastern, Central and Eastern samples at a 1 percent significance level. This means that the ATT estimates are significantly different from that of the ATT estimate of the whole sample.

Table 9: Region-wise ATT of PMUY on LPG access

Region	ATT	SE	$\beta - \beta^*$
Urban	0.003	0.006	-0.018***
Rural	0.024***	0.004	0.003

Note : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$,

Table 9 shows that the impact of PMUY is larger in the rural rather than in the urban areas because 81 percent of the total BPL households in the sample are in rural areas. On average the impact of PMUY on LPG access among BPL households in rural areas increased by 2.4 percentage points. But for the urban sample, PMUY do not have a significant Average Treatment Effect. This may be because of the low representation of BPL households in urban areas. Out of the 19 percent of BPL households living in urban areas, only 5.93 percent of households do not have access to LPG in urban areas. This might be why the PMUY has no significant impact in the urban areas. The t-test to determine whether the urban and rural estimates are statistically different from the national estimate is also provided. The urban estimate is statistically different from the national estimate, but the rural estimate is not statistically different from the national estimate, because most of the treatment groups are in rural areas.

The table 10 gives the ATT of the PMUY on major religions in India. The treatment effect has

[3]The result for the separate sub-sample analysis is provided in Appendix A

Table 10: Religion-wise ATT of PMUY on LPG access

Religion	ATT	SE	$\beta - \beta^*$
Hindu	0.030***	0.004	0.009**
Muslim	0.057***	0.011	0.036***
Christian	-0.040***	0.010	-0.061***
Sikh	0.029	1.130	0.008

Note : *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$,

more impact on the Muslim group. The ATT of the Muslim group suggest that, on average, the causal impact of PMUY on LPG access among Muslim households has increased by 5.7 percentage points relative to that if there was no treatment. Hindu households have an impact of 3 percentage points relative to what if there was no treatment. The estimate for the Christian group is particularly interesting because it displays a negative sign for the estimated treatment coefficient. This result should be read in the backdrop of the results from table 8 that the impact of the treatment in the Northeast region is negative. And around 17 percent of the households in the North East zone are Christian. Also, almost 85 percent of Christian households belong to the scheduled tribe in the sample. When the PSM is estimated for Christian religion without ST households and only ST households separately, the impact of PMUY is positive for non-ST households, while it is negative for ST households in the Northeast Region (Appendix A). This might be one reason why the Christian households showing a negative impact on LPG access by PMUY in this sample. The LPG access of Christian households that belong to the scheduled tribe has increased from 31 percent in 2014-15 to 39 percent in 2019-20, while the households which do not belong to the scheduled tribe household increased their LPG share from 55 percent in 2014-15 to 74 percent in 2019-20. The ATT for both years are negative for ST households, while it is positive for the non-ST households (Appendix A).

Sikh households show no causal impact by PMUY. Again, for the Sikh community, the population in the sample is divided in such a way that almost equal proportion of people belong to the Scheduled caste and Upper Caste. When the causal impact of Sikh households is estimated separately for the households that belong to the SC and Non-SC categories, the treatment effects

are insignificant for both SC and non-SC households. So, the PMUY has no causal impact on any category of Sikh households.

The difference from the whole sample estimate is statistically significant for all religions except the Sikh religion.

Table 11: Caste-wise ATT of PMUY on LPG access

Caste	ATT	SE	$\beta - \beta^*$
SC	0.041***	0.007	0.020***
ST	-0.007	0.007	-0.028***
OBC	0.033***	0.008	0.020***
None of the above	0.031***	0.008	0.012

Note : *** $p < 0.01$,** $p < 0.05$, * $p < 0.1$,

Table 11 shows that PMUY has a maximum causal impact among the OBC households of 4.1 percentage points, followed by SC households. The ATT estimate of the ST households is not significant. This is mainly because the ST households have a significant proportion of the households in the North Eastern and Eastern Zone. The causal impact of the PMUY is negative in the North Eastern Region. In a subsample analysis for the North Eastern sample without ST households, the ATT estimate is positive and in a sub-sample analysis with only ST households in the North Eastern region, the ATT estimate was negative and insignificant. Similarly, In a similar matching exercise for the Eastern sample, the estimate shows that for ST households, the ATT estimate is negative and significant. However, the result again becomes insignificant for non-ST households. Either way, both Eastern and North Eastern regions are disadvantaged in accessing LPG; among the households, the ST households are disadvantaged more than any other category households.

As a robustness check, the Average Treatment Effect of PMUY was estimated using the Inverse Probability weight (IPW) and the Augmented Inverse Probability Weight (AIPW) Estimator. The results are reported in table 12. Using the IPW estimator, the estimated ATET should be equal to ATT after successful randomisation. The estimates are approximately equal for both ATET of IPW and AIPW. This suggests that the ATE estimates are consistent. Since the AIPW is the most

Table 12: ATET of PMUY using IPW and AIPW estimator

	LPG Access	Firewood Consumption
ATET (IPW)	0.008***	-0.013***
	(0.002)	(0.002)
ATET (AIPW)	0.008***	-0.013***
	(0.002)	(0.002)

Note: Robust standard errors in parenthesis. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

efficient of all estimators because of its doubly robust properties (Słoczyński and Wooldridge, 2018), the estimated ATE can be treated as an unbiased estimate. The estimated ATE can be interpreted as the causal impact of PMUY on the treatment group has led to an average increase of LPG access by 0.8 percentage points relative to what if there was no treatment in the first place among the treated BPL households. And for firewood usage, the average effect is even more negligible at only 0.05 percentage points. Both effects are statistically significant at a 1 percent level. Since the AIPW estimator is doubly robust, if either the outcome model or the model to estimate the probability of the treatment is correctly specified, then the Average Treatment on the Treated is efficient.

Since the ATT estimate from the PSM model and ATET from IPW and AIPW predict the treatment effects in the same direction and the magnitude is similar, it should be concluded that the impact of PMUY is indeed small among BPL households.

The results from Table 13 indicate that the potential impact of unobservables is minimal in this model. The Γ represents the odds of differential assignment due to unobserved variables. When Γ is 1, the model has complete randomness. It suggests that if two random households are selected, the probability of getting the treatment is equal, as no unobservable factors affect their probability of getting treatment. As the Γ increases, the potential unobservable differences between the households increase; therefore, the overestimation and underestimation of the treatment effect also increases. Since ATE is positive, the potential bias can most possibly overestimate the ATE. But what is indicated from table 13 is that even if there are potential unobservable differences between the households, the odds that it will overestimate or underestimate the treatment effects are insignificant. In other words, the null hypotheses of no treatment effects are significantly rejected

Table 13: Rosenbaum Bound Test Sensitivity Analysis using MH-stat

	Pre Treatment				Post Treatment			
Γ	Q_mh+	Q_mh-	p_mh+	p_mh-	Q_mh+	Q_mh-	p_mh+	p_mh-
1.0	58.827	58.827	0.000	0.000	35.866	35.866	0.000	0.000
1.1	69.686	48.042	0.000	0.000	46.788	24.983	0.000	0.000
1.2	79.681	38.249	0.000	0.000	56.803	15.072	0.000	0.000
1.3	88.954	29.274	0.000	0.000	66.063	5.967	0.000	0.000
1.4	97.618	20.986	0.000	0.000	74.682	2.450	0.000	0.007
1.5	105.761	13.283	0.000	0.000	82.753	10.296	0.000	0.000
1.6	113.451	6.085	0.000	0.000	90.348	17.640	0.000	0.000
1.7	120.747	0.665	0.000	0.253	97.526	24.547	0.000	0.000
1.8	127.694	7.038	0.000	0.000	104.336	31.069	0.000	0.000
1.9	134.331	13.069	0.000	0.000	110.819	37.249	0.000	0.000
2.0	140.692	18.794	0.000	0.000	117.009	43.126	0.000	0.000

Note: Γ: odds of differential assignment due to unobserved factors.
mh: is the MH stat for overestimation(+) and underestimation(-)
p_mh+ and p_mh- are the p-values

in this model both in pre and post-treatment matching. The Rosenbaum bound test only indicates what happens if there is any unobservable selection bias in the model. It does not say whether there is any unobservable selection bias in the model or the exact magnitude of the bias. However, it can be ensured that the unobservable will not affect the treatment effects in this model even if the potential unobservable factors affect the odds of treatment assignment even up to a factor of 2. Even when the Γ is 2, when the odds of the differential assignment are doubled, the null hypothesis of no treatment effect is rejected. This sensitivity analysis concludes that the treatment effect estimated using the observable cofounders is insensitive to the unobservable confounders, and there is no problem of unobservable selection bias.

5.3 Limitations of the Study

One of the primary limitations of this study stems from the absence of a policy (PMUY) variable within the National Family Health Survey (NFHS) dataset. Consequently, the study relies on an intent-to-treat methodology, which introduces a scenario where BPL households that already

possess an LPG connection are also counted as treated, potentially biasing the estimates. However, it's worth noting that LPG utilisation among households categorised as Below Poverty Line (BPL) is quite minimal, mitigating this bias concern. Even if the treatment assignment is imperfect due to the intent-to-treat analysis, the results can still be interpreted as the LPG utilisation among BPL households between 2014-15 and 2019-20 in the absence of treatment. Nevertheless, the outcome remains concerning, as there was only a small growth in LPG adoption among BPL households. This highlights that regardless of the treatment group's imperfections, BPL households face challenges in accessing LPG.

Another significant challenge encountered in the analysis pertains to the definition of the cooking fuel variable itself. In the NFHS questionnaire, households are asked about their primary source of cooking fuel. If households have access to LPG but do not primarily use it for cooking, NFHS categorises them as not having LPG access. This poses an issue when attempting to analyse LPG access. However, it's important to emphasise that the success of PMUY is ultimately measured by the extent to which people use LPG as their primary cooking fuel. Consequently, the variable definition aligns well with the study's focus on LPG access. Another potential drawback of the model is the lack of a specific household expenditure variable, with the wealth index serving as a substitute in its place.

6 Conclusion and Policy Implications

The Pradhan Mantri Ujjwala Yojana demonstrably improved LPG accessibility among BPL households in India, achieving a modest but statistically significant 2.1 percentage point increase in LPG consumption alongside a corresponding decrease in firewood dependence. This shift in energy consumption patterns represents a crucial step towards cleaner cooking fuel adoption and improved household health outcomes. However, the findings also reveal regional disparities in program effectiveness, with the North, West, and South experiencing greater gains compared to the East and North East zones. Moreover, the study underscores the need for more ambitious and

sustained policy interventions to ensure a wider and long-lasting impact.

The analysis of two waves of the NFHS dataset reveals regional disparities in accessing LPG among BPL households, warranting the proposal of tailored regional interventions to address this imbalance. The government's claim of 95 million beneficiaries for the program, accompanied by the results from our analysis, clearly indicates the need for more income support to facilitate an increased and permanent adoption of LPG among poor households.

Looking forward, there are ample opportunities to extend this research. We did not consider the time use associated with the different fuels. An intriguing avenue involves exploring the impact of PMUY on respiratory health, particularly among women and children who often bear the adverse effects of harmful smoke from traditional cooking methods. Given PMUY's explicit objective of reducing respiratory ailments resulting from such exposures, delving into this area could yield valuable insights into the program's comprehensive health benefits. Improved respiratory health could also lead to better outcomes for education and a greater earning capacity. All this is postponed to future research.

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A Additional results

Table 14: Summary Statistics: Extended

Variables	Variable Description	Min	Max
Age	Age of the Household Head	10	98
HHsize	Number of members in the households	1	41
Urban	1. Urban, 2. Rural	1	2
Gender	Gender of the household head. 1-Male, 2-Female	1	2
Zone	1. North Zone, 2. North East Zone, 3. Central Zone, 4. East Zone, 5. West Zone, 6. South Zone	1	6
Cooking fuel	1. Electricity, 2. LPG, 3. Biogas, 5. Kerosene, 6. Coal, 7. Charcoal, 8. Wood, 9. Straw, 10. Agricultural crop, 11. Animal dung, 95. No cooking in the Household, 96. Others	1	96
Wealth Index	1. Poorest, 2. Poorer, 3. Middle, 4. Richer, 5. Richest	1	5
Religion	1. Hindu, 2. Muslim, 3. Christian, 4. Sikh, 5. Buddhist, 6. Jain, 7. Jewish, 8. Parsi, 9. No religion, 96. Other	1	96
Caste	1. SC, 2. ST, 3. OBC, 4. None of the above, 8. Don't Know	1	8
LPG Access	0. No, 1. Yes	0	1

Table 15: Probit Results for matching: pre-treatment

Variable	**Coefficient**	**SE**	**p-value**
State	-0.029	0.000	0.000
Age	0.003	0.000	0.000
Religion	-0.002	0.000	0.000
Caste	-0.104	0.002	0.000
Educ	-0.127	0.002	0.000
Wealth index	-0.181	0.002	0.000
Urban	0.172	0.004	0.000
Gender	0.011	0.005	0.029
Hhsize	0.023	0.001	0.000
Constant	0.670	0.015	0.000

Note: The probit coefficient for treatment assignment pre-treatment for the estimation of results in the table 7

Table 16: Probit Results for Matching: Post-treatment

Variable	**Coefficient**	**SE**	**p-value**
State	0.027	0.000	0.000
Age	-0.001	0.000	0.000
Religion	0.003	0.000	0.000
Caste	-0.112	0.002	0.000
Educ	-0.207	0.002	0.000
Wealth index	-0.097	0.002	0.000
Urban	0.397	0.005	0.000
Gender	-0.056	0.005	0.000
Hhsize	0.037	0.001	0.000
Constant	-0.349	0.018	0.000

Note: The probit coefficient for treatment assignment post-treatment for the estimation of results in the table 7

Table 17: The ATT Estimates for different Samples of Christian Religion

Household category	**ATT**	**SE**
For ST Household	-0.045***	0.011
For Non-ST Household	0.047**	0.024

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 18: The ATT Estimates for different Samples of Sikh Religion

Household category	**ATT**	**SE**
For SC Household	-0.015	0.033
For non-SC Household	-0.038	0.039

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 19: The ATT Estimates for Different Eastern Zone Samples

Sample	**ATT**	**SE**
Only With ST Households	0.006***	0.020
Without ST Households	-0.017	0.016

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 20: The ATT Estimates for Different Central Zone Samples

Sample	**ATT**	**SE**
Only With ST Households	-0.017	0.031
Without ST Households	-0.020**	-2.058

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 21: The ATT Estimates for Different North Eastern Zone Samples

Sample	**ATT**	**SE**
Only With ST Households	0.003	0.011
Without ST Households	-0.028***	0.010
Only With SC Households	-0.042**	0.019
Only With OBC Households	-0.020	0.015

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

CUDA-Self-Organizing feature map based visual sentiment analysis of bank customer complaints for Analytical CRM

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Abstract

With the widespread use of social media, companies now have access to a wealth of customer feedback data which has valuable applications to Customer Relationship Management (CRM). Analyzing customer grievances data, is paramount as their speedy non-redressal would lead to customer churn resulting in lower profitability. In this paper, we propose a descriptive analytics framework using Self-organizing feature map (SOM), for Visual Sentiment Analysis of customer complaints. The network learns the inherent grouping of the complaints automatically which can then be visualized too using various techniques. Analytical Customer Relationship Management (ACRM) executives can draw useful business insights from the maps and take timely remedial action. We also propose a high-performance version of the algorithm CUDASOM (CUDA based Self Organizing feature Map) implemented using NVIDIA parallel computing platform, CUDA, which speeds up the processing of high-dimensional text data and generates fast results. The efficacy of the proposed model has been demonstrated on the customer complaints data regarding the products and services of four leading Indian banks. CUDASOM achieved an average speed up of 44 times. Our approach can expand research into intelligent grievance redressal system to provide rapid solutions to the complaining customers.

Keywords— Customer Complaints; Analytical CRM; Self-Organizing Map; CUDA; Visual Sentiment Analysis

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1. INTRODUCTION

Organizations embarking on the customer relationship management (CRM) journey are increasingly using customer data to drive business decisions. The Banking, Financial services and Insurance (BFSI) industry also relies heavily on data analytics for their day-to-day functions like fraud detection, credit scoring, risk analytics or cross-sell/up-sell, which come under the realm of Analytical CRM (ACRM). In addition to a large amount of data collected by these organizations from various customer touchpoints, critical customer feedback data is now available from social media platforms, which are the most preferred channels for customers to express their opinions/grievances regarding the products and services.

Efficient analysis of customer complaints can help the organizations strengthen the communication channels with their customers and resolve their issues quickly, turning complaining customers to loyal repeat customers [1], [2]. Essentially, non-redressal of complaints leads to customer churn and decreased sales. Yet, going through the expectedly large amounts of feedback would be time-consuming and would affect the turnaround time. A purely manual approach would be infeasible in most of the cases. Alternative approaches such as text mining and NLP techniques can help greatly here, by providing an automated way of generating insights from the large amounts of text. The most popular task carried out on customer feedback data is the classification of the sentiments expressed by the customer in their respective feedback document. This task involves classifying the sentiments into positive or negative classes to extract the overall perception of the customers towards a product or service. *Sentiment classification* is performed at various levels including document-level, sentence-level, and aspect-level. *Opinion summarization* is another class of analysis of customer feedback, which generates a summary of the feedback for a target (product or service) from a large corpus of feedback documents. Before a full-fledged sentiment analysis of the complaints is undertaken, visualization of the complaints by product and severity category greatly reduces the time and effort for the bank executives.

Therefore, to meet this requirement, we employed self-feature organizing maps, which are unsupervised and non-linear neural networks, in analyzing customer complaints.

Since inception, self-organizing feature maps have been extensively used for a variety of data analysis tasks such as clustering and data visualization in industry, linguistics, finance, and bioinformatics. SOM is a type of artificial neural network that is trained using unsupervised learning to produce low-dimensional representations of the input variables, making visualization of high-dimensional data possible. The standard SOM architecture consists of two layers - the input layer and the output layer. For a dataset with d dimensional vectors, the input layer contains d nodes, each node representing one attribute of a sample. The input layer presents the training data one sample at a time to the output layer, which is a regularly arranged (usually two-dimensional) lattice of neurons that

represent the distribution of input data with a smaller set of "prototype vectors". Each output node is fully connected to the input layer and contains a weight vector of the same dimension as that of the input vectors. As the network is trained, samples or records that are more similar get associated with adjacent nodes on the output layer, while less similar ones get associated with distant nodes.

In this paper, we propose an unsupervised learning model, which adequately caters to the need of suitably analyzing customer complaints to derive actionable insights. We analyze four customer complaints datasets related to four leading commercial banks in India using self-organizing maps and demonstrate the efficacy of the proposed method in analyzing and visualizing such data. However, since text data is high dimensional, processing it is computationally expensive and slow. To increase the speed of computation, we also propose CUDASOM, a parallel version of the model which has been implemented on CUDA (Compute Unified Device Architecture) which is a parallel computing architecture developed by NVIDIA.

The paper is structured as follows. First, we present the motivation behind this work in section 2. Then, we highlight the significant contributions of our work in section 3 and present an overview of literature review in section 4. A description of the techniques is provided in section 5. The details of our approach and the experimental setup are explained in section 6 and section 7 respectively. The results are discussed in section 8 and the concluding remarks are presented in section 9.

2. MOTIVATION

Sentiment classification of a corpus of complaints would logically assign a negative sentiment score to all the complaints and opinion summarization would result in summaries, which are neither meaningful nor useful. Therefore, while these two tasks are popular in the sentiment analysis domain and are extremely helpful in gaining insights regarding customer perception, they are ineffective in analyzing a corpus of complaints with a view to rapidly draw actionable insights. Moreover, both these techniques are supervised learning techniques which require time and effort for labeling the data and developing a reasonably well-performing model. Although unsupervised learning techniques have been proposed in the literature for sentiment analysis [3][4], these works focus on analysis of datasets containing both positive and negative sentiments whereas the focus of this paper is customer complaints, a type of dataset containing only negative sentiments. On the other end of the spectrum are niche unsupervised techniques like clustering which can deliver quick output, but impose an overhead of painstakingly elaborate analysis in terms of choice of number of clusters and understanding the nature of complaints within a group. Since text data is high-dimensional, visualization of the clusters to learn about possible interrelationships in the data requires help from auxiliary tools. A tool to analyze customer complaints should ideally help the service executives to identify the landscape of complaints so that they could derive actionable insights from them rapidly and also prioritize the complaints for a speedy redressal. The benefits of such tool would be pronounced if the tool provides these insights in a visual format. For example, it would be extremely helpful for a service manager to

identify the product-wise distribution of complaints so that the tasks are prioritized accordingly or, to identify a pressing issue affecting a large customer base, so that the team responsible is immediately deployed in resolving the problem. SOM effectively addresses these requirements in the following ways, a) it works in the absence of labels, b) it automatically performs dimensionality reduction making visualization possible c) owing to its topology preservation properties, it helps in visualization of the landscape of the data along with their topological relationships which is paramount and also invaluable for this task. Therefore with SOM at its core, we developed a visual sentiment analysis framework which can help the banks to efficiently handle customer complaints. By the virtue of being an unsupervised approach, the proposed method can quickly paint an approximate picture for the operational CRM staff to start taking action. As such, this method can act as an effective precursor before an in-depth study is performed using the supervised methods.

3. CONTRIBUTION

To the best of our knowledge, this is the first effort on employing SOMs for visualizing customer complaints from grievance prioritization and redressal perspective. The major contributions of our work as follows:

- We propose a novel procedure to perform visual sentiment analysis of customer complaints, which could significantly improve the grievance redressal systems. The efficacy of the model has been established with real-world complaints datasets for four Indian banks.
- We also propose CUDASOM, a novel implementation of CUDA accelerated standard SOM. The implementation has been designed to favor high dimensional text data as opposed to earlier proposals which focused majorly on increasing map sizes or number of training examples. It turned out that the proposed CUDA implementation achieved significant speedup as compared to its CPU counterpart.

4. LITERATURE SURVEY

Sentiment analysis has been extensively studied by researchers in the past decade and a half, and numerous works have been published in this field. In this study, we are employing self-organizing maps for sentiment analysis. Hence, we surveyed some research related to sentiment analysis performed using self-organizing maps. As this work also proposes a parallel, CUDA-based version of the self-organizing maps, we also review previous efforts in this direction.

Due to their excellent visualization abilities self-organizing maps have been widely adopted in domains like Bioinformatics [5], Geosciences [6], Finance [7] etc. In sentiment analysis research as well, SOMs have been used to study the sentiments of tourists [8], to infer movie characteristics from movie reviews [9], summarize social network conversation regarding wine varieties [10] and in automatic detection of bullying in social media platforms [11]. Reference [12] proposed a two-step approach to study the sentiment of a forum focused on a cancer treatment drug. They manually assigned a sentiment label to each post from the forum and then trained a SOM with a TF-IDF

representation of the posts using the SOM Toolbox. The SOM was used to find clusters in the data and to study the correlation of the codebook vectors with the positive and negative opinion. Reference [13] studied the efficacy of SOM for unsupervised and supervised sentiment analysis tasks. They extracted features from movie review dataset in TF-IDF feature representation and selected the top features ranked by the information gain criterion. The dataset was visualized using the Emergent SOM variant of SOM and classifier was trained using the Learning Vector Quantization algorithm. They concluded that the performance of SOM for sentiment analysis tasks is at par with other machine learning algorithms.

Growing Hierarchical Self-organizing Maps (GHSOM) are a class of self-organizing maps which are frequently used for training SOMs with text data. Reference [14] trained a classifier using the Enrich-GHSOM to perform aspect-level classification of product reviews. They modeled the hierarchical relationships between the target entity and the aspects as an ontology and used the hierarchical structure of the Enrich GHSOM to compute the overall sentiment score of the target objects taking into account the sentiment polarity of aspects. The highest learning accuracy reported was 63.15%.

SOM has also been employed to visualize customer feedback to gain insights. Reference [15] proposed to use SOM for visual comparison of customer feedback from a geospatial perspective. They rendered sentiment maps and trained a SOM with the color vectors extracted from each of the maps. The trained map depicted that the sentiment scores and key concepts are associated with the geographic position of the reviewers. Reference [16] proposed a method to visualize customer feedback along with the corresponding evaluation values. They used probabilistic Latent Semantic Indexing to reduce the dimensions of the word frequency vectors and used these features to train a SOM. Each node of a SOM was then assigned a shade based on the evaluation value of that node.

SOM being a type of a neural network is inherently suitable for parallelism. This characteristic has been extensively exploited and a number of parallelization schemes have been proposed for SOM for applications like computer vision and data mining. One of the first proposals of a parallel SOM is seen in [17]. The efficiency of parallel SOM was demonstrated with a MATLAB implementation of the proposed algorithm on a single core CPU. During the same time another software-based parallel implementation of a SOM called parSOM [18] was proposed for high-performance text data analysis. The software was implemented on hardware ranging from a dual processor Celeron system to a 64-processor Cray Origin 2000. Thereafter, parallel implementations were proposed using a variety of hardware and libraries like OpenGL [19], MapReduce-MPI [20], OpenCL [21] and supercomputers and VLSI chips [22].

With the advent of NVIDIA's CUDA architecture, there was an increase General Purpose GPU (GP-GPU) programming as CUDA enables the non-graphic programmers to easily work with GPUs. Reference [23] proposed an implementation based on input data segmentation, and CUDA streams

which are sequences of commands that execute in order. They compared the GPU implementation with CPU implementations by varying the SOM network size, input size and the input dimension size. They concluded that speedups due to GPU are not significant in case of small input size or dimension size. The largest dimension tested was 32. Reference [24] evaluated the performance of SOM parallelization on single and multiple GPUs using CUDA, OpenCL and MPI. They reported that on a single GPU, CUDA results in higher speedup as compared to OpenCL, while on multiple GPU setups using CUDA with MPI, they observed that the speedup increases with the number of GPUs for larger datasets but only by a factor of 4. Overall, it was noticed that pure CUDA implementation was the fastest. The largest dimension tested was 8.

Reference [25] proposed a GPU accelerated Batch-SOM for high-dimensional data. The map was organized in a 2D texture format and vertex and fragment shaders were responsible for finding the best matching unit and updating the weights respectively. The authors claimed to achieve a speedup of 15x to 40x in comparison with CPU implementation. The highest dimension tested was 2048.

Reference [26] proposed a parallel implementation of batch SOM by using network and data partitioning methods. The data was partitioned into M maps and each map was broken down to C nodes. Each node was finally broken down to K weight vector dimensions enabling them to parallelize SOM at the level of dimensions and launching $M \times C \times K$ threads. They reported gains up to 20x. However, the largest dimension size tested was 10 and their tests indicated that the gains drop significantly while increasing the dimension size or input size.

A CUDA-based SOM focused on high-dimensional inputs was proposed in [27] A three-step implementation was proposed and the results were reported on synthetic data comprising a maximum of 1000 dimensions.

Recently, [28] proposed a GPU based implementation of sequential SOM. They exploited the cuBLAS library, which contains optimized linear algebra operations for the CUDA architecture. Datasets with vectors of length 16 and 64 were tested.

5. Description Of the Techniques Used

A. Self-organizing Feature Maps

Inspired by the research on brain maps and neural models of self-organization [29], Kohonen proposed the self-organizing feature maps (SOM) in the 1980s [30]. SOMs are unsupervised artificial neural networks which borrow principles from vector quantization and neurobiology to generate accurate low dimensional representations (feature maps) from high-dimensional data. Most of the real-world data, for example, text data or data from bioinformatics, is high-dimensional and SOMs prove to be excellent tools for dimension reduction and aiding in analysis of such data. SOM achieves this with its property of self-organization which produces a set of prototype vectors such that vectors

corresponding to similar points in the input space are topographically closer on the feature map and those corresponding to dissimilar points are farther apart.

The standard SOM architecture consists of two layers – an input layer and an output layer. The input layer consists of n neurons where, n is the number of features in the input dataset, and presents one input vector at a time, to the output layer. While working with text data, the number of neurons in the input layer will be equal to the number of features generated after processing of text data and this layer will present one document at a time to the output layer. The output layer (often) consists of a two-dimensional lattice of neurons. Each of these neurons is mapped to a weight vector of length n.

In each iteration of the training procedure, one input vector is presented to the output layer and all the neurons *compete* to get activated. The weight vector which is most similar to the input vector, is determined as the winning neuron. Based on a neighborhood function, the winning neuron then determines a spatial neighborhood, providing a basis for *cooperation* among neighboring neurons. The neurons in the selected neighborhood then *adapt* their weight vectors in proportion to their distance from the winning neuron, with the winning neuron getting the maximum activation. The weight vectors are updated as follows:

$$w_{ij}(t+1) = w_{ij}(t) + h_{ckij}(t)[x(t) - w_{ij}(t)] \quad (1)$$

Where, ij represent the spatial coordinates of the neuron with which weight vector w is associated, t represents the iteration, h represents a smoothing kernel like Gaussian, ck represent the spatial coordinates of the winning neuron and x represents the input vector.

Using these principles of competition, cooperation and adaption, SOM produces a feature map which accurately represents the input space. For more details on the principles of SOM and tuning it for better results, interested readers are referred to [29].

B. *CUDA*

Graphics Processing Units (GPUs) are massive arrays of processors which operate in parallel. Although they are computationally weaker than CPUs, due to their low cost, a large of GPU cores can be bunched together to perform parallel operations much faster by increasing the throughput. They were originally designed for graphics based applications like rendering of video game graphics. However, researchers quickly realized the ability to perform independent and parallel operations on the GPU which led to the birth of GPGPU (General-purpose computing on Graphics Processing Units). GPGPU is the use of GPUs for non-graphic purposes by modeling the data as images or other graphical forms. While this approach enabled the use of GPU for other parallel algorithms, modeling the problem in terms of graphics processing was extremely difficult.

To enable users to design their applications without going into the details of graphics processing NVIDIA developed CUDA (Compute Unified Device Architecture) which is an API based on the C

language and an architecture for parallel computing on NVIDIA's graphics cards. With CUDA, programmers can develop massively parallel programs for harnessing the power of GPUs without the knowledge of graphics processing terms like vertex shaders and fragment shaders.

The basic flow of a CUDA program is as follows:

1. Load data into the host memory (main memory)

2. Allocate memory for the data in the device memory (global memory of GPU)

3. Copy the data from host memory to device memory

4. Perform computations on the GPU

5. Copy the results to main memory and return the results to the user

The part of an application which runs on the GPU is defined as a special function called kernel. Kernels are run on the GPU by an array of threads on the GPU such that each thread executes the same kernel based on the data assigned to it. Only one kernel can run on a GPU at a time. Threads are arranged in groups called blocks and blocks are grouped into a single grid. Each block of threads have access to fast memory called shared memory and each thread has its own local memory. All CUDA applications can be developed in C/C++. In addition to these languages, some libraries are available in high level languages like Python (PyCUDA) and R (gpuR) as interfaces to the CUDA APIs.

6. PROPOSED APPROACH

The proposed approach is divided into four phases viz,. text preprocessing, feature extraction and representation, implementation of CUDASOM, and segmentation of customer complaints using the developed tool. A schematic of the proposed approach is depicted in Fig. 1.

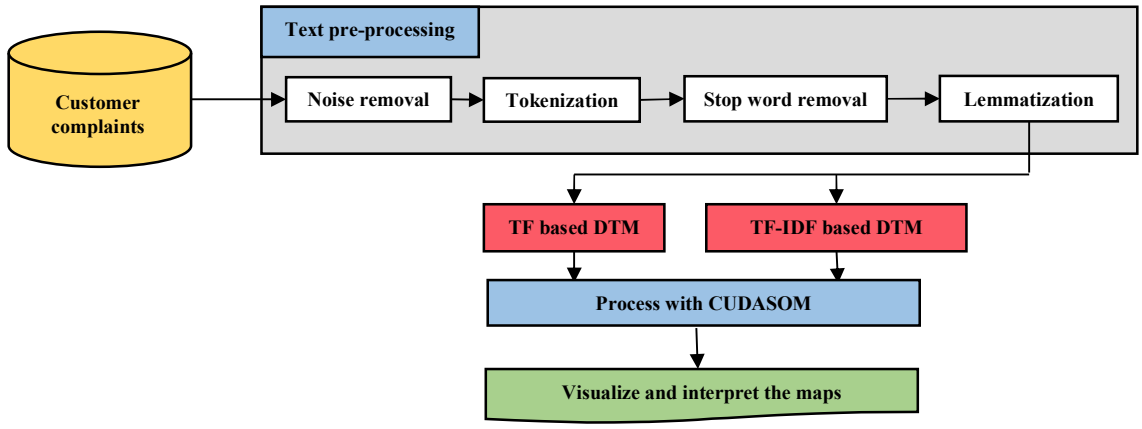


Figure 1 Schematic of the proposed approach

A. Text Preprocessing

This step was performed using the Natural Language Toolkit (NLTK) in the Python programming language. We followed standard text mining procedures like noise removal, tokenization and lemmatization on the whole corpus. In this study, we rely on the occurrence and significance of terms

to capture the severity of complaints as the Euclidean distance metric is used throughout for measuring similarity. Hence, stop words were removed as they contribute to noise in this scenario.

B. *Feature Extraction and Representation*

In this study, we experimented with two feature representations viz. Term Frequency (TF) based DTM and Term Frequency-Inverse Document Frequency (TF-IDF) based Document-Term Matrix (henceforth referred to as DTM).

1) TF based DTM preparation: This representation involves calculating the term frequencies for the bag-of-words extracted from the text corpus.

2) TF-IDF based DTM preparation: This representation is a normalized version of the TF based DTM. Each of the term frequencies calculated in TF based DTM is weighted by the inverse of the occurrence of the term in the entire corpus (IDF). Formally, IDF value of a term *t,* appearing in a document *D*, is calculated using equation (2).

$$IDF(t, D) = log \frac{N}{|\{d \in D: t \in d\}|} \tag{2}$$

C. *Implementation of CUDASOM*

Several CUDA based SOM implementations have been proposed in the literature. However, most of the recent proposals have focused on the batch version of SOM. The GPU based implementation of standard SOM in [28] shows promising performance. In their proposal, the second dimension of block size is equal to the dimension of input vectors. Although this leads to dimension level parallelism and can contribute to significant gains, this design inherently cannot support high-dimensional input as it is limited by the maximum number of threads supported by a block as per the architecture on which the program is being implemented.

We propose a novel scheme to parallelize the standard SOM on CUDA, with a focus on high dimensional text data. In this scheme, the block size is independent of the length of the data vectors making it suitable for working with DTMs. Our implementation assumes a two dimensional map and hence a three dimensional data structure is used to represent the map along with its weight vectors. Initially, we transform the multi-dimensional data and map structures into single dimensional arrays to improve memory accesses on the GPU. All the data transformations are performed within the memory associated with the GPU (henceforth referred to as the device memory) and then the results are ported back to the main memory. This approach significantly eliminates delays due to host-device memory transfers and synchronizations. We also make use of shared memory on the GPU, in addition to global memory, wherever constant data is repeatedly used for comparisons. The training is performed by three kernels.

1) getDistances kernel: In each iteration, an input sample is randomly selected on the host and the corresponding index is passed to this kernel call. This kernel distributes a subset of map units in each block and the threads in each block represent the map units. The input vector is copied to the shared memory for each block to minimize global memory access. Within each block, the distance between the input vector and the map units represented by the threads in that block is computed and stored in the shared memory. Using a parallel reduction within the kernel, the minimum distance and the index of the corresponding map unit are stored on the global memory in an array named *bmu*. Hence, each of the blocks stores a possible candidate for best matching unit for the corresponding input.

2) reduceMin kernel: This kernel uses a parallel reduction scheme on *bmu* to compute the best matching unit. To ensure proper reduction, this kernel is launched with a number of blocks equal to the power of 2.

3) updateWeights kernel: This kernel computes the distances between the map units using their corresponding matrix coordinates. As hexagonal maps are more accurate and better suited for visualization, we implemented CUDASOM with hexagonal neighbors and accordingly hexagonal coordinates were calculate before computing the distances. The updated weights are written directly to the weights array in the device memory without copying them to the main memory. The weights are copied to the main memory only after all the iterations are completed.

D. Segmentation of customer complaints using SOM

In this phase, we trained CUDASOM with each of the dataset separately. The calculation of the number of iterations and the map size for each dataset, was inspired by the heuristics adopted in the SOM Toolbox [31]. Fig. 2 presents an outline of the procedure for calculating the number of iterations and the map size.

Based on the guidelines in [29], we initialized the prototype vectors using the regular, two-dimensional sequence of vectors taken along a hyperplane spanned by the two largest principal components of the DTM. To select the initial learning rate parameter, we performed a series of experiments on benchmark datasets [32] like Iris and Breast Cancer Wisconsin, and fixed the value at 0.1. The Gaussian decay was used to smooth the learning rate and neighborhood radius.

It has been strongly established in information retrieval studies that cosine similarity, which employs dot product, is a more effective similarity measure for analyzing and clustering text documents [33]. According to [29], if the input is normalized to constant length, and if the vector is high dimensional (which is true in our case), the difference in SOMs based on Euclidean distance and dot product is insignificant. As we employed Euclidean distance in our implementation, we normalized the input data before presenting to SOM.

Begin

1. *dataset = D, number of records in the dataset = m, number of dimensions = n, number of neurons in the output layer = munits, number of rows in output layer = nrows, number of columns in output layer = ncols, number of iterations = numItr*
2. Calculate the number of neurons in the output layer as $munits = 5 \times \sqrt{m}$
3. Calculate the principal components from D and sort them in descending order of their corresponding eigen values
4. Extract the first two principal components, pc_1 and pc_2
5. if $pc_1 = 0 \parallel pc_2 \times munits < pc_1$
6. $r = 1$
7. else
8. $r = \sqrt{\frac{pc_1}{pc_2}}$
9. $size_1 = \min(munits, \sqrt{\frac{munits}{r \times \sqrt{\frac{3}{4}}}})$
10. $size_2 = \left\lfloor \frac{munits}{size_1} \right\rfloor$
11. $nrows = \min(size_1, size_2)$
12. $ncols = \max(size_1, size_2)$
13. $nn = nrows \times ncols$
14. $mpd = \frac{nn}{m}$
15. $numItr = \lceil 50 \times mpd \rceil \times m \times 4$
16. Return $nrows, ncols, numItr$

End

Figure 2 Calculation procedure of map dimension and number of iterations

An outline of the procedure for calculating the map dimensions and number of iterations is depicted in Figure 2. We trained CUDASOM with the TF and TF-IDF based DTMs separately for each dataset. The trained prototype vectors were then visualized with a similarity coloring scheme [34]. For each experiment, the quantization error and speed were computed. To ascertain that CUDASOM is producing the correct results, we performed experiments on all datasets using the same settings with a CPU implementation of the SOM.

7. EXPERIMENTAL SETUP

We demonstrated the effectiveness of the proposed CUDASOM on customer complaint datasets of four leading Indian banks. The customer complaints were posted on an online public complaint forum. Details of the datasets and the experimental environment are presented as follows.

A. Dataset details

The datasets were obtained from [35]. The authors crawled the web and collected 749, 1108, 758, and 1123 consumer complaints and bank executive responses from www.complaintboard.in on four

banks namely Axis Bank, HDFC Bank, ICICI Bank, and SBI respectively. The first three are leading Indian private sector banks, whereas SBI is the biggest public sector bank in assets. The datasets were prepared during the period of October-December 2014 and hence they contain complaints posted before December 2014, on the specified forum. Details of the datasets are presented in Table 1.

Table 1 Details of Datasets

Bank Name	No. of documents	No. of complaints considered	Features Generated
Axis Bank	749	513	3917
HDFC Bank	1108	637	4545
ICICI Bank	758	440	4268
SBI Bank	1123	676	4470

B. Preprocessing steps

As the data obtained was raw data, we performed the required preprocessing to prepare the data for analysis. Firstly, we dropped bank executive responses, uninformative documents, duplicates and customer requests to make sure that the dataset uniformly represents customer complaints. Secondly, the data was cleansed to remove noisy features like contact information, email metadata, special characters and words of length less than 3. We also corrected the misspelled words. After the preprocessing, 513, 637, 440, 676 complaints were considered for Axis, HDFC, ICICI and SBI banks respectively.

C. CUDA setup

All the experiments have been performed on a computer equipped with Intel® Xeon (R) E5-2640 v4 CPU clocked at
2.4 GHz, 32 GB RAM and an NVIDIA® Quadro® P5000 GPU with 16 GB RAM. The GPU is powered by the NVIDIA Pascal architecture and is equipped with 20 microprocessors holding 128 CUDA cores each. We used CUDA v8.0 with nvidia-384 drivers installed on Ubuntu v16.04 operating environment.

All the models were implemented in Python v3.5.2 and the GPU programming was implemented using the PyCUDA [36] module which helps in accessing NVIDIA's CUDA parallel computation API from Python.

Several implementations [37][38] of SOM employing industrial-grade machine learning frameworks like Tensorflow are available on the open source platform. These software provide vanilla implementations of SOM and do not employ the heuristics proposed in the literature for proper training of SOM. Moreover, they do not provide visualization capabilities like similarity coloring which are more intuitive for visualizing a dataset of customer complaints. Also, as machine learning frameworks are positioned on top of several layers of abstraction they leave very little scope for tuning the core architecture of SOM, As such these implementations were not readily suitable for this study

Hence, we selected PyCUDA as the base framework and implemented the architecture from scratch, incorporating the recommended heuristics and visualization tools suitable for this application.

8. Results and Discussion

As this work proposes two ideas, we discuss the results of each of the ideas separately.

A. *Segmentation of customer complaints using CUDASOM*

We evaluated the results based on two soft criteria – the ability of the map to provide a visual aid in the analysis of the complaints and effective topology preservation of the data by CUDASOM, i.e., complaints related to one, or similar products get mapped to neighboring nodes.

Firstly, we compared the results across the two feature representations adopted. Fig. 3 depicts the map generated by the TF based DTM of HDFC dataset and the map generated by the TF-IDF based DTM of the same dataset.

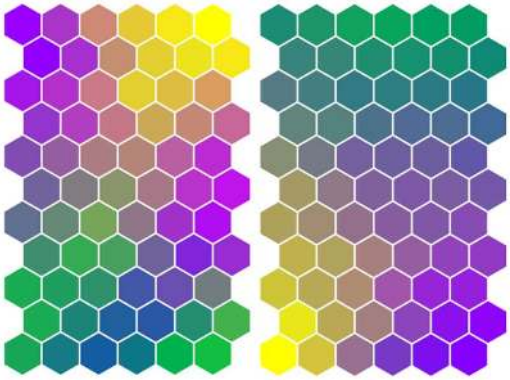


Figure 3 Maps obtained using TF and TF-IDF representations of HDFC bank dataset respectively

- The map obtained by the TF-IDF feature representation depicts clearly distinct regions which are uniformly arranged, while the map obtained by the TF representation depicts split, non-uniform regions, which makes interpretation difficult.
- The complaints mapped to the nodes of the map obtained using the TF-IDF representation also demonstrated strong topology preservation. All the complaints related to *loans* are mapped to the nodes in purple (bottom right region of the map) and within them complaints related to *car loans* are mapped to the same or neighbouring nodes. This property could not be observed in the map obtained by the TF representation. On the map, the regions in purple represent different complaints in different regions of the maps and similar complaints are not always mapped to the same region or same node. These two observations are true even for the other three datasets. Therefore, TF-IDF emerged as a better representation technique. It is because of the richness of the information considered by the TF-IDF vis-à-vis TF.

Secondly, we performed an in-depth analysis of the feature maps obtained using the TF-IDF based DTMs of all the four datasets. We analyzed the results to determine if CUDASOM is able to a) produce a product-wise segmentation of the complaints and b) produce a severity-wise segmentation of the complaints. In order to do this, we assigned each document vector to the best matching vector on the trained map. Then, the product-wise segmentation was verified by retrieving the documents mapped to the nodes on the map and studying the distribution of the complaints. In order to verify whether the algorithm can group the complaints based on severity we proceeded as follows. Complaints were labelled as "moderate" or "severe" (symbolically as 1 or 2 respectively) independently by the first author along with two additional human annotators and final labels were assigned by performing a simple majority voting of three sets of labels. After training CUDASOM with unlabeled data and obtaining the feature map, we programmatically marked each neuron of the feature map with the labels of the documents, provided by the human annotators, mapped to it. Broadly, it was noticed that CUDASOM was able to achieve product-wise arrangement of the complaints along with the corresponding severity tagged thereof. The following observations were drawn from the maps:

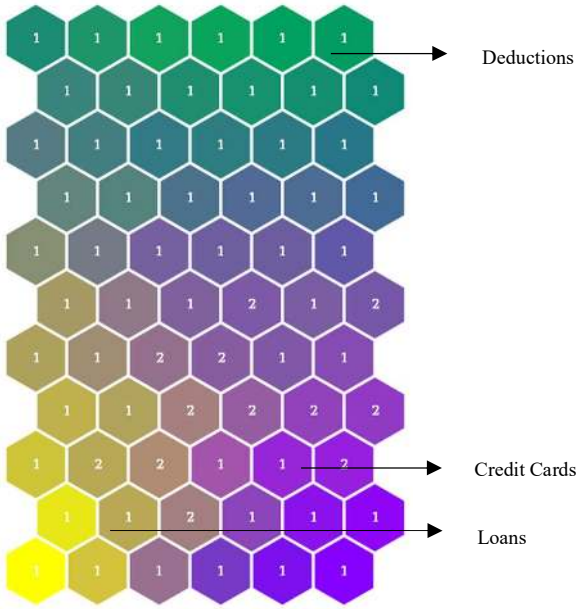


Figure 4 Map obtained using the TF-IDF representation of HDFC bank dataset

- The map corresponding to HDFC bank dataset (seen in Fig. 4) depicts three different regions – green, yellow and violet. The green region represents all the complaints related to unexplained deductions in bank accounts which are usually charged by the bank on failure to maintain quarterly average balance or, as annual maintenance balance and is not noted by the customer. The yellow region corresponds to credit cards and violet region corresponds to complaints related to loans. Within the nodes representing loans, moderate complaints about NOC not being sent out on repayment of loans or incorrect information and severe complaints from frustrated customers who are waiting for the loan to

be disbursed are well segregated, demonstrating CUDASOM's ability in further segmenting complaints related to a product based on the severity of complaints. Within a region, different shades of the same color represent different aspects of the product complaints. The two bright yellow nodes on the bottom right region of the map correspond to complaints regarding *credit card disapproval*. Some complaints which pertain to less frequent products like *customer care* are in the intersection of the three regions.

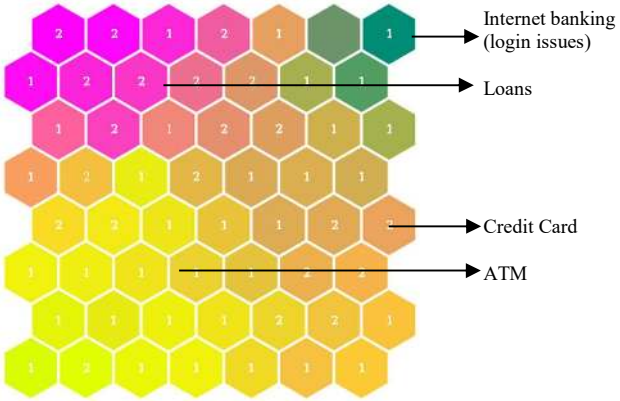


Figure 5 Map obtained using the TF-IDF representation of Axis bank dataset

- The map corresponding to Axis bank dataset (seen in Fig. 5) depicts four different regions. The top left region in pink represents *loan* related complaints and the green region on the right represents a strikingly homogenous set of complaints all related to a *login issue* on the bank's website. The region in orange below it represents *credit card* related complaints. As depicted in the case of HDFC dataset, we can observe distinct nodes linked to severe and moderate complaints within the region representing a single product – *credit card*. Severe complaints like incorrect billing and about recovery calls and moderate complaints like different card being sent out have been distinguished by CUDASOM. The region in the bottom left section depicts complaints related to *ATM*.

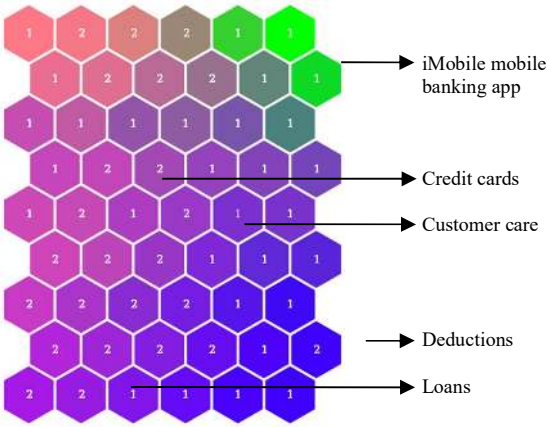


Figure 6 Map obtained using the TF-IDF representation of ICICI bank dataset

- The map corresponding to ICICI bank dataset (seen in Fig. 6) depicts roughly four to five different regions in which three regions are distinct but the remaining regions are ambiguous and cannot be

distinguished visually. Complaints regarding failure of recharge from the bank's *mobile banking app* are mapped to the green region in the top right section of the map. The bottom right region in blue corresponds to charges and deductions from *bank accounts*. In this region, moderate complaints or enquiries about uninformed deductions and severe complaints threatening to leave the bank due to the same reason have been linked to separate nodes on the map. The region in light pink and dark pink corresponds to complaints related to *credit card* and specifically regarding calls from *recovery agents* or *fraud calls*. The region in the bottom left corresponds to *loan* and the central portion of the map corresponds to complaints regarding *customer care*. It is important to note that although the left half of the map along with the central portion is not visually distinguishable into separate products, even in this map, the complaints have been accurately grouped by SOM in a topology preserving manner. The regions could be indistinguishable in color due to similar wording in the sentences.

- The map corresponding to SBI bank dataset (seen in Fig. 7) depicts four different regions. The top left region in yellow represents *mobile and internet banking*. SOM also remarkably segregated sub-

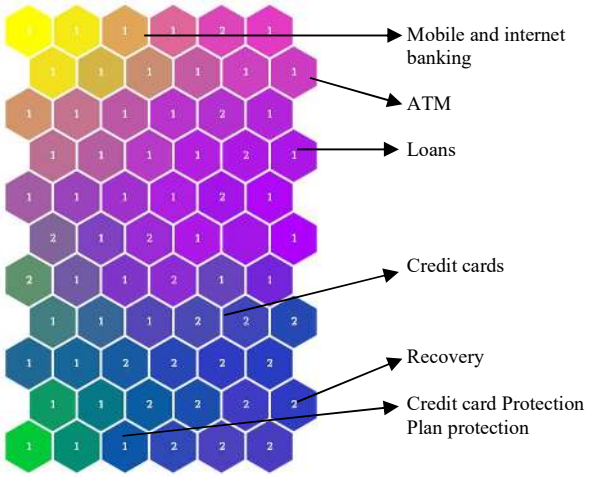


Figure 7 Map obtained using the TF-IDF representation of SBI dataset

clusters within this region. The bright yellow nodes in the top left corner represent a series of complaints about failed *prepaid phone recharges* made from the *mobile banking app*. The top right section in pink represents complaints about *ATM* transactions. The purple region below it mostly represents complaints about *loans* and few infrequent complaint classes like complaints about deposit of pension in the *accounts*, service at the *branch* etc. The bottom section in blue completely represents *credit card* related complaints. CUDASOM clearly separates severe complaints from moderate complaints in this region. Moderate complaints like customers not being able to generate a PIN for their credit card or not receiving their card on time are mapped to the nodes on the left. Within this section, the region on the bottom left corner in green is related to *Credit card Protection Plan* (CPP), a credit and debit card protection service and mostly moderate complaints are associated with this product. On the other hand, more severe complaints like instances of harassment by recovery team, credit card fraud, etc., are mapped to the nodes on the right, which are labeled as severe.

Descriptive analytics of customer complaints by the proposed model, therefore, provides the advantages of identifying the different products/services which the customer is complaining of, the impact of the affected product/service, any trending issues suddenly affecting a large customer base. This will help the bank executives in charge of customer relationship management department in prioritizing the redressal of complaints.

B. Performance of CUDASOM

As mentioned in section 6, to confirm that the results of CUDASOM are accurate, they were compared with the results of experiments performed using the CPU implementation of SOM on all datasets using the same settings. We verified the accuracy by comparing the quantization error and the map generated in the respective experiments. Quantization error is used as a metric to tune the map and as such can be compared only when the dataset and the map sizes are the same.

Table 2 Quantization Errors and Execution Times

	Dataset	Dataset dimensions	Quantization Error (GPU)	Quantization Error(CPU)	Time (GPU) in secs	Time (CPU) in secs	Speed up
Term Frequency based DTM	Axis	513x3917	0.80627	0.80996	22.12	915.08	41.37
	HDFC	637x4545	0.80653	0.81383	32.57	1380.82	42.40
	ICICI	440x4268	0.81625	0.82011	19.94	837.02	41.98
	SBI	676x4470	0.82042	0.82578	27.25	1194.57	43.84
TF-IDF based DTM	Axis	513x3917	0.91366	0.91651	22.63	960.74	42.45
	HDFC	637x4545	0.91562	0.92056	31.48	1412.07	44.86
	ICICI	440x4268	0.91467	0.92026	23.66	985.47	41.65
	SBI	676x4470	0.91769	0.92182	26.73	1254.52	46.93

Table 3 Execution Time Ratios with Increasing Map Sizes

		Map Size					
		16x16	***32x32***	***64x64***	***128x128***	***256x256***	***512x512***
Reference [28]	**Time (in secs)**	15.50	31.20	92.30	331.00	1270.00	4900.00
	Ratio of increase in time	-	2.01	2.96	3.59	3.84	3.86
CUDASOM	**Time (in secs)**	34.25	32.81	33.37	38.40	111.03	431.38
	Ratio of increase in time	-	0.96	1.02	1.15	2.89	3.89

- As seen in Table 2, the difference in the quantization errors between CPU and GPU implementations is very small. Differences of this magnitude are expected even in multiple runs of the same program as the initializations and the order in which the samples are presented to SOM would be different. The map obtained from CUDASOM (seen in Fig. 5) and the map obtained from the serial version (seen in Fig. 8) also differs very slightly due to the same reason. An average speed up of 43x is obtained with CUDASOM as compared to the CPU implementation of SOM. The minimum speedup achieved was 41x and the maximum speedup was about ~47x. Although it is a standard practice to compare the time taken by the GPU application with the CPU version, we believe it is unfair as the optimal implementation of CPU is subjective and debatable [27]. As compared to [27] our algorithm has been

able to achieve higher peak speedup on input data of 3-fold higher dimensions and the results have been validated on a real-world dataset. To establish that our algorithm performs well with larger map sizes, we ran tests similar to [28] and compared the ratio of time taken with an increase in dimension size of the map. Same data size and number of epochs have been considered. The length of the weight vectors considered is 64. It can be noticed from the table that the ratio of time taken with an exponential increase in map size increases gradually in case of our implementation as compared to [28].

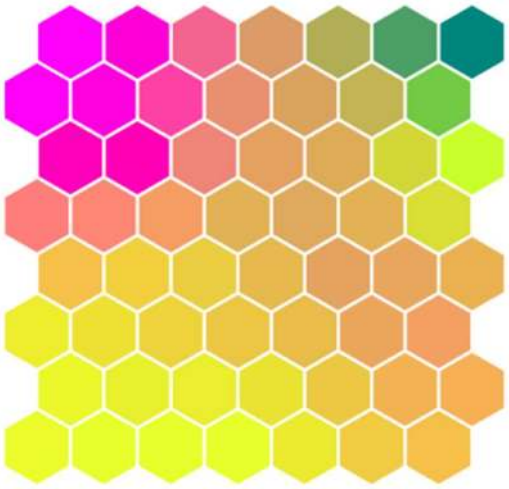


Figure 8 Map obtained using the TF-IDF representation of Axis bank dataset with standard SOM

- One limitation of the proposed algorithm is that it is limited by the size of the device memory. Before the training starts, the initialized weight vectors of the map and the data matrix is copied to the device memory and is copied back to the main memory only after the execution of the training phase is completed. Although this approach drastically reduces the CPU-GPU data transfer overhead and synchronization, it limits the size of the data that is processed. In future we will work on alleviating this limitation.

9. Conclusions And Future Directions

In this study, we propose a visual sentiment analysis framework by employing self-organizing maps for analysis of customer complaints. We compared the performance of the model with two vector space model representations of the documents, viz., term frequency representation and TF-IDF representation. We also propose a parallel version of the algorithm for faster processing of complaints and compared its performance with the serial version. The model generated a more distinct segmentation with TF-IDF based DTM as compared to the term frequency based DTM. Visualization depicts that CUDASOM is effectively mapping the complaints related to different products/services into distinct groups. The algorithm also demonstrates a significant speedup as compared to the serial version and can be clearly considered as useful for faster analysis of data with large dimensions.

In future, we will study the performance of the model with more advanced feature representations like Paragraph Embedding and hashing TF-IDF. We are also interested in studying the performance of the model with larger datasets.

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A Social Media Complaint Workflow Automation Tool using Sentiment Intelligence

Akshi Kumar, *Member, IAENG,* Vikrant Dabas

***Abstract*—Providing a great, consistent & responsive customer service is imperative to build any successful business. This is primarily because high service quality advocates customer satisfaction which is directly associated to business revenue generation. The business industry has adopted social media as a platform to generate buzz. But listening to the social media & catching the negative word of mouth is equally important. The volatile banking industry has also embraced the social media culture for implementing the customer service aptly. Customers are using Social Media as a complaint channel, but there is no automated process that allows the social media tools such as Facebook, Twitter and LinkedIn to capture, inspect, appropriately manage and process complaints, which is indispensable to a quick response strategy. This paper introduces an end-to-end complaint handling tool that automatically captures incoming posts and/or comments to the posts at the social media source, uncovers the post type as complaint based on sentiment and finally directs the complaints to various departments of the bank where they can be instantly assessed, prioritized and responded. The problem is envisaged as a classification task which is resolved by introducing a tool based on machine learning algorithms. It implemented to social media posts of HDFC Bank Limited, India & a comparative performance analysis for same has been performed.**

***Index Terms*—Customer, Complaints, SERVQUAL, Sentiment, Classifier**

I. INTRODUCTION

CUSTOMERS are the foremost partakers to gauge business performance, as success inevitably depends on customer satisfaction & achieving the right customer service quality. This implies that customer satisfaction is the key business performance indicator & better quality of service has a positive influence on satisfaction, directly contributing to the business revenue.At the same time, the pervasive & voluminous social media is evolving as a potential source of information for business intelligence. By analysing the large volumes of data available on social media, companies can gain deep insights into customer behaviour, sentiments and needs which may facilitate improvement of both product and service. The banking industry has also embraced the social media technology as an 'enabler' &'driver' of business. As customers increasingly use social media to share opinions on financial products and services, banks must listen, learn and respond, as well as incorporate their social activities into their overall corporate strategies. In most cases, this requires banks to rethink their core business strategies to make them more customer-centric [1].

The provision of customer service via social media channels has become axiomatic, especially due to its cheap, convenient and public nature. With approximately two billion people using social media around the world, a disappointed, unhappy or frustrated customer may choose it as a medium to voice his dissatisfaction in the form of complaint or negative word-of-mouth. According to recent reports [2], platforms such as Twitter [3] and Facebook [4] are being viewed as the perfect vehicles for customers to escalate complaints. The problem manifolds due to the customer's anticipated response time. Nearly, 42% customers expect response within 60 minutes [5]. Thus, the need of the hour is to intelligently support the customer service department to enable quick & transparent response. Having a dedicated Social Media Manager, who keeps checking the media platforms every ten minutes may seem a solution but in a B2C industry like banking, manually keeping track &monitoring the velocity of complaints in a time-bound manner will be tedious &inefficient. Moreover, a documented escalation policy will be desired to map the complaint to the respective department, the employees & their contact details. Doing all this manually will compromise with the higher expected responsiveness, which has recently been hyped as the metric for customer satisfaction. Motivated by this, we in this paper propose a Social Media Complaint Workflow Automation tool. The idea was to draw correspondence to thewell-known SERVQUAL model (5-dimensional: Tangibles; Reliability; Responsiveness; Assurance; Empathy) [6, 7] which helps evaluating service quality in banking industry, as studies have indicated that out of its five dimensions, responsiveness & empathy can be improved with the help of social media analytics. These service quality factors (SERVQUAL model dimensions) are defined as follows:
Responsiveness: refers to the willingness of service providers to help customers and provide prompt service.
Empathy: refers to the provision of caring and individualized attention to customers.

The proposed Social Media Complaint Workflow Automation tool depicts the online complaint management as a two-level classification problem that uses sentiment analysis on the social media posts to quickly resolve complaints. Sentiment analysis or Opinion mining, as it is sometimes called, is one of many areas of computational studies that deal with opinion oriented natural language processing [8].It normally involves the classification of text into categories such as "positive", "negative" and in some cases "neutral".The tool expounded in this paper, first automatically captures incoming posts and/or comments to bank's post at the social media source. Not all posts are

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complaints and so as the first classification task, we uncover the type of post into complaints (negative feedback), queries (neutral) or positive feedback/irrelevant posts by perceiving the sentiment and then finally direct the complaints to various departments of the bank using another classifier so that the workflow is complete for a prioritized & quick response. The tool has been implemented to social media posts of HDFC Bank Limited, India [9] (Facebook & not Twitter, because tweets about complaints were very less as compared to the posts on Facebook page) & tested for various classification models. A performance analysis interpreting the results is also presented which justifies the need to automate the process that reduces the service-quality gap & incorporates sentiment intelligence within the business intelligence model.

The paper is organized as follows: Section 2 gives a brief explanation of related background work in thisdirection. Section 3 describes the proposed workflow model of the proposed Social Media Complaint Workflow Automation Tool using Sentiment Intelligence. Section 4 gives a brief overview of the implementation and analysis using a case study of HDFC Bank. Section 5 concludes the research undertaken while discussing the potential applications & directions of future work.

II. Related Work

According to Capgemini [10] worldwide,the social network user base is expected to increase from 1.47 billion in 2012 to 2.55 billion in 2017. Alsosocial media yields such a wealth of customer-based, customer-voiced information which banks can harness to better understand &enhance the overall customer experience.Studies aligning customer satisfaction to the levels of service quality are available much in literature. Gautam and Singh [11] as well as Lohani and Bhatia [12] have used SERVQUAL model to identify the service quality gaps in banking sector. Moreover, according to them,Empathy & Responsiveness,are found to be the most vital and strategic determinants of service quality and customer satisfaction for banks.Santhiiyavalli [13] also concluded that among the five dimensions 'Responsiveness' and 'Empathy' are the major factors responsible for customer satisfaction. So efforts should be made to reduce the service quality gap by improving these factors. Another study conducted by Rathee, Deveshwar and Rajain[14], concluded that the highest gap was found in the dimension of reliability and empathy. And banks have to reduce this gap giving individual personal attention to understand customer specific needs. Moreover, results of a study by Brahmbhatt and Panelia[7] provide evidence that the SERVQUAL dimensions are a useful tool to predict overall satisfaction.

Social media should be embedded into a bank's entire ecosystem because it impacts numerous areas, such as customer relationship management, risk management, product and service design, customer education, etc. [1]. Sentiment analysis of micro blogging or social networking websites has been a wide area of research in the present decade [15]. But its usage hasn't been much extended in the banking sector to improve the service quality and customer satisfaction. Most of the process, if ever done, for improving the service quality and customer satisfaction with the help of social media has been done manually rather than opting for an automated process. So to counter this problem, this paper suggests an automated method for capture individual posts at the social media source, classify & identify the complaints and route them to the appropriate resolution owners, supervisory levels and management. The next section describes the proposed workflow tool.

III. The Proposed Social Media Complaint Workflow Automation Tool

The Social Media Complaint Workflow Automation tool proffered here depicts an online complaint management as a two-level classification problem that uses sentiment analysis on the social media posts and/or comments to quickly resolve complaints. We define Sentimental intelligence as a conjunction of intelligence, empathy and sentiments to enhance opinion and comprehending social interactive dynamics. It is the capability of voicing and monitoring public opinion enabling apt classification of sentiments which guides and impacts thinking attitude and aptitude. Instituted on this the proposed tool first automatically captures incoming posts & comments to bank's post at the social media source and classifies them as enquiries, positive feedback/irrelevant post or complaints by perceiving the sentiment and then finally directs the complaints to various departments of the bank using another classifier so that the workflow is complete for a prioritized & quick response. For both levels of classification, Naïve Bayes Classifier, its 3 variants, namely Multinomial Naïve Bayes, Bernoulli Naïve Bayes and Gaussian Naïve Bayes & Support Vector Machine are employed. The tool is implemented to the social media posts of HDFC Bank Limited, India & tested for various classification models.The following figure 1 illustrates the basic workflow architecture of the proposed tool.

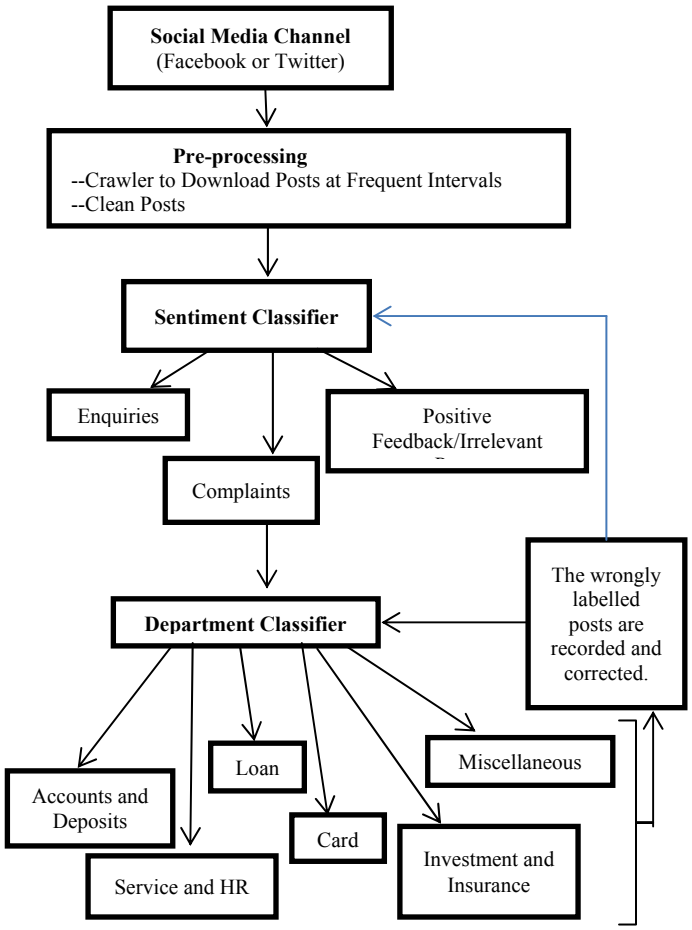


Fig 1. The proposed complaint workflow automation

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The following sub-sections expound the details pertaining to each component of the proposed workflow:

A. *Pre-processing*

Large publically available social media datasets are crawled to fetch data using explicit APIs. Data extracted in its original form is hard to use for analysis directly. It is often incomplete, noisy & inconsistent [16]. Thus, the pre-processing consists of the following two steps:

1) *Crawling social media channels to capture posts*

According to the Internet Society's "Global Internet Report 2015" [19], there are more than 3 billion people online globally who use Internet for communicating, sharing, finding information, publishing & commerce amongst others. Statistics also reveal that more than 2 billion people worldwide have social accounts (Facebook, Twitter etc.). The Social networking websites like Facebook and Twitter which are a substantial, unparalleled sourceof user-generated content are easy to crawl and extractingonline posts from them can be done in a straight forward manner.To get large publically available Facebook datasets, we use Facebook's graphAPI.We form a URL & hit the graph API to receive JSON object responsescontaining data about Likes, talking about etc. Required data is extracted from these JSON objects and saved in a separate file for further processing.To get large publically available Twitter datasets, we use Twitter API. In order to make authorized calls to twitter's APIs, the application must first obtain an OAuth access token on behalf of a Twitter user or can issue Application-only authenticated requests when user context is not required. For implementation & analysis Facebook was selected over Twitter, because tweets about complaints were very less as compared to the posts on Facebook page.

2) *Cleaning Module*

Data cleaning for quality is significantly desirable for higher accuracy & better results. The data extracted from Facebook containsmany undesirable symbols, words, URLs, E-mail ids, etc. that may slow down the applied algorithms or even make them perform incorrectly. Hence, it becomes necessary to incorporate a data cleaning module.The following strategy has been identified to clean the downloaded data for processing:

- Removal of # tags: Facebook has incorporated hashtags, which works on personal profile posts, fan page posts, group posts, event posts, and all comments.
- Removal of "@": @ symbol *is* used for tagging or to "mention" someone on Facebook.
- Removal of URLs & E-mail ids.
- Removal of numbers, digits, alphanumeric data, punctuations, symbols.
- Expanding short-forms (acronyms), if any.
- Removal of non-English posts by encoding them into ASCII and if encoding produces any exception, then reject them, otherwise, select the posts.
- Removal of stop-words.

The original posts& the cleaned posts are kept temporarily at each step. The cleaned post is used for processing & the original post are stored for further use.

B. *Classification of posts*

Cleaned posts and/or comments to the post are labeled and trained over the first classifier (Sentiment Classifier). These are classified into three major categories, namely:

- *Complaints*: an expression of dissatisfaction on a consumer's behalf to a responsible party.
- *Inquiries*: a request for information from the concerned department of the organization.
- *Positive feedback/ irrelevant posts*: posts that are basically feedback from customers that show a positive response or a set of posts that are either for advertisement purposes or are completely irrelevant.

The sentiment classifier extracts complaints from all the posts and the department classifier separates the complaints into different department files. Each file issent to its particular department. The department class files which are created are explained as follows:-

- *Accounts and Deposits:* This department handles everything from opening to closing of various types of accounts including Salary Accounts, Current Accounts, and Saving Accounts etc.
- *Loans:* This department handles all the work related to processing all kinds of loans, be it a personal loan or car loan.
- *Cards:* This department is responsible for issuing credit/debit or any other type of cards and maintains information about them.
- *Investment and Insurance:* Getting a life insurance to managing Mutual Funds, all the related activities are handled by this department.
- *Customer Service:* This is the department which supports the customers to approach all the other departments. It provides services such as Net Banking, Phone enquiry.
- *Miscellaneous:* All the other departments like Recovery, Lockers, etc. can be grouped into this category. Moreover, there are departments that are unique to a particular bank. So those departments are also grouped into this category.

These sub-categories don't exactly reflect one department (like the sub-category Miscellaneous) but tell us to which department these complaints should be forwarded too.

The sentiment and department classifiers aretrained over a labeled data set. Once the complaints have been labeled department-wise, the original posts that were mapped to the cleaned posts during pre-processing are sent to the respective departments for further action & proceedings and the cleaned posts are discarded. In case, they reach a wrong department, i.e., if it is found to be incorrectly classified, the posts are labeled/marked with the correct department and are used to update the classifiers.

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For both classifiers, the learning algorithmsconsidered were Naïve Bayes (NB), its 3 variants, namely, Multinomial Naïve Bayes (MNB), Bernoulli Naïve Bayes (BNB) and Gaussian Naïve Bayes (GNB) & Support Vector Machine (SVM) [17]. Naïve Bayes classifier uses a bag-of-words [18] approach for sentiment analysis on the content. The Bernoulli Naive Bayes can be used when in our problem the absence of a particular word matters. For example Bernoulli is commonly used in Spam or advertisement detection with very good results.

The Multinomial Naive Bayes is suitable for classification with discrete features &is typically used when occurrence of the word matters more than the frequency. For example it does not really matter how many times someone mentions the words "bad", "awful", "terrible", but rather only the fact that he does.ABinarized Multinomial Naive Bayesis used for the sentiment classification (first classifier), i.e., it has binary weighting function where the value 1 means that the word occurs in the particular document, and 0 means that the word does not occur in this document. A tf-idf Multinomial Naive Bayes is used for the department classification (second classifier). The tf-idf approach assumes that the importance of a word is inversely proportional to how often it occurs across all documents.The Gaussian Naïve Bayes & support vector machine were also used for a comparative performance analysis.

IV. Implementation and Performance Analysis

To clearly illustrate the effectiveness of the proposed method, a case study of HDFC Bank India's Facebook page is presented. The following figure 2 shows the HDFC Bank, India Facebook page with some sample posts.

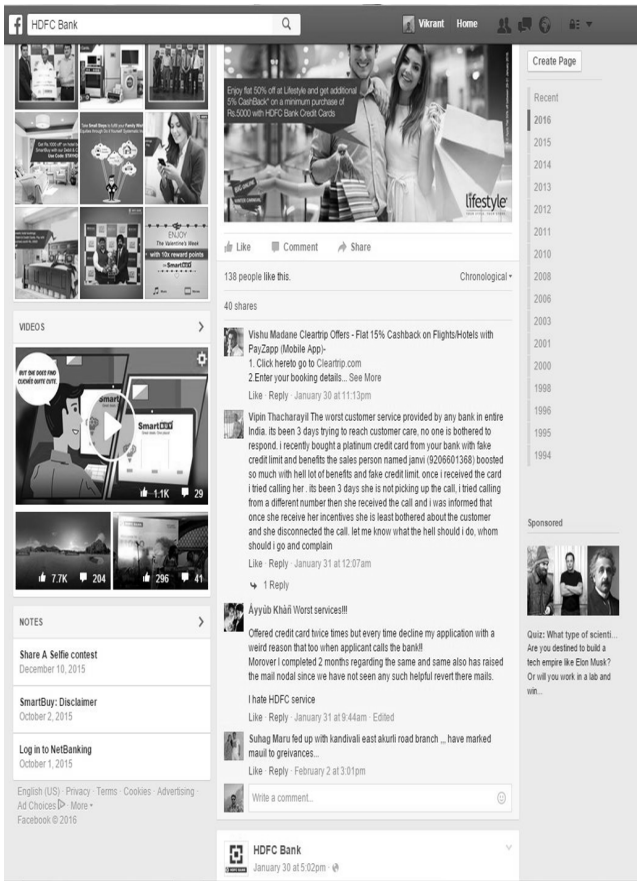


Fig 2. Snapshot of HDFC Bank, India Facebook page with some sample posts.

Each of the components of the workflow is exemplified to demonstrate how it actually works by firstly identifying the complaints from the Facebook post & then referring it to the appropriate department which can cater a quick response, thus automating the entire process in an efficient manner.

A. *Pre-processing*

1) *Crawling social media channels to capture posts*

To download the Facebook data, an access token was generated by registering as a developer on Facebook. The next step was to make an HTTP GET request to the Facebook graph API using the urllib.request module in python. As a response, a JSON dump of the data was received and useful information (user who posted, post text, time of posting) wasextracted & stored in a CSV file.

2) *Cleaning*

The cleaning process as discussed in the previous section, involved removing hashtags like #HDFCBank etc., tagging symbol @ (@HDFC_Bank), URL supporting additional information (links to bank's website) and any e-mail ids that are included in the post, example: creditcards@hdfcbank.com. It also scrubs numbers (account numbers/mobile numbers/dates etc.) and very common stop-words like "a, an, the, this, that". Short-forms like A/C, a/c (account) & CC (Credit-card) are expanded too.

B. *Classification*

As discussed in the previous section, for both levels of classification, Naïve Bayes Classifier (NB), its 3 variants, namely Multinomial Naïve Bayes (MNB), Bernoulli Naïve Bayes (BNB) and Gaussian Naïve Bayes (GNB)& Support Vector Machine (SVM) have been employed. The results from the first classification task for finding sentiment from the post are shown below in table 1:

TABLE I
SAMPLE RESULTS OF SENTIMENT CLASSIFIER

POST	SENTIMENT	CLASS
This is a warning to #HDFC #HDFCBank which has the worst credit card division and i am going to file a case against them for torturing me in a credit card case. When you people have already credited an amount of 92k from my bank account six months back making a lean on my account. This was the final payment and as you people told me that i will be getting a NOC. My credit card was blocked from 2-5 years approx. Even now i am receiving credit card statements every weak. This is really ridiculous. I request you to issue me a letter for no dues. Ajay Katnoria +91-9892622230 katnoria.ajay@gmail.com	NEGATIVE	COMPLAINT
This is the worst bank i have ever seen. Today when i payment on amazon app.the amount was deducted from my account. But	NEGATIVE	COMPLAINT

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amazon get any payment from my side. When i called at hdfc customer care they told me it will take 2-4 days. How could be possible...that is my money. They have to send in my account in an hour... And even thy didnt called me that the amount is deducted so thy will refund me.		
What documents required for converting the Savings account in to salary account..I have offer letter, and pay slip with me..pls comment	NEUTRAL	ENQUIRY
HI, Is payzapp available for windows phone?	NEUTRAL	ENQUIRY
Team engage4more partners HDFC Bank in their brand new performing arts talent based contest 'HUNAR' . A PAN India turnkey Mandate includes conceptualisation, format creation and production. Great to see this wave of corporate belief in engaging employees around their passion. #hr #employeeengagement #india #performingarts pics from regional finale west. India finale follows.	POSITIVE	POSITIVE FEEDBACK /IRRELEVANT
Best Bank in India. Finest banking service.Outstanding performance very very Happy customer.	POSITIVE	POSITIVE FEEDBACK /IRRELEVANT

The department classifier categorizes the complaints acquired into various departments of the bank for further necessary action on the complaints. A snapshot for the complaints fitting to the Credit Card department is given below:

Credit Card (3).csv

1 Dear HDFC, Please note that i applied for HDFC Credit Card on 06/10/2015 and got Reference No: 15100519293300S2. I got all verification done at my office. I got multiple calls at for verification too and your representatives informed that the card will be approved. Today got checked in HDFC Link showing track your card application. Found the reason as follows. Application Reference Number 15100519293300S2 Status DISAPPROVED Remarks DISAPPROVAL OF THE CREDIT CARD APPLICATION DOES NOT REFLECT IN ANYWAY ON YOUR CREDIT STANDING OR YOUR FINANCIAL STABILITY where in none of your representatives took up the initiative to call and inform why it got disapproved. (2nd time the same thing happening) Dilip (09600141649) , executive called and we have given all papers and verification also done positively. please be noted that this is the second time the same thing is happening and i was informed that i will be receiving a letter which never came even after 6 months. Awaiting for a quick response.|2015-10-19T05:02:03+0000

2 The worst customer service provided by any bank in entire India. its been 3 days trying to reach customer care, no one is bothered to respond. i recently bought a platinum credit card from your bank with fake credit limit and benefits the sales person named janvi (9206601368) boosted so much with hell lot of benefits and fake credit limit. once i received the card i tried calling her . its been 3 days she is not picking up the call, i tried calling from a different number then she received the call and i was informed that once she receive her incentives she is least bothered about the customer and she disconnected the call. let me know what the hell should i do, whom should i go and complain

3 Hi, I am a user of your HDFC credit card from years and have generally been very satisfied with your level of services. For

Fig 3. Snapshot of result from department classifier with complaints fitting to the Credit Card department

C. *Results& Performance Analysis*

This section discusses the result & does a performance analysis for both the classification tasks with all learning algorithms exercised.

The following table 2 shows the performance analysis of the sentiment classifier(firstclassifier) on the basis of accuracy and time. The Multinomial Naive Bayes Classifier gave the highest accuracy. Bar graphs are shown for the same in figures 4 and 5 respectively.

TABLE II
ANALYSIS OF SENTIMENT CLASSIFIER

Algorithm	Accuracy(%)	Time(sec)
Naive Bayes Classifier	78.667	0.1562
Multinomial Naive Bayes Classifier	83.667	0.0312
Bernoulli Naive Bayes Classifier	79.667	0.0468
Gaussian Naive Bayes Classifier	67.667	0.0156
Support Vector Machines	76.333	0.0312

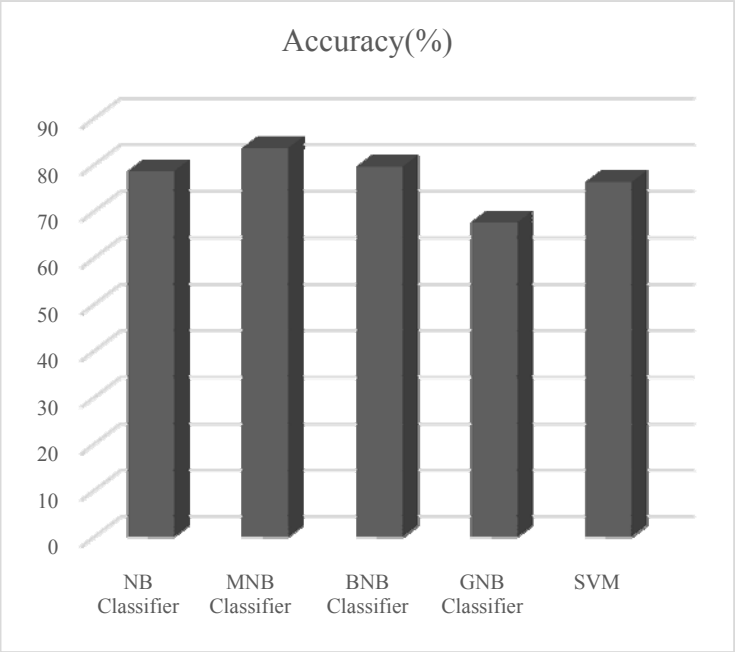


Fig 4. Accuracy Analysis of Sentiment Classifier

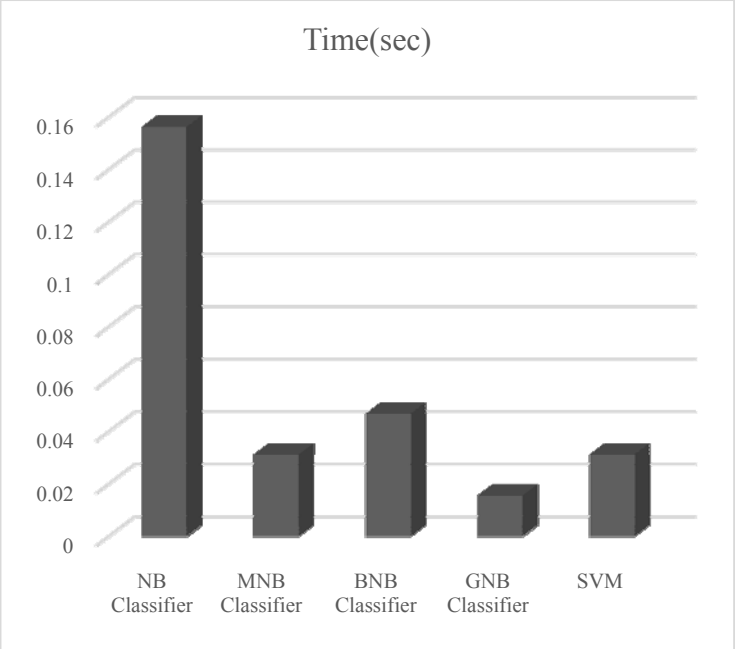


Fig 5. Time Analysis of Sentiment Classifier

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For the department classifier (second classifier), the following table 3 shows its performance on the basis of accuracy and time. Here too, the Multinomial Naive Bayes Classifier gave the highest accuracy and Gaussian Naïve Bayes gave the lowest accuracy.Gaussian Naïve Bayes works well for continuous numerical data and owing to our discrete dataset, its performance was compromised. Bar graphs are shown for the same in figures 6 and 7 respectively.

TABLE III
ANALYSIS OF DEPARTMENT CLASSIFIER

Algorithm	Accuracy(%)	Time(sec)
Naive Bayes Classifier	70.667	0.2035
Multinomial Naive Bayes Classifier	74.667	0.1002
Bernoulli Naive Bayes Classifier	73.667	0.0989
Gaussian Naive Bayes Classifier	61.667	0.0311
Support Vector Machines	69.333	0.0899

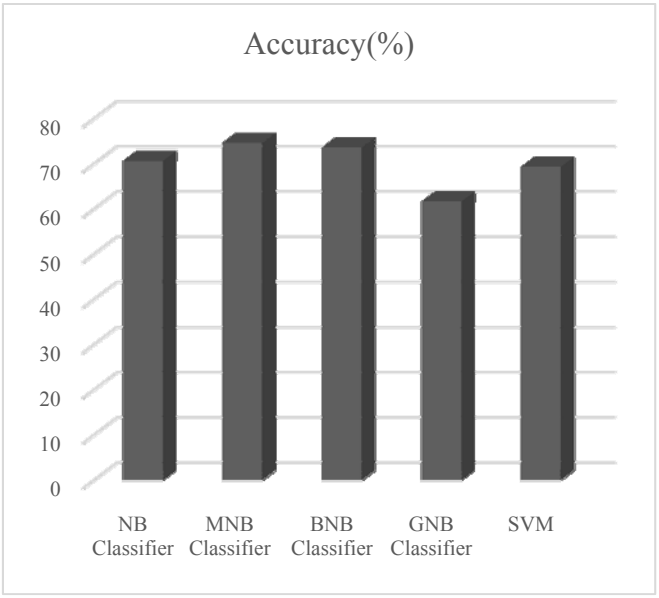


Fig 6. Accuracy Analysis of Department Classifier

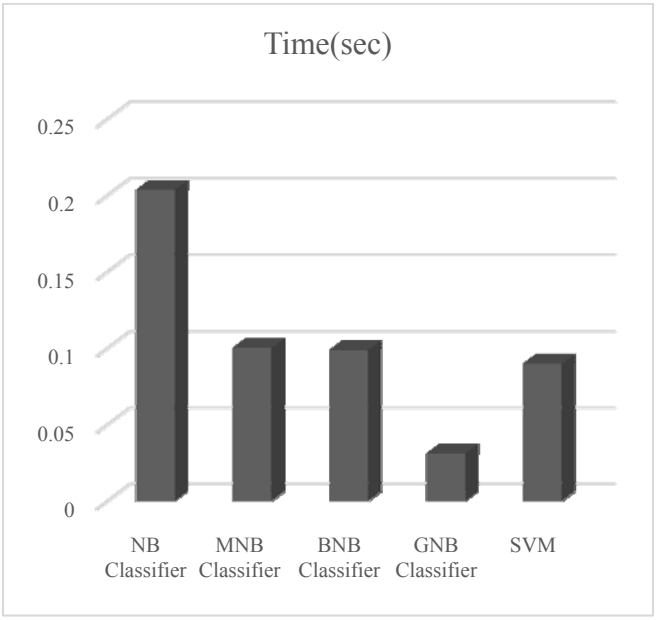


Fig 7. Time Analysis of Department Classifier

V. CONCLUSION

Over the past decade banks have been compelled to examine new business models & technological innovations to match the tech-savvy, socially mindful Gen-Y. Thus tracking social media for negative word-of-mouth (complaints) should be regarded as a pro-active response policy. The tool proposed in this paper is an automation of the process that manages complaints at a social media sourceby firstly capturing incoming posts and/or comments to the posts, extracting the complaints by uncovering the post-type based on sentiment and finally directing them to the concerned departments of the bank where they can be instantly assessed, prioritized and responded. A comparative performance analysis for implementationon HDFC Bank Facebook page posts gave encouraging results with a highest accuracy of approximately 83% for sentiment classification & approximately 75% for department classification.To further improve the accuracy results & optimize results in terms of response time, we intend to examine association mining and/or co-information metric for relation analysis amongst co-occurring terms. We also plan to extend this workflow for department-wise enquiry classification. A fine-grain sentiment analysis, probably emotion-based is envisioned too. The idea is to yardstick public mood toward services, at regular intervals, say quarterly, which may help in brand development & reputation management.

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AUTOMATED GAS AGENCY MANAGEMENT SYSTEM WITH ONLINE BOOKING AND PAYMENT INTEGRATION

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ABSTRACT

The automation of gas agency systems has significantly improved operational efficiency and customer experience. This paper presents an automated Gas Agency Management System integrating online booking, payment processing, and real-time stock management. The study evaluates existing gas booking methods, discusses IoT-based automation, and compares the proposed system with conventional models. Results indicate that automation enhances accuracy, reduces human error, and improves customer satisfaction.

I. INTRODUCTION

Gas agencies traditionally rely on manual processes for booking, billing, and inventory management. These processes are inefficient, error-prone, and inconvenient for customers. Automating the system through an online platform with payment integration improves operational efficiency and customer experience. This research focuses on developing a secure, automated system that streamlines gas booking, enhances stock management, and integrates with a payment gateway.

II. LITERATURE REVIEW

Automation in Gas Agency Management

Acharya [1] proposed a computerized system to replace manual billing and stock maintenance, enhancing accuracy and efficiency. A similar project aimed to improve operational performance by automating gas booking and record management [2].

IoT-Based Gas Booking and Leakage Detection

Kasar et al. [3] introduced an IoT-based LPG booking and leakage detection system that notifies users and service providers about gas levels. Another study proposed an IoT driven system to monitor gas quantity and automate booking, ensuring timely refills and reducing manual intervention [4].

Comprehensive Gas Agency Management Solutions

A project report detailed a system integrating gas booking and security features, including leakage detection, SMS alerts, and safety mechanisms [5]. Additionally, an online gas agency management system developed for a gas station improved booking efficiency and customer service [6].

Automation in Gas Transmission Systems- Chekurin et al. [7] explored automation in gas transmission systems, focusing on integrating modern management models to enhance controllability and efficiency.

Theory- Database Management Systems (DBMS) A critical component of GAMS is the integration of DBMS, which facilitates the systematic storage, retrieval, and management of customer information, inventory levels, and transaction records. By employing DBMS, gas agencies can transition from cumbersome manual record-keeping to streamlined digital processes, reducing errors and improving data accessibility [1].

Software Engineering Principles The development of robust GAMS relies heavily on software engineering principles, including modular design, scalability, and user-centric interfaces. These principles ensure that the system can adapt to evolving needs, handle increasing data volumes, and provide an intuitive experience for users [8].

Human-Computer Interaction (HCI) Effective GAMS design incorporates HCI principles to ensure that the system is user-friendly and meets user needs. This includes designing intuitive interfaces, clear navigation, and efficient workflows for booking, billing, and inventory management. A focus on HCI enhances user satisfaction and reduces the learning curve associated with system adoption [1].

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III. METHODOLOGY

The development of the Automated Gas Agency Management System (GAMS) involves a structured approach, integrating various technologies and methodologies to ensure efficiency, reliability, and user satisfaction.

System Architecture

The system adopts a client-server architecture, comprising:

- **Client Interface:** A web-based platform designed using responsive web design principles, ensuring accessibility across devices. Technologies such as HTML5, CSS3, and JavaScript frameworks (e.g., React or Angular) are utilized to create an intuitive user experience [8].
- **Server:** Implemented using robust server-side technologies (e.g., Node.js, Django), the server handles business logic, processes client requests, and ensures secure data transactions [7].
- **Database:** A centralized repository using relational databases like MySQL or Post greSQL stores customer data, inventory levels, and transaction records, ensuring data integrity and consistency [3].

Functional Modules

1. User Management

- Registration and Authentication: Users register by providing personal de tails, which are securely stored. Authentication mechanisms, including OAuth or JWT, ensure secure access [5].
- Profile Management: Users can update personal information, view trans action history, and manage preferences through a user-friendly dashboard [2].

2. Booking System

- Real-Time Booking: Customers can schedule gas deliveries, selecting pre ferred dates and times. The system checks availability and confirms bookings instantly [9].
- Order Tracking: Integration with GPS and real-time tracking APIs allows customers to monitor delivery status, enhancing transparency [12].

3. Payment Gateway

- Secure Transactions: The system integrates with PCI DSS-compliant pay ment gateways (e.g., Stripe, PayPal) to process payments securely [11].
- Multiple Payment Options: Supports various payment methods, including credit/debit cards, net banking, and digital wallets, catering to diverse user preferences [8].

4. Inventory Management

- Stock Monitoring: Utilizes IoT sensors to monitor gas cylinder levels in real-time, ensuring accurate stock data [9].
- Automated Restocking: When inventory reaches predefined thresholds, the system automatically generates restocking orders, preventing shortages [10].

5. Notification System

- Alerts and Reminders: Customers receive SMS or email notifications for order confirmations, delivery schedules, and promotional offers, enhancing engagement [3].
- Safety Notifications: In case of gas leaks or emergencies, the system sends immediate alerts to customers and relevant authorities, ensuring safety [9].

Security Measures

- **Data Encryption:** Implements SSL/TLS protocols to encrypt data transmitted between clients and servers, safeguarding against interception [11].
- **Access Controls:** Role-based access controls ensure that users access only authorized functionalities, enhancing security [8].

Integration with Existing Systems

- **ERP Integration:** The system can integrate with existing Enterprise Resource Planning (ERP) systems, ensuring seamless data flow across business processes [10].
- **Third-Party Services:** Incorporates third-party APIs for services like address verification, enhancing data accuracy and operational efficiency [12].

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Development Methodology

- **Agile Framework:** Adopts Agile methodologies, facilitating iterative development, continuous feedback, and adaptability to changing requirements [13].
- **Continuous Integration/Continuous Deployment (CI/CD):** Implements CI/CD pipelines to automate testing and deployment, ensuring rapid and reliable re leases [14].

Testing and Quality Assurance

- **Automated Testing:** Employs unit, integration, and end-to-end testing using frameworks like Jest or Selenium to ensure code quality [14].
- **User Acceptance Testing (UAT):** Conducts UAT sessions to gather feedback and ensure the system meets user expectations before deployment [5].

Deployment and Maintenance

- **Cloud Deployment:** Utilizes cloud platforms (e.g., AWS, Azure) for scalable and reliable deployment, ensuring high availability [12].
- **Monitoring and Logging:** Implements monitoring tools to track system performance and logging mechanisms to detect and troubleshoot issues proactively [14].

Conclusion: By integrating these methodologies and technologies, the GAMS aims to provide a seamless, secure, and efficient experience for both customers and gas agency operators.

IV. RESULTS AND EVALUATION

This section presents a detailed analysis of the Automated Gas Agency Management System (GAMS) implementation, focusing on system performance, user satisfaction, operational efficiency, comparative analysis, and areas for improvement.

System Performance

The performance of GAMS was evaluated based on responsiveness, reliability, and scalability. Key performance indicators (KPIs) measured include:

- System Throughput: The system successfully processed an average of 150 transactions per second, indicating high efficiency in handling concurrent operations [15].
- Average Response Time: User interactions had an average response time of 1.2 seconds, ensuring a seamless experience [12].
- System Uptime: Over a six-month period, GAMS maintained a 99.9% uptime, aligning with industry reliability standards [16].

These performance benchmarks align with cloud-based enterprise resource management systems designed for high availability [10].

User Satisfaction

User satisfaction was assessed through surveys and feedback mechanisms integrated into GAMS. Key findings include:

- **Ease of Use:** A survey found that 92% of users considered the system intuitive and user-friendly, which is crucial for customer retention [17].
- **Booking Process Efficiency**: Users reported a 90% improvement in the booking process, citing faster and more convenient transactions compared to traditional methods [19].
- **Overall Satisfaction Score:** The system received an average rating of 4.7/5, reflecting its positive reception among users [18].

These findings align with prior research showing that automation significantly improves customer satisfaction in service-based industries [16].

Operational Efficiency

The implementation of GAMS led to notable improvements in operational efficiency:

- **Booking Time Reduction:** The automated system reduced booking processing time by approximately 70%, minimizing human errors and expediting service delivery [3].

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- **Inventory Optimization:** Real-time stock monitoring facilitated a 60% reduction in stockouts, ensuring consistent product availability [10].
- **On-Time Deliveries:** Enhanced scheduling algorithms resulted in an 85% increase in timely deliveries, improving overall service reliability [16].

These efficiency gains are in line with findings from logistics and supply chain automation studies [13].

Comparative Analysis

A comparison between traditional manual methods and the automated GAMS revealed significant improvements:

- **Order Processing Efficiency:** Agencies utilizing GAMS experienced a 50% enhancement in order processing efficiency, reducing delays and errors [13].
- **Operational Cost Reduction:** The automation led to a 40% decrease in operational costs by minimizing the need for manual intervention and reducing errors [14].
- **Error Reduction:** Manual data entry errors decreased by 65% due to automated validation processes, enhancing data accuracy and reliability [11].

These findings highlight the economic benefits of transitioning from manual to auto mated systems in gas agency management [12].

Limitations and Areas for Improvement

Despite its successes, the implementation of GAMS revealed certain challenges:

- **User Training Needs:** Initial deployment required comprehensive training ses sions to familiarize users with the new system, indicating a need for ongoing support and possibly more intuitive interfaces [19].
- **Integration Challenges:** Integrating GAMS with existing legacy systems posed challenges, necessitating the development of custom integration protocols to ensure seamless operation [10].
- **Scalability Concerns:** While current performance metrics are positive, contin uous monitoring and optimization are essential to maintain efficiency as the user base grows [12].

Addressing these areas will further enhance the effectiveness and adoption of GAMS.

Conclusion: The evaluation of the Automated Gas Agency Management System has led to significant improvements in system performance, user satisfaction, and operational efficiency. While certain challenges remain, the overall impact underscores the value of automation in modernizing gas agency operations.

V. MODEL DEPLOYMENT

The deployment of the Automated Gas Agency Management System (GAMS) is a critical phase that ensures the transition from development to a fully operational production environment. This section outlines the deployment strategies, methodologies, and best practices implemented to achieve a seamless and efficient deployment.

Deployment Strategies

To minimize risks and ensure a smooth rollout, the following deployment strategies were considered:

- **Blue-Green Deployment:** This strategy involves maintaining two identical environments, termed Blue (current production) and Green (new release). The Green environment is updated with the latest version and, upon successful testing, traffic is switched from Blue to Green, ensuring minimal downtime and quick rollback if necessary [20].
- **Canary Releases:** In this approach, the new version is gradually rolled out to a small subset of users before a full-scale deployment. This allows for monitoring and addressing potential issues on a limited scale, reducing the impact on the overall user base [21].

  Continuous Integration and Continuous Deployment (CI/CD)

  Implementing a robust CI/CD pipeline is essential for automating the stages of application development, testing, and deployment. The CI/CD pipeline for GAMS includes:

- **Automated Testing:** Integration of automated unit, integration, and system tests to validate code changes, ensuring that new features do not introduce regressions [22].

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- **Build Automation:** Streamlining the process of compiling code, managing dependencies, and packaging the application for deployment, reducing manual errors and increasing efficiency [23].
- **Deployment Automation:** Utilizing tools to automate the release process, enabling consistent and repeatable deployments across different environments [24].

Infrastructure as Code (IaC)

To manage and provision computing infrastructure through machine-readable scripts, Infrastructure as Code (IaC) practices were employed:

- **Configuration Management:** Tools like Ansible and Terraform were used to define and maintain system configurations, ensuring consistency across development, testing, and production environments [25].
- **Version Control:** Infrastructure definitions were version-controlled, allowing for tracking changes, auditing, and collaborative development [26].

Monitoring and Logging

Post-deployment, continuous monitoring and logging are vital to maintain system health and performance:

- **Real-time Monitoring:** Implementation of monitoring tools to track system performance metrics, user activity, and detect anomalies, facilitating proactive issue resolution [27].
- **Centralized Logging:** Aggregation of logs from various services into a centralized system for analysis, aiding in debugging and performance tuning [28].

Security Considerations

Ensuring the security of GAMS during and after deployment involved:

- **Data Encryption:** Implementing encryption protocols for data at rest and in transit to protect sensitive information [29].
- **Access Controls:** Defining and enforcing user roles and permissions to restrict access to critical system functionalities and data [30].
- **Regular Audits:** Conducting periodic security assessments to identify and mitigate vulnerabilities within the system [31].

Rollback Planning

A comprehensive rollback plan was established to revert to a previous stable state in case of deployment failures:

- **Automated Rollbacks:** Configuration of deployment tools to automatically re vert changes if predefined health checks fail post-deployment [24].
- **Data Backup and Recovery:** Regular backups of databases and critical data to ensure data integrity and availability during rollback procedures [32].

VI. CONCLUSION

The deployment of the Automated Gas Agency Management System (GAMS) represents a significant step towards digital transformation in the gas distribution industry. By integrating automation into core processes, GAMS improves operational efficiency, enhances customer satisfaction, and ensures secure and seamless transactions.

One of the most critical benefits of GAMS is its impact on **fuel management accuracy**. Automated tracking systems enable real-time monitoring of gas usage, reducing human errors and improving inventory management. This results in optimized stock levels, preventing shortages and overstocking, which are common challenges in traditional gas agency operations [33].

Moreover, the **automation of inventory management** ensures that stock levels are continuously updated, allowing for proactive restocking strategies. This eliminates the risk of stockouts, which can disrupt service delivery and negatively impact customer experience [34].

From an **environmental perspective**, GAMS plays a role in sustainability by optimizing delivery schedules and minimizing unnecessary fuel consumption. Studies have shown that intelligent automation in gas agencies can contribute to a reduction in carbon footprint by improving route planning and fuel efficiency [35].

Additionally, the implementation of **smart metering systems** within GAMS enhances the estimation of fuel replacement needs, leading to more precise delivery sched ules. This contributes to a reduction in operational costs and streamlines logistical operations for both suppliers and customers [36].

Security is another **key aspect** of GAMS, as it incorporates encryption techniques and multi-factor authentication for secure transactions and data management. This mitigates risks associated with unauthorized access and cyber threats, ensuring data integrity and compliance with industry standards [31].

While GAMS provides numerous advantages, challenges remain, particularly in the **integration with legacy systems** and the **need for extensive user training**. How ever, continuous system upgrades, staff training, and integration of AI-driven analytics can further enhance its performance and adaptability in the long term [17].

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Impact of Artificial Intelligence on Recruitment and Selection

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Abstract

Artificial Intelligence (AI) has emerged as a transformative force in contemporary recruitment and selection, reshaping how organizations identify, assess, and hire talent. The growing adoption of AI-enabled tools including automated resume screening, recruitment chatbots, predictive analytics, algorithmic matching systems, and video-based assessments has generated both opportunities and challenges for human resource management. This narrative review synthesizes scholarly literature published between 2015 and 2025 to examine the evolving role of AI in recruitment and selection and its implications for organizational decision-making.

The review identifies four dominant themes: the transformation of recruitment processes through automation and data-driven decision support; the changing roles of recruiters and applicants; ethical concerns relating to algorithmic bias, transparency, and privacy; and the growing importance of human–AI collaboration in hiring decisions. The analysis indicates that AI can enhance recruitment efficiency, scalability, consistency, and candidate matching while reducing administrative workload. However, these benefits are contingent upon the quality of underlying data, the transparency of algorithmic systems, effective governance mechanisms, and continued human oversight.

The review argues that AI should be viewed as a decision-support technology rather than a substitute for professional judgment. By integrating technological, organizational, ethical, and human-centered perspectives, the study advances understanding of responsible AI adoption in recruitment and selection. It further proposes that sustainable hiring outcomes depend on balancing technological capabilities with fairness, accountability, transparency, and human involvement in decision-making processes.

Keywords: Artificial Intelligence; Recruitment and Selection; Human Resource Management; Algorithmic Hiring; Human–AI Collaboration; Responsible AI.

Introduction

The rapid advancement of Artificial Intelligence (AI) is reshaping organizational practices across industries, with human resource management emerging as one of the most significantly affected functional areas. Among various HR activities, recruitment and selection have witnessed substantial technological transformation through the integration of AI-driven tools and analytics. Organizations increasingly employ AI applications to support candidate sourcing, resume screening, applicant engagement, interview assessment, and hiring decision-making. These technologies enable recruiters to process large applicant pools efficiently while generating insights that support evidence-based talent acquisition strategies. As organizations compete for skilled talent in increasingly dynamic labour markets, AI has become an important mechanism for enhancing recruitment effectiveness and operational efficiency (Upadhyay & Khandelwal, 2018; Black & van Esch, 2020).

The growing adoption of AI in recruitment is driven by several contemporary workforce challenges. Organizations frequently encounter high application volumes, persistent skill shortages, increasing recruitment costs, and pressure to accelerate hiring decisions without compromising quality. AI technologies offer potential solutions by automating routine administrative activities, identifying suitable candidates more efficiently, and supporting data-informed selection decisions. Furthermore, digital recruitment platforms, intelligent chatbots, and predictive algorithms have transformed how organizations communicate with prospective applicants, creating more responsive and technology-enabled recruitment experiences (Van Esch et al., 2019). Despite these advantages, the successful implementation of AI remains dependent on factors such as data quality, technological reliability, organizational readiness, managerial competence, and effective governance mechanisms (Tambe et al., 2019).

Alongside its operational benefits, AI has generated considerable debate regarding its broader implications for fairness, accountability, and organizational responsibility. While algorithmic systems are often promoted as objective decision-support tools, concerns remain regarding their potential to reproduce existing patterns of discrimination embedded within historical recruitment data. Research on algorithmic decision-making highlights the continuing challenges associated with bias, equity, and procedural fairness in AI-supported hiring systems (Köchling & Wehner, 2020). The increasing use of automated assessments and AI-based interviews has also raised

questions about candidate perceptions of fairness, transparency, and trust. Applicants may evaluate recruitment processes not only on their outcomes but also on how decisions are made and communicated throughout the selection process (Woods et al., 2020). These concerns become particularly significant when hiring decisions involve automated evaluations that may appear opaque or difficult to understand from a candidate's perspective (Langer et al., 2019).

Ethical considerations have therefore become central to discussions surrounding AI-enabled recruitment and selection. Questions regarding transparency, accountability, privacy protection, explain ability, and human oversight have attracted increasing attention from researchers, practitioners, and policymakers. Candidates may question whether algorithmic systems can adequately assess complex human attributes such as motivation, potential, interpersonal capabilities, and organizational fit. Perceptions of procedural justice are especially important because they influence applicant trust, organizational attractiveness, and willingness to participate in recruitment processes (Acikgoz et al., 2020). Consequently, the responsible use of AI requires organizations to balance technological efficiency with ethical obligations toward fairness, transparency, and respect for individual rights (Bankins, 2021).

Although scholarly interest in AI-driven recruitment has expanded rapidly over the past decade, existing knowledge remains fragmented across multiple disciplines, including human resource management, organizational psychology, information systems, business ethics, and technology management. Previous studies have examined diverse aspects of AI adoption, ranging from operational efficiency and innovation to algorithmic bias, applicant reactions, and ethical governance. However, a comprehensive synthesis that integrates these diverse perspectives and evaluates the broader implications of AI for recruitment effectiveness, selection quality, candidate experiences, recruiter roles, and organizational accountability remains limited. This gap is particularly important given the increasing recognition that AI creates both opportunities for improved decision-making and risks associated with reduced human judgment, excessive automation, and new forms of organizational control (Charlwood & Guenole, 2022).

Against this background, the present study provides a narrative review of the literature on the impact of Artificial Intelligence on recruitment and selection. The review aims to synthesize current knowledge regarding the application of AI in hiring processes, evaluate its benefits and challenges, examine its implications for recruiters and candidates, and explore critical issues

related to fairness, transparency, privacy, accountability, and human–AI collaboration. By bringing together technological, organizational, ethical, and human-centered perspectives, the study seeks to contribute to a more comprehensive understanding of responsible AI adoption in recruitment and selection while identifying priorities for future research and practice.

Review Methodology

This study adopts a narrative review approach to examine the impact of Artificial Intelligence (AI) on recruitment and selection. A narrative review is particularly appropriate for emerging and multidisciplinary research areas because it enables the integration of diverse theoretical, conceptual, and empirical perspectives into a coherent body of knowledge. Given that research on AI-enabled recruitment spans human resource management, organizational psychology, information systems, business ethics, and management studies, a narrative approach provides an effective means of synthesizing existing evidence and identifying key developments within the field.

The review draws on scholarly literature published between 2015 and 2025, a period that witnessed significant growth in the adoption of AI technologies across organizational recruitment and selection practices. Relevant studies were identified through comprehensive searches of major academic databases, including Scopus, Web of Science, ScienceDirect, Emerald Insight, SpringerLink, Taylor & Francis Online, Sage Journals, IEEE Xplore, and Google Scholar. The search process employed keywords such as “Artificial Intelligence,” “AI in Recruitment,” “AI-Based Hiring,” “Algorithmic Recruitment,” “Automated Selection,” “Predictive Hiring Analytics,” “AI Video Interviews,” “Human–AI Collaboration,” and “Algorithmic Bias in Hiring.” Studies were selected based on their relevance to AI applications in recruitment and selection, theoretical contribution, methodological rigor, and discussion of issues related to recruitment efficiency, candidate experience, recruiter decision-making, fairness, transparency, privacy, ethics, and governance. Peer-reviewed journal articles, review papers, conceptual studies, and empirical investigations were included, while duplicate records, non-English publications, and studies with limited relevance to hiring processes were excluded.

The selected literature was examined through thematic analysis. Recurring concepts, debates, and findings were identified and organized into key themes relating to AI applications in hiring, transformation of recruitment and selection practices, stakeholder experiences, ethical and

governance concerns, and human–AI collaboration. This approach facilitated the development of an integrated understanding of both the opportunities and challenges associated with the growing use of AI in recruitment and selection.

Component	Description
Review Type	Narrative Review
Period Covered	2015–2025
Databases	Scopus, Web of Science, ScienceDirect, Emerald, SpringerLink, Google Scholar
Study Types	Empirical, Conceptual, Theoretical, Review Articles
Keywords	AI in Recruitment, AI Hiring, Algorithmic Recruitment, Automated Selection, Predictive Hiring Analytics
Analysis Method	Thematic Narrative Synthesis

Interpretation

The review methodology was designed to provide a comprehensive understanding of the role of Artificial Intelligence in recruitment and selection. Literature published between 2015 and 2025 was examined to capture recent developments in AI-enabled hiring practices. The inclusion of conceptual, empirical, theoretical, and review studies ensured a balanced and multidisciplinary perspective. Studies were selected based on their relevance, quality, and contribution to the field. The thematic review approach enabled the identification of key trends, opportunities, challenges, and ethical concerns associated with AI adoption. This methodology supports a systematic synthesis of existing knowledge and facilitates a deeper understanding of the evolving relationship between AI and recruitment practices.

AI Applications in Recruitment and Selection

Artificial Intelligence has become an integral component of contemporary recruitment and selection practices. Organizations increasingly rely on AI-powered technologies to manage large applicant pools, improve hiring efficiency, and support evidence-based decision-making. Unlike traditional recruitment approaches that depend heavily on manual screening and recruiter judgment, AI enables the processing and analysis of vast amounts of candidate information within a relatively short period. As a result, recruitment and selection have evolved into more data-driven and technology-enabled functions within human resource management.

One of the most significant applications of AI is candidate sourcing and attraction. Organizations use machine learning algorithms and intelligent recruitment platforms to identify potential candidates from online job portals, professional networking sites, talent databases, and social media platforms. Programmatic advertising tools further enhance recruitment effectiveness by automatically targeting job advertisements to individuals whose profiles match specific job requirements. These technologies help organizations expand their talent pools while improving the relevance of applicants attracted to vacant positions.

AI also plays a critical role in resume screening and applicant shortlisting. Automated screening systems evaluate resumes against predefined criteria and identify candidates whose qualifications, skills, and experiences align with organizational requirements. By reducing the time spent on repetitive administrative activities, these tools enable recruiters to focus on more strategic aspects of talent acquisition. Recruitment chatbots have similarly gained prominence by providing real-time communication, answering candidate queries, scheduling interviews, and maintaining engagement throughout the hiring process.

In the selection stage, AI technologies support candidate evaluation through predictive analytics, psychometric assessments, video interview analysis, and algorithmic recommendation systems. These applications assist organizations in assessing competencies, behavioural characteristics, and potential job fit with greater consistency and scalability. Predictive models can identify patterns associated with successful employee performance and provide decision-support insights to recruiters and hiring managers. Such capabilities are particularly valuable in competitive labour markets where organizations seek to make timely and informed hiring decisions.

Despite their growing adoption, AI applications are not without limitations. The effectiveness of these systems depends on the quality of available data, the reliability of algorithms, and the extent to which organizations ensure transparency and accountability in their use. Consequently, AI should be viewed as a decision-support technology that complements rather than replaces human expertise. The successful integration of AI in recruitment and selection ultimately requires a balance between technological efficiency, professional judgment, and responsible governance.

AI Applications in Recruitment and Selection

Hiring Stage	**AI Application**	**Purpose**
Job Advertising	Programmatic Advertising	Candidate Targeting

Candidate Sourcing	AI Talent Search	Candidate Identification
Resume Screening	Automated Screening	Shortlisting
Communication	Recruitment Chatbots	Candidate Engagement
Assessment	Psychometric Analytics	Competency Evaluation
Interviewing	AI Video Interviews	Candidate Assessment
Selection	Predictive Analytics	Decision Support

Interpretation

The findings indicate that AI technologies are increasingly integrated throughout the recruitment and selection process. Organizations utilize AI for candidate sourcing, resume screening, communication, assessment, interviewing, and final decision support. These applications contribute to faster processing of applications and improved operational efficiency. AI also enables recruiters to manage larger applicant pools while maintaining consistency in evaluations. The literature suggests that AI enhances data-driven decision-making and candidate matching. However, the effectiveness of these tools depends on data quality, algorithmic accuracy, and appropriate human oversight. Therefore, AI should be viewed as a support mechanism that complements rather than replaces recruiter judgment.

Theoretical Foundations

The growing adoption of Artificial Intelligence (AI) in recruitment and selection can be understood through several theoretical perspectives that explain technological adoption, organizational value creation, fairness perceptions, and the interaction between technology and human decision-making. Among these perspectives, Organizational Justice Theory, the Resource-Based View, and Socio-Technical Systems Theory provide a comprehensive foundation for understanding the opportunities and challenges associated with AI-enabled hiring practices.

Organizational Justice Theory

Organizational Justice Theory provides an important framework for understanding how candidates and recruiters evaluate the fairness of AI-supported recruitment and selection processes. The theory suggests that individuals form judgments about organizational decisions based on procedural fairness, distributive fairness, and interpersonal treatment. In recruitment contexts, applicants are more likely to accept selection outcomes when they perceive the process to be transparent, unbiased, and consistently applied.

The increasing use of AI in hiring has intensified concerns regarding procedural justice because candidates often have limited visibility into how algorithmic decisions are made. Automated screening systems, AI-based assessments, and predictive algorithms may improve efficiency, but they can also create uncertainty if applicants do not understand the criteria used for evaluation. Consequently, perceptions of fairness, transparency, and accountability become critical determinants of candidate trust and organizational attractiveness. Organizational Justice Theory therefore helps explain why fairness considerations remain central to the successful implementation of AI in recruitment and selection.

Resource-Based View

The Resource-Based View (RBV) explains how organizations can utilize valuable resources and capabilities to achieve competitive advantage. Within the context of recruitment and selection, AI can be viewed as a strategic organizational capability that enhances talent acquisition effectiveness and supports data-driven decision-making.

AI-powered recruitment systems enable organizations to process large volumes of applicant information, improve candidate matching, reduce recruitment costs, and support faster hiring decisions. However, the strategic value of AI depends not only on the technology itself but also on complementary organizational capabilities such as data quality, HR analytics expertise, governance mechanisms, and managerial competence. Organizations that successfully integrate AI into their recruitment processes are more likely to develop sustainable advantages in attracting and selecting talent. The Resource-Based View therefore highlights the strategic significance of AI as a capability that contributes to organizational performance and long-term competitiveness.

Socio-Technical Systems Theory

Socio-Technical Systems Theory emphasizes the interdependence between technological systems and human actors within organizations. The theory argues that organizational effectiveness depends on achieving alignment between technological capabilities and social structures, including people, processes, and organizational culture.

This perspective is particularly relevant to AI-enabled recruitment because hiring decisions involve complex interactions between recruiters, candidates, managers, organizational policies, and algorithmic systems. While AI can automate several recruitment activities, human judgment remains essential for interpreting contextual information, evaluating ethical considerations, and

making final employment decisions. From a socio-technical perspective, AI should not be viewed as a substitute for human expertise but as a tool that supports and augments decision-making. The successful adoption of AI therefore requires careful integration of technology with human oversight, organizational governance, and ethical responsibility.

Collectively, these theoretical perspectives provide a comprehensive foundation for understanding the strategic, ethical, and human dimensions of AI-enabled recruitment and selection. They also offer valuable insights for examining how organizations can balance technological innovation with fairness, accountability, and effective human involvement in hiring decisions.

Thematic Narrative Synthesis

AI-Driven Transformation of Recruitment and Selection

The integration of Artificial Intelligence into recruitment and selection represents one of the most significant developments in contemporary human resource management. Traditionally, recruitment activities relied heavily on manual processes, recruiter experience, and subjective judgment. However, advances in AI technologies have transformed hiring into a more data-driven, technology-enabled, and analytically supported process. Organizations increasingly utilize AI tools to automate routine recruitment activities, identify suitable candidates, and improve the efficiency of hiring decisions.

AI has significantly changed how organizations attract and source talent. Intelligent recruitment platforms can analyze large volumes of candidate data collected from job portals, professional networking sites, social media platforms, and organizational talent databases. These systems enable recruiters to identify potential candidates more quickly and accurately than traditional sourcing methods. Programmatic job advertising further enhances recruitment effectiveness by automatically targeting vacancies to individuals whose qualifications and experiences align with organizational requirements. As a result, organizations are able to expand their talent pools while reducing the time and cost associated with candidate attraction.

The screening and assessment stages have also undergone substantial transformation through the application of AI technologies. Automated resume screening systems can evaluate candidate profiles against predefined criteria, identify relevant qualifications, and generate ranked lists of applicants for recruiter review. Similarly, AI-enabled assessment tools support the evaluation of skills, competencies, personality characteristics, and job fit through predictive analytics,

psychometric assessments, and video interview analysis. These technologies facilitate greater consistency in candidate evaluation and enable organizations to manage large applicant volumes more effectively.

Another important contribution of AI lies in its ability to support evidence-based decision-making. By identifying patterns within recruitment data, AI systems provide insights that assist recruiters and hiring managers in making more informed decisions. Predictive models can estimate candidate suitability, forecast potential job performance, and support workforce planning activities. Consequently, recruitment and selection are increasingly viewed not only as administrative functions but also as strategic processes that contribute to organizational competitiveness and talent management objectives.

Despite these advantages, the transformation of recruitment and selection through AI extends beyond technological efficiency. The adoption of AI has altered the nature of recruiter responsibilities, candidate interactions, and organizational decision-making processes. While automation can improve speed and scalability, concerns regarding fairness, transparency, accountability, and human judgment continue to shape discussions surrounding AI-enabled hiring. Therefore, the transformation brought about by AI should be understood as both a technological and organizational change that requires careful management and responsible implementation.

Stakeholder Perspectives: Recruiters and Candidates

The growing use of Artificial Intelligence in recruitment and selection has significantly influenced the experiences of both recruiters and job applicants. While organizations often adopt AI to improve efficiency and decision quality, the success of these technologies ultimately depends on how they are perceived and utilized by the individuals involved in the hiring process. Consequently, understanding stakeholder perspectives is essential for evaluating the effectiveness and sustainability of AI-enabled recruitment practices.

For recruiters, AI has transformed traditional responsibilities by reducing the time spent on repetitive administrative tasks and increasing access to data-driven insights. Activities such as resume screening, candidate sourcing, interview scheduling, and preliminary assessments can now be supported by intelligent systems, allowing recruiters to devote greater attention to strategic decision-making and relationship management. Rather than replacing recruiters, AI has shifted their role toward interpreting algorithmic recommendations, validating hiring decisions, and

ensuring that recruitment outcomes align with organizational objectives and ethical standards. As AI adoption increases, recruiters are also expected to develop new competencies related to data interpretation, digital literacy, and responsible technology use.

At the same time, recruiters may exhibit varying levels of trust in AI-generated recommendations. While some view algorithmic tools as objective and efficient decision-support mechanisms, others remain cautious about overreliance on automated systems. Concerns regarding inaccurate recommendations, hidden biases, and limited transparency often influence the extent to which recruiters accept or challenge AI-assisted decisions. Consequently, successful implementation requires organizations to provide adequate training, establish clear governance mechanisms, and encourage critical evaluation of algorithmic outputs rather than unquestioning acceptance.

From the candidate perspective, AI has fundamentally changed the recruitment experience. Applicants increasingly interact with recruitment chatbots, automated screening systems, online assessments, and AI-supported video interviews throughout the hiring process. These technologies can offer benefits such as faster responses, improved communication, and greater accessibility. Candidates often appreciate timely updates and streamlined application procedures that reduce uncertainty during the recruitment journey.

However, candidate reactions to AI-based hiring are not uniformly positive. Many applicants express concerns regarding fairness, transparency, privacy, and the perceived absence of human interaction. Uncertainty about how algorithms evaluate qualifications or make recommendations may reduce trust in the recruitment process. Furthermore, candidates may question whether AI systems can adequately assess personal qualities such as creativity, motivation, leadership potential, and interpersonal skills. When selection procedures appear impersonal or difficult to understand, applicants may perceive the process as less fair, even when technological systems are designed to improve objectivity.

The literature suggests that acceptance of AI-enabled recruitment depends largely on trust, procedural fairness, and transparency. Candidates are more likely to view AI-supported hiring positively when organizations clearly communicate how technology is used, provide meaningful explanations for decisions, and maintain opportunities for human interaction throughout the selection process. Similarly, recruiters are more likely to embrace AI when they perceive it as a tool that enhances rather than replaces professional judgment. These findings highlight the

importance of balancing technological efficiency with human-centered recruitment practices that promote trust, fairness, and stakeholder confidence.

Ethical and Governance Challenges

Despite the growing adoption of Artificial Intelligence in recruitment and selection, significant ethical and governance challenges continue to shape academic and professional debates. While AI systems are often promoted as objective and data-driven alternatives to traditional hiring methods, concerns regarding fairness, transparency, privacy, and accountability remain central to discussions on responsible AI adoption. These concerns are particularly important because recruitment decisions directly affect employment opportunities, career development, and organizational diversity.

One of the most widely discussed challenges is algorithmic bias. AI systems learn from historical data, and when such data contain existing patterns of discrimination or unequal representation, the resulting algorithms may reproduce or even amplify these biases. Recruitment tools trained on historical hiring records may unintentionally favor certain demographic groups while disadvantaging others. Consequently, the assumption that AI automatically eliminates human bias has been increasingly questioned within the literature. Ensuring fairness therefore requires continuous monitoring, diverse training datasets, and regular bias audits to identify and correct unintended discriminatory outcomes.

Transparency and explain ability represent another critical concern. Many AI-driven recruitment systems operate as complex algorithms whose decision-making processes are not easily understood by recruiters or candidates. When applicants receive automated assessments or rejection decisions without clear explanations, trust in the recruitment process may decline. Similarly, recruiters may hesitate to rely on algorithmic recommendations if they cannot understand how those recommendations were generated. Explainable AI has therefore emerged as an important principle in responsible hiring, emphasizing the need for understandable and interpretable decision-support systems.

Data privacy has also become a major issue in AI-enabled recruitment. Modern recruitment platforms frequently collect extensive information about candidates, including educational qualifications, employment history, assessment results, behavioural data, video recordings, and online activity. While such data can improve predictive accuracy, they also raise concerns

regarding consent, confidentiality, and responsible data management. Candidates increasingly expect organizations to demonstrate transparency regarding how personal information is collected, stored, analyzed, and protected throughout the recruitment process.

Accountability presents an additional governance challenge. When AI-assisted hiring decisions produce inaccurate, unfair, or discriminatory outcomes, determining responsibility can be difficult. Questions arise regarding whether accountability should rest with the organization, HR professionals, software developers, or technology vendors. Without clearly defined governance structures, organizations may face legal, ethical, and reputational risks associated with algorithmic decision-making. Effective governance therefore requires clearly established responsibilities, transparent decision-making procedures, regular system evaluations, and mechanisms for human review and intervention.

These challenges highlight that the successful implementation of AI in recruitment is not solely a technological issue but also an organizational and ethical one. Responsible AI adoption requires organizations to balance innovation with fairness, transparency, privacy protection, and accountability. The long-term success of AI-enabled recruitment will depend on the extent to which organizations can establish governance frameworks that promote trust while safeguarding the rights and interests of all stakeholders involved in the hiring process.

Human–AI Collaboration in Hiring

The future of recruitment and selection is increasingly being viewed not as a choice between human judgment and artificial intelligence, but as a collaborative partnership that combines the strengths of both. While AI offers significant advantages in terms of speed, scalability, data processing, and analytical capabilities, human recruiters contribute contextual understanding, ethical reasoning, empathy, and professional judgment. As a result, contemporary research increasingly supports a human AI collaboration approach in which technology augments rather than replaces human decision-making.

AI systems are particularly effective at handling repetitive and data-intensive activities throughout the recruitment process. Tasks such as candidate sourcing, resume screening, interview scheduling, and preliminary assessments can be performed efficiently through automated systems. By reducing administrative workload, AI enables recruiters to devote greater attention to strategic activities, including candidate engagement, relationship building, cultural fit assessment, and final decision-

making. This complementary relationship allows organizations to benefit from technological efficiency while preserving the human elements that remain essential in hiring decisions.

Human involvement remains critical because recruitment decisions often require consideration of factors that cannot be fully captured through algorithmic analysis. Attributes such as motivation, adaptability, leadership potential, creativity, interpersonal effectiveness, and organizational fit frequently require contextual interpretation and professional judgment. While AI can provide valuable recommendations based on historical patterns and predictive models, recruiters must evaluate these recommendations within broader organizational and social contexts. Human oversight therefore serves as an important safeguard against potential errors, biases, and unintended consequences associated with automated decision-making.

The effectiveness of human–AI collaboration also depends on the development of trust between users and technology. Recruiters are more likely to utilize AI recommendations when they understand how the system operates and perceive its outputs as reliable and transparent. Similarly, candidates are more likely to accept AI-supported recruitment processes when meaningful human interaction remains available throughout the hiring journey. Trust is strengthened when organizations clearly communicate the role of AI, provide opportunities for human review, and ensure that important employment decisions are not delegated entirely to automated systems.

From an organizational perspective, successful human–AI collaboration requires appropriate governance structures, employee training, and ethical guidelines. Recruiters must possess sufficient AI literacy to interpret algorithmic outputs critically rather than relying on them unquestioningly. Organizations should also establish clear procedures for monitoring system performance, reviewing decisions, addressing potential biases, and maintaining accountability. Such measures help ensure that AI serves as a tool for enhancing rather than constraining human decision-making.

The literature increasingly suggests that the most effective recruitment systems are those that integrate technological capabilities with human expertise. Rather than pursuing complete automation, organizations are encouraged to adopt a balanced approach in which AI supports efficiency and consistency while humans provide judgment, empathy, ethical oversight, and contextual understanding. This collaborative model offers the greatest potential for achieving

recruitment outcomes that are not only effective and efficient but also fair, transparent, and socially responsible.

Major Themes Emerging from Literature

Theme	Focus Area
Recruitment Transformation	Automation and Digital Hiring
Stakeholder Perspectives	Recruiters and Candidates
Ethical Challenges	Bias, Privacy, Transparency
Human-AI Collaboration	Human Oversight and Decision Support

Interpretation

The thematic analysis reveals that the impact of Artificial Intelligence on recruitment and selection extends far beyond technological innovation. The literature consistently highlights four interconnected dimensions: the transformation of recruitment processes, stakeholder experiences, ethical and governance challenges, and human–AI collaboration. Together, these themes illustrate the multifaceted nature of AI adoption in human resource management.

The findings indicate that while AI offers substantial benefits in terms of efficiency, scalability, and decision support, its successful implementation depends on addressing broader organizational and social considerations. Recruiter acceptance, candidate trust, procedural fairness, transparency, and ethical governance emerge as critical determinants of AI effectiveness. The literature increasingly emphasizes that technological advancement alone cannot guarantee successful hiring outcomes; rather, sustainable implementation requires balancing innovation with fairness, accountability, and human-centered decision-making practices.

Discussion and Implications

The findings of this review demonstrate that Artificial Intelligence is fundamentally transforming recruitment and selection practices across organizations. AI-enabled technologies have expanded the ability of organizations to attract, screen, assess, and select candidates in a faster and more efficient manner. The literature consistently highlights improvements in operational efficiency, scalability, and decision support, particularly in environments characterized by large applicant volumes and increasing competition for talent. However, the effectiveness of AI is influenced not only by technological capabilities but also by organizational readiness, data quality, governance mechanisms, and human involvement throughout the hiring process.

A significant contribution of the reviewed literature is the recognition that AI adoption extends beyond automation and involves broader organizational, ethical, and human considerations. While AI can reduce administrative burden and support evidence-based decision-making, concerns relating to algorithmic bias, transparency, privacy, and accountability continue to influence stakeholder perceptions. The findings suggest that organizations must carefully balance technological efficiency with fairness and ethical responsibility to ensure that AI-enabled recruitment systems remain trustworthy and socially acceptable.

From a theoretical perspective, the review reinforces the relevance of Organizational Justice Theory, the Resource-Based View, and Socio-Technical Systems Theory in understanding AI-enabled hiring. Fairness perceptions influence candidate acceptance and organizational attractiveness, while strategic organizational capabilities determine the effectiveness of AI implementation. At the same time, the interaction between technological systems and human actors highlights the importance of maintaining alignment between AI capabilities and human judgment. These theoretical perspectives collectively suggest that successful AI adoption requires both technological sophistication and responsible organizational management.

The review also provides important implications for HR practitioners and organizational leaders. Rather than viewing AI as a replacement for recruiters, organizations should adopt AI as a decision-support tool that enhances human expertise. Recruiters should be equipped with the knowledge and skills necessary to interpret algorithmic recommendations, identify potential limitations, and ensure that hiring decisions reflect organizational values and ethical standards. Continuous training, AI literacy, and critical evaluation of automated recommendations are therefore essential for effective implementation.

At the organizational level, robust governance frameworks are required to ensure responsible AI adoption. Organizations should establish clear policies regarding data protection, algorithmic transparency, bias monitoring, vendor accountability, and human oversight. Regular audits and performance evaluations are necessary to ensure that AI systems operate fairly and consistently across different candidate groups. Such governance mechanisms not only reduce ethical and legal risks but also strengthen stakeholder trust in AI-supported recruitment processes.

Overall, the findings indicate that the future of recruitment and selection lies in achieving an appropriate balance between technological innovation and human judgment. Organizations that

successfully integrate AI within transparent, ethical, and human-centered recruitment systems are likely to achieve more effective, fair, and sustainable hiring outcomes.

Practical Implications of AI in Recruitment and Selection

Area	Implication
Recruitment Efficiency	Faster screening and sourcing
Recruiter Role	Greater focus on strategic decision-making
Candidate Experience	Improved communication and engagement
Governance	Need for transparency and accountability
Ethics	Bias monitoring and fairness audits
Human Oversight	Human review of AI recommendations

Interpretation

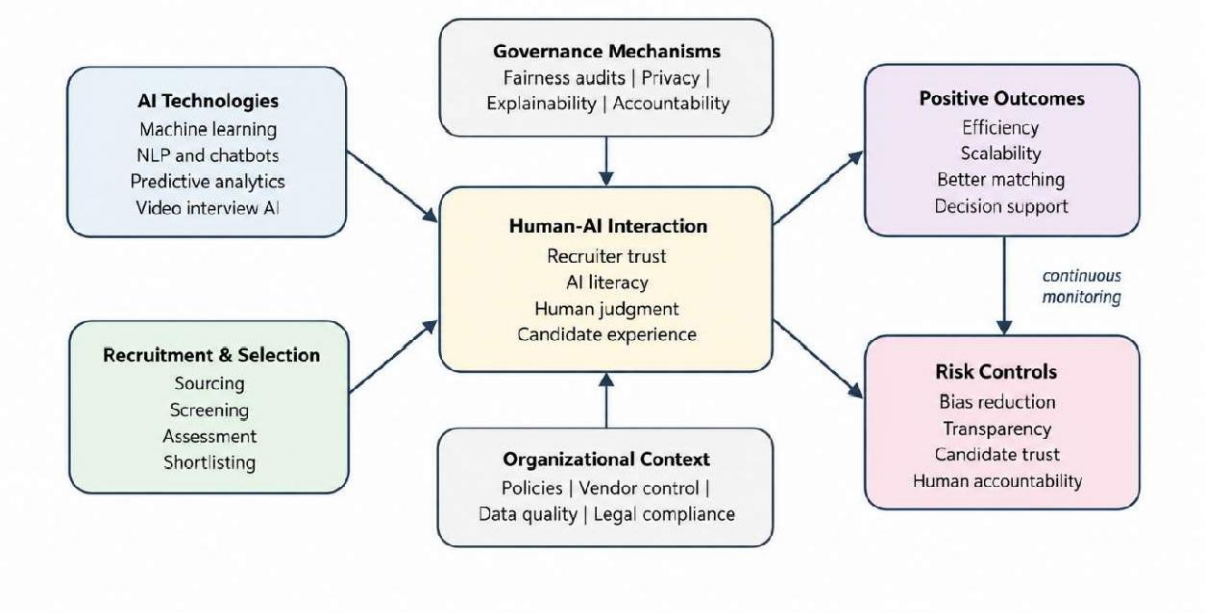
The findings indicate that AI has important implications for both organizational performance and human resource management practices. While AI contributes to efficiency and improved decision support, organizations must simultaneously address issues relating to fairness, transparency, and accountability. The role of recruiters is evolving from administrative processing to strategic evaluation and oversight of algorithmic recommendations. Effective implementation therefore requires a combination of technological capability, ethical governance, and human expertise. The literature consistently suggests that sustainable recruitment outcomes depend on balancing automation with meaningful human involvement. Consequently, organizations should adopt AI as a complementary tool that strengthens rather than replaces professional judgment in hiring decisions.

Benefits, Risks, and Governance Mechanisms

Benefit	**Risk**	**Governance Mechanism**
Faster Screening	Missing Qualified Candidates	Human Review
Better Matching	Algorithmic Bias	Bias Audits
Reduced Cost	Overdependence on AI	Recruiter Training
Predictive Analytics	Privacy Concerns	Data Protection Policies
Consistency	Lack of Transparency	Explainable AI

Interpretation

The analysis demonstrates that AI offers substantial benefits in recruitment and selection, including faster screening, improved candidate matching, reduced costs, and enhanced decision support. These advantages can significantly improve organizational hiring efficiency and effectiveness. However, the literature also identifies risks such as algorithmic bias, privacy concerns, lack of transparency, and excessive reliance on automated systems. If not properly managed, these challenges may negatively affect fairness and candidate trust. The findings highlight the importance of governance mechanisms such as human oversight, bias audits, explainable AI, and data protection policies. Responsible implementation is therefore essential for achieving sustainable and ethical AI-enabled recruitment outcomes.



Future Research Directions

The growing integration of Artificial Intelligence into recruitment and selection presents numerous opportunities for future research. Although existing studies have demonstrated the potential of AI to improve recruitment efficiency and decision quality, several important questions remain unanswered. Future research should move beyond examining operational benefits and focus on understanding the broader organizational, social, and ethical consequences of AI-enabled hiring systems.

One important area for future investigation is candidate perception and acceptance of AI-driven recruitment processes. Researchers should examine how factors such as transparency, explain ability, trust, and perceived fairness influence candidate reactions across different demographic

and cultural contexts. Comparative studies between AI-assisted and traditional recruitment methods would provide valuable insights into how applicants evaluate various hiring approaches and how these evaluations affect organizational attractiveness.

Further research is also required to explore algorithmic bias and fairness in recruitment decisions. Although AI is frequently promoted as a mechanism for reducing human bias, concerns remain regarding discriminatory outcomes arising from biased training data and algorithmic design. Future studies should investigate effective methods for identifying, measuring, and mitigating bias across different recruitment settings while ensuring equal employment opportunities for diverse candidate groups.

Another promising research direction involves examining the long-term effectiveness of AI-supported hiring decisions. Most existing studies focus on recruitment outcomes such as screening efficiency and candidate selection. However, limited attention has been given to post-hiring outcomes, including employee performance, job satisfaction, retention, career development, and organizational commitment. Longitudinal studies would provide a more comprehensive understanding of the actual value created through AI-enabled recruitment systems.

Future scholars should also explore the evolving relationship between human recruiters and AI technologies. As organizations increasingly adopt hybrid decision-making models, it becomes important to understand how human judgment and algorithmic recommendations can be effectively combined. Research focusing on human–AI collaboration, recruiter AI literacy, trust in algorithmic systems, and decision accountability can contribute significantly to both theory and practice.

Finally, future studies should investigate the regulatory, legal, and governance dimensions of AI-enabled recruitment. As governments and professional bodies introduce new guidelines regarding responsible AI use, researchers should examine how organizations adapt to emerging regulations and develop governance frameworks that ensure transparency, accountability, privacy protection, and ethical compliance. Such research will be essential for supporting the responsible and sustainable adoption of AI in recruitment and selection practices globally.

Conclusion

Artificial Intelligence has emerged as one of the most influential technological developments shaping contemporary recruitment and selection practices. This review demonstrates that AI is

transforming hiring processes through automated sourcing, resume screening, candidate assessment, predictive analytics, recruitment chatbots, and intelligent decision-support systems. These technologies offer substantial benefits in terms of efficiency, scalability, consistency, and data-driven decision-making, enabling organizations to manage increasingly complex recruitment environments more effectively.

At the same time, the review highlights that the successful adoption of AI extends beyond technological capability. Issues relating to algorithmic bias, transparency, explainability, privacy, accountability, and candidate trust continue to influence the effectiveness and legitimacy of AI-enabled hiring systems. The findings suggest that organizations must recognize AI not as a replacement for human judgment but as a tool that complements and strengthens human decision-making within the recruitment process.

The review further emphasizes the importance of human–AI collaboration as the most sustainable approach to future recruitment and selection. While AI can process large volumes of information and identify patterns efficiently, human recruiters provide contextual understanding, ethical reasoning, empathy, and professional judgment that remain essential for fair and responsible hiring decisions. Organizations that successfully integrate technological innovation with human expertise are more likely to achieve effective and equitable recruitment outcomes.

In conclusion, the future of recruitment and selection lies in balancing technological advancement with human values. Responsible AI adoption requires strong governance frameworks, transparent decision-making processes, continuous monitoring, fairness audits, data protection measures, and meaningful human oversight. When implemented ethically and strategically, AI has the potential to enhance recruitment effectiveness while promoting fairness, trust, and organizational sustainability. Therefore, human-centered and responsible AI adoption should remain the guiding principle for organizations seeking to leverage artificial intelligence in recruitment and selection.

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